

**NEGATION IN KHUZESTANI ARABIC AND SADAT TAWAHER SIGN
LANGUAGE**

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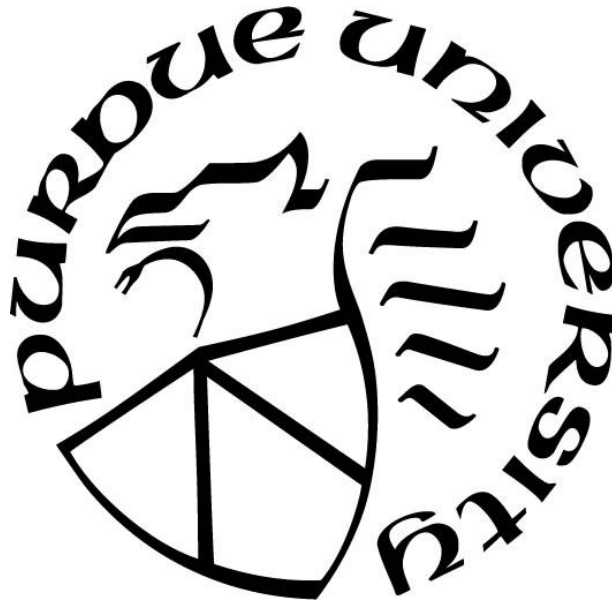
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In loving memory of my father (1945-2021)

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ABBREVIATIONS

abo	Air blown out
ACC	Accusative
ADV	Adverb
AdvP	Adverb Phrase
Affirm	Affirmative
ASL	American Sign Language
Asp	Aspect
Aux	Auxiliary
bht	Backward head tilt
bl	brow lowering
br	Brow raising
C	Complementizer
CL	Classifier
CSL	Chinese Sign Language
COND	Conditional
Det	Determiner
DGS	German Sign Language
DN	Double Negation
Emph	Emphatic
Exist	Existential
FinSL	Finnish Sign Language
GEN	Genitive
GSL	Greek Sign Language
hs	Headshake
ht	Head turn
HZJ	Croatian Sign Language (Hrvatski Znakovni Jezik)
IMPF	Imperfective
IPSL	Indo-Pakistani Sign Language
IX-1	Index 1 (First person)

IX-2	Index 2 (Second person)
IX-3	Index 3 (Third person)
JSL	Japanese Sign Language
KhA	Khuzestani Arabic
le	Lowered eyebrows
lhm	leftward hand movement
LIS	Italian Sign Language (Lingua Italiana dei Segni)
LIU	Jordanian Sign Language (Lughat il-Ishaara il-Urdunia)
LSB	Brazilian Sign Language (Língua de Sinais Brasileira, Libras)
LSC	Catalan Sign Language (Llengua de Signes Catalana)
NC	Negative Concord
Neg	Negative
NegP	Negative Phrase
NGT	Sign Language of the Netherlands
NZSL	New Zealand Sign Language
Poss	Possessive
PP	Preposition Phrase
PRF	Perfective
Proh	Prohibitive
PRTC	Participle
PST	Past
re	Raised eyebrows
RSL	Russian Sign Language
SJ	Signed Japanese
SASL	South African Sign Language
SF	Singular feminine
SL	Sign Language
SM	Singular masculine
STSL	Sadat Tawaher Sign Language
T	Tense
t	topic

TİD	Turkish Sign Language (Türk İşaret Dili)
TP	Tense Phrase
V	Verb
VGT	Flemish Sign Language (Vlaamse Gebarentaal)
VP	Verb Phrase

TRANSLITERATION CONVENTIONS FOR KHUZESTANI ARABIC

Consonants

IPA	Romanization	English approximation
ʔ	'sæʔæl 'he asked'	The pause in uh-oh!
b	'baʔʃir 'tomorrow'	bird
p	'pitzæh 'pizza	pizza
t	'tæmir 'date'	sit
θ	θar 'revenge'	three
dʒ	'idʒazæh 'permission'	jam
ʧ	'ʧibi:r 'big'	church
ħ	ħar 'hot'	hat , but pharyngeal
x	'xæbær 'news'	no equivalent, German <i>Bach</i>
d	dar 'room'	door
ð	'iðræh 'corn'	they
r	'mærræh 'turn'	Scottish <i>curd</i> , Spanish <i>perro</i>
ɾ	'rimæh 'he shot'	better
z	'zilmæh 'man'	zigzag
ʒ	ʒar 'neighbor'	pleasure
s	'saʃæh 'hour'	sun
ʃ	'ʃæmis 'sun'	shout
sʕ	'sʕæbær 'patience'	emphatic /s/, no equivalent
dʕ	'dʕæyətʕ 'pressure'	emphatic /d/, no equivalent
tʕ	tʕe:r 'bird'	emphatic /t/, no equivalent
zʕ	'zʕalæm 'oppressor'	emphatic /z/, no equivalent
ʕ	ʕæm 'uncle'	no equivalent
ʁ	ʁe:r 'other'	no equivalent, Spanish <i>fuego</i>
f	ə'fraʃ 'carpet'	fruit
q	'qasəm [proper name]	no equivalent
k	'kærim 'generous'	park
g	'gimær 'moon'	groom
l	'lazəm 'necessary'	lazy
ʈ	'aʈhæh 'God'	tool
m	'mæθæl 'proverb'	moon
n	nær 'fire'	name
h	'hælbæt 'maybe'	hire
w	'wæræ 'behind'	wall
v	'virus 'virus'	voucher
y	yæm 'beside'	yes

Vowels – Monophthongs

IPA	Romanization	English approximation
ɑ	'amær 'he ordered'	car
æ	ʃæm 'how many'	family
ʊ	mʊx 'brain'	put
u	ruħ 'spirit'	ruse
ɪ	'inti 'you'	pit
e	ræd' det 'I returned'	set
i	'mɑʃɪ 'left'	happy
o	'nobæh 'turn'	mourn
i:	di:n 'religion'	reed
e:	be:t 'house'	bird
ə	'ð'ahər 'appearance'	about

Vowels – Diphthongs

IPA	Romanization	English approximation
ɑɪ	maɪ 'water'	tie
ɑʊ	'ʒællɑʊ [proper name]	now
æʊ	næʊnæʊ 'kitty'	meow
æj	ħæj 'neighborhood'	

ABSTRACT

Human language is expressed mostly through either the aural-oral modality or the visual-gestural modality. In the aural-oral modality, humans encode information through speech via sound in the case of spoken languages while in the visual-gestural modality, information is encoded through gesture/sign as in the case of sign(ed) languages. Over time, attempts have been made to delineate the differences and similarities between spoken and sign languages on one hand, and between and across sign languages themselves on the other. These attempts fall within the field of cross-modal typology where attempts are made to propose a theory that would account for the differences and similarities found among various spoken and sign languages (Zeshan & Palfreyman, 2017). This dissertation brings spoken and sign language data together.

I report on a young sign language called Sadat Tawaher Sign Language (STSL) that emerged naturally, in complete isolation from any other sign language / manual communication system, and without any external intervention such as deaf education. This incident transpired around sixty years ago in a small village named Sadat Tawaher in southwestern Iran after a man lost his hearing completely. Given that STSL emerged and matured within the vicinity of the spoken language Khuzestani Arabic (KhA), I attempted to answer the question whether STSL is a copied version of KhA. STSL users mostly consist of the family and close friends of the deaf individual who use this sign system only to communicate with him, but not among each other. Since STSL users speak KhA as their native language and verbally say what they sign, I hypothesized that STSL would logically conform in its grammatical structure to that of KhA. I narrowed down this question by examining negation in both KhA and STSL and gathered signed productions, story-telling data, and grammaticality judgments from five participants. Thus, the participants were bimodal bilinguals since they were fluent in a spoken and a signed language. Multiple manual signs were attested in the data which we glossed as NEG_{basic}, NEG_{proh}, NEG_{poss}, NEG_{exist}, and another category, Other. Except for Other, each negator consists of two elements: a manual element and a non-manual element. These non-manual elements are as follows: backward head tilt for NEG_{basic}, side-to-side headshake for NEG_{proh} and NEG_{exist}, and air blown out of mouth for NEG_{poss}. STSL has a SVO basic word order but also allows SOV which seems to be a derived order. KhA has a rather flexible word order and, under certain circumstances, allows all six possible word orders. However, SVO is the most common word order in KhA.

The findings showed negation is strictly sentence-final in STSL whereas it is preverbal in KhA (i.e., S-Neg-VO in KhA and SVO-Neg in STSL). The results showed that in STSL the manual negators are required while the non-manual negators are optional. Thus, the manual negators may optionally be accompanied by non-manual negators. However, the non-manual markers were very frequently left out and the sentences were negated by use of manual negators only. Moreover, when present, the non-manual negator accompanies the manual negator only and never spreads on adjacent signs, constituents (e.g., VP) or the whole clause. Furthermore, in STSL the manual negator can negate the sentence on its own whereas the non-manual negator cannot do so. Thus, given these two facts, we classify STSL as typologically manual dominant which patterns with Italian Sign Language (LIS) and Turkish Sign Language (TİD) (Zeshan, 2006c).

In analyzing the relationship between the manual and non-manual negators in STSL, we argue that the non-manual negator is part of the lexical description of the manual negator. Thus, since the non-manual marker constitutes part of the phonological make-up of the negative manual sign, in the absence of the manual negator, the non-manual negator cannot negate the sentence on its own as it cannot be phonologically realized. Therefore, the manual negator, being obligatory, along with the manual negator which may or may not be phonologically realized, occupies the head position of NegP. Spec,NegP, on the other hand, is occupied by a covert negative operator. Given that in STSL the manual negator cannot be absent and the non-manual negator never spreads onto the VP or whole clause, we argue that there is not a c-command relation. Instead, a spec-head relation is established between the manual negator and the covert negative operator. Furthermore, the findings clearly showed that more than one manual negator may be used in STSL without changing the polarity of the sentence; i.e., the sentence remains negative. This indicates that STSL is a negative concord (NC) language. However, the use of two negative particles in the same clause results in an affirmative reading in KhA, indicating that KhA is a double negation (DN) language. In STSL, NC serves an emphatic purpose only. As a result, the first manual negator in NC constructions serves sentential negation while the other manual negator(s) are negative adverbials. Therefore, STSL conforms to a different word order in its negative constructions from that of KhA.

Furthermore, while producing negative sentences in speech and sign, the participants *simultaneously* adhered to the KhA grammar and STSL grammar, producing S-Neg-VO and SVO-Neg orders. This finding confirms that the human brain is capable of the simultaneous production of two different grammars when physical constraints are removed, that is, when language users

can encode information through two different channels, e.g., auditory-oral modality and visual-gestural modality. In conclusion, STSL has its own grammar independent of KhA likely due to modality effects of the channel through which it communicates information.

CHAPTER 1. INTRODUCTION

Human language is expressed mostly through either the auditory-oral modality or the visual-gestural modality. In the former, human beings encode information through speech via sound in the case of spoken languages such as English, Arabic, or Navajo. In the latter modality, information is encoded through gesture/sign as in the case of signed languages such as Chinese Sign Language (CSL), American Sign Language (ASL), or Georgian Sign Language (GESL). Thus, over time, scholars have attempted to delineate the differences and similarities between spoken and sign languages on one hand, and between and across sign languages themselves, on the other. Such attempts fall within the field of linguistic typology where attempts are made to propose a theory that would account for the differences and similarities found among various spoken and sign languages. Comparisons of this type are motivated by ‘modality effects’ which are defined as “differences between signed and spoken languages due to the way they are produced and perceived” (Zeshan & Palfreyman, 2017, p. 178). Zeshan and Palfreyman (2017) further divide such effects into two types: ‘absolute modality effects’ and ‘relative modality effects’. The former are found only in one modality, i.e., either spoken or signed, while the latter might be found more often in one modality than in the other.

Zeshan and Palfreyman (2017) also mention cases in which modality seems to have little impact on the grammatical structure. Woll and Sutton-Spence (2011) explain that while early modern research focused on the structural similarities between spoken languages and sign language, more recent researchers have emphasized the underlying typological differences between the two types of languages. Woll and Sutton-Spence (2011) explain that sign languages are characterized by two features: (i) “produced and perceived in the visual modality”, and (ii) “unrelated to the spoken languages which surround them” (p. 359). The authors further contend that the systematic typological differences between spoken languages and sign languages are believed to stem from the latter’s use of the visual modality, signing space, types of articulators (manual and non-manual), and gestures. In the same vein, Sadat Tawaher Sign Language (STSL), the sign language that is the focus of this dissertation, uses the gestural-visual modality although it emerged and grew in the vicinity of Khuzestani Arabic (KhA) which uses the aural-oral modality.

In the next sections, I will briefly describe some terminology that is of importance to this dissertation such as gesture, sign, and sign language typology since the current work incorporates spoken and signed language.

1.1 Speech, Gesture, and Sign

Gestures are “meaningful movements of the body, the hands, and the face during communication, which accompany the production of both spoken and signed utterances” (Özyürek, 2012, p. 626). It has long been known that speech is produced alongside and combined with gesture by humans. All human beings, hearing and deaf, produce gesture when they talk even when their addressee cannot see them (e.g., when on a phone conversation) or when there is no addressee in the first place (e.g., when they talk to themselves or think out loud). Blind people, who have never seen other individuals gesture when they speak, also gesture when they talk.

Research on gesture is highly interdisciplinary and borrows ideas and insights from such fields as anthropology, psychology, linguistics, among others. Another definition of gestures is “movements of the hand that occur in the context of communication” which can also include head and body movements (Cartmill, 2022, p. 456). According to Cartmill, gesture is located somewhere between what he terms ‘manual action’ such as adjusting one’s body for comfort and ‘manual language’ by which he means ‘signed language’ (p. 456). Goldin-Meadow and Brentari (2017) use ‘consistency in form’ as a criterion by which to distinguish between a gesture and a sign: if a consistent form is used among signers of the same language, then it is a sign; with a gesture, such consistency is not expected. Spontaneous co-speech gestures used frequently by hearing speakers are not the same as signs used by deaf signers in sign language. Co-speech gestures, which are found in all languages and cultures and which are used by every hearing person, are used subconsciously for communicative purposes (Fitch, 2010).

However, gestures from the hearing community do frequently get borrowed and incorporated into a sign language to become signs later in which case and as a result they become subject to grammatical constraints of that sign language (Zeshan & Palfreyman, 2017). Thus, while sign languages possess linguistic features of spoken languages, gesture lacks such features. Nevertheless, this does not mean that gesture lacks any structure. In fact, quite the contrary, gesture has some complex features. As Cartmill (2022) explains, gesture can become conventionalized in which case cultural knowledge is required to understand its associated meaning (e.g., ‘thumbs-up’

for approval). Some functions of gesture include the ability to point things not present, highlighting speech by indicating stress for example, and mapping the physical world onto abstract concepts, for example, by opening and moving one's hands away from each other to demonstrate expansion of an idea. Following McNeill (1992), Cartmill (2022, p. 457) mentions the following types of gesture:

- (a) Emblems, also called emblematic or conventional are gestures that have acquired conventionalized meanings, for example, the gesture for OK or thumbs-up.
- (b) Deictic or indexical gestures function as attention-getters by directing the addressee's attention to a specific location or in a certain direction in the environment by holding or pointing at something.
- (c) Beat (also known as batonic, rhythmic, or emphatic) gestures are characterized by rhythmic hand or body movements aligned with speech.
- (d) Representational gestures are divided into two types: iconic gestures represent some aspect of their referent such as shape or size, e.g., fingers wiggling and moving down to represent 'snow'. Metaphoric gestures represent abstract concepts or ideas, e.g., pointing downwards with the hand cupped to indicate 'today'.

Since William Stokoe's (1960) seminal work on American Sign Language (ASL), a sign traditionally consists of three main elements, namely, handshape, location, and movement. Later decades, however, witnessed considerable modifications to this view. Currently, a sign is seen as sublexically composed of five parameters: handshape, hand orientation, location, movement, and non-manual marker(s) if any (Zeshan & Palfreyman, 2017). Changing any of these parameters can potentially result in a new sign with a different meaning and thus create a minimal pair comparable to minimal pairs in spoken language although not identically because we deal with sounds in spoken language which are not the same as the parameters mentioned above for signs in signed language. The relationship between a sign and what it means can be non-arbitrary, i.e., iconic. The extent to which a sign is visually motivated depends on the context in which it is used and the referent that it represents. For example, in British Sign Language (BSL) and American Sign Language (ASL), the sign for TEA reflects the action of drinking from a cup and dipping a teabag, respectively (Woll & Sutton-Spence, 2011).

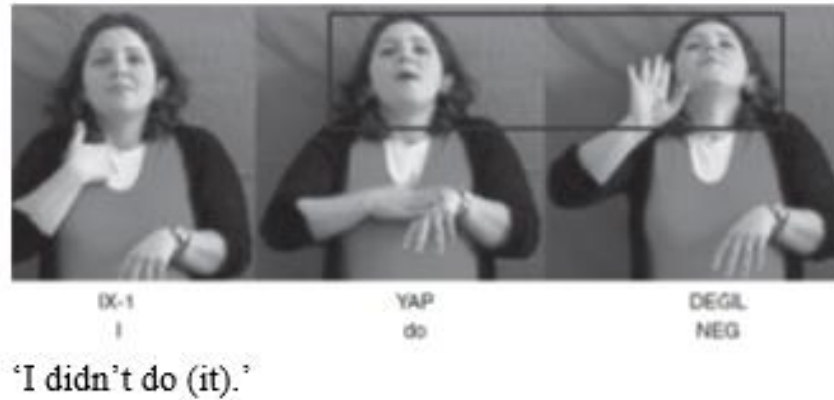
1.2 Manual and Non-Manual Markers

Manual markers (MMs) refer to the use of hands while signing whereas non-manual markers (NMMs) are facial expressions, head and body movements and postures such as brow raise, puffed cheeks, wrinkled nose, headshake, amongst others, that are used in all sign languages (Wilbur, 1979). It has long been observed and argued that non-manual markers have not only expressive and emotive functions such as displaying anger, happiness, etc., but also grammatical functions (Sandler & Lillo-Martin 2006). The non-grammatical functions of non-manual behaviors have been likened to suprasegmental features such as intonation and pitch which are found in spoken language. An example is that of the typically rising intonation found in yes/no questions in spoken language which might be taken as equivalent of brow raise, along with other non-manuals, in signed yes/no questions.

- (1.1) brow raise
 IX-2 BUY CAR
 ‘Did you buy a car?’

As Gökgöz (2021) points out, it is common to find the same non-manual marker(s) in (an)other sign language(s) and in the same grammatical structure. However, this fact should not be taken to mean that that NMM behaves the same way in all the sign languages in which it is found. For instance, Gökgöz (2021) mentions the backward head tilt (bht) which accompanies the manual negative marker in Turkish Sign Language (Türk İşaret Dili; TİD). The same NMM is found in Greek Sign Language (GSL) in the same grammatical structure, i.e., negation. However, in TİD, bht accompanies the manual negator and often spreads onto the verb to which the manual marker clicitizes (2) while in GSL, bht might spread beyond the verb (3).

(2.1)



(Gökgöz, 2021, p. 276)

(3.1)

bht

INDEX- 1 AGAIN GO NOT-B

‘I won’t go (there) again.’

(Gökgöz, 2021, p. 276, from GSL, Antzakas 2006, p. 265)

Gökgöz (2021) contends that such precaution in dealing with and interpreting NMMs is important because different NMMs play different functions in different languages and thus analyzing one and the same NMM should be done on a case-by-case basis.

1.3 Spoken Languages vs. Sign Languages

Sign languages are “the class of natural human languages which have arisen spontaneously within Deaf communities” (Woll & Sutton-Spence, 2011, p. 359). Brentari (2019) defines a sign language, for the purposes of her work on phonology, as any manual language that is used in and shared within a deaf community, produced on the hands, face, and body, is passed down generationally, and serves the linguistic and cultural purposes expected of a language (p. 2).

Thus, sign languages are the primary means of communication among deaf¹ people and are now treated as legitimate languages with their own unique linguistic structures. For a long time, however, sign language was not considered ‘language’ and was simply treated as gesture as reflected by the International Congress of the Educators of the Deaf’s decision in 1880 to condemn

¹ For the purposes of this dissertation, I avoid the ‘deaf’ / ‘Deaf’ distinction and will only use the term ‘deaf’ to refer to the person whose deafness motivated the current study.

the teaching of sign language to deaf children. A similar view of sign languages is even seen in Hockett's (1960) design features of language the first one of which, the vocal-auditory channel, ruled out sign language as language. One major reason for the reluctance to accept sign languages as languages was the fact that sign languages were compared to and seemed to be qualitatively different from spoken languages (Goldin-Meadow & Brentari, 2017). In other words, spoken language was considered the norm and sign languages were not 'real' languages because they 'defied' the norm by using a different modality than the one used in spoken language.

Although sign languages might be in close contact with the surrounding spoken language(s), they are not derived from the spoken language(s). Rather, sign languages possess their own independent linguistic structure. Additionally, although both spoken and sign languages transfer information sequentially, sign languages are capable of transmitting information simultaneously because signers have access to more than one articulator at the same time, i.e., the two hands. For instance, a signer can use his/her hands to sign two different referents in the signing space and the relationship between them. However, since humans have one vocal apparatus, they can only speak one word at a time though they can add additional information by resorting to such suprasegmental features as tone, prosody, etc. (Woll & Sutton-Spence, 2011).

Nevertheless, research has shown that spoken language(s) surrounding a sign language might strongly influence the development of various components of homesign systems such as word order (Fischer, 2014; Meir, Sandler, Padden, & Aronoff, 2010). Nyst (2007), for instance, reports that Adamorobe Sign Language (AdaSL), a village sign language used in Adamorobe, an Akan village in eastern Ghana, is highly influenced in its structure from the spoken language, i.e., Akan, as manifested in the loan translations and word order borrowed from this spoken language, to the extent that she refers to AdaSL as 'sign-supported Akan'. Similarly, Fischer (2014) has argued for the influence of surrounding spoken language on the constituent order of an emerging sign language.

Nevertheless, spoken and signed languages exhibit a good deal of similarities on all levels of linguistic analysis, e.g., phonology, morphology, and syntax (Goldin-Meadow & Brentari, 2017). For instance, consider English and Italian Sign Language (LIS). On a syntactic level, both languages use complementizers in relative clauses but in different ways: in English, the complementizer *who* appears at the left edge of the relative clause (4a) while PE (the complementizer in LIS) takes a right edge position in LIS (4b):

- (4.1) a. The man [**who** works downstairs] is my friend. [English]
b. UOMO [LAVORA DI SOTTO **PE**] AMICO MIO. [LIS]
(Examples b & c, respectively, from Goldin-Meadow & Brentari, 2017, p. 5)

1.4 Context of the Current Study

In this section, I will provide information about the context of the study including the province of Khuzestan, Khuzestani Arabic (KhA), the village of Sadat Tawaher, and the family in which STSL spontaneously emerged and grew.

1.4.1 Khuzestan

Khuzestan, indicated with number 15 on the map below, is one of the 31 provinces of Iran (official name: Islamic Republic of Iran) with a population of 4.5 million inhabitants. Spanning 63,213 square kilometers (24,407 square miles), Khuzestan is the largest province in the western side of Iran and is in the southwest of Iran. It borders Iraq and the Persian Gulf and its capital city is Ahvaz. Some major cities are Abadan, Behbahan, Dezful, Khorramshahr, Bander Imam, Shoush (Susa), Masjed Soleiman, Andimeshk, Mah Shahr, among others.



Figure 1.1. Province of Khuzestan, Iran
(https://www.iranchamber.com/provinces/15_khuzestan/15_khuzestan.php)

Khuzestan enjoys highly fertile soil, making it one of the most profitable regions for agriculture. Furthermore, as a result of its vicinity to the Persian Gulf, the seafood industry is another popular profession. Khuzestan also has a long history of civilization, dating back four thousand years BCE. Geographically, Khuzestan has both deserts and mountains, hence another reason for the booming tourism industry in this region.



Figure 1.2. Khuzestan, Iran (<https://miraskhz.ir/rha/>)



Figure 1.3. Khuzestan, Iran (<https://miraskhz.ir/rha/>)

1.4.2 Khuzestani Arabic (KhA)

Khuzestani Arabic (KhA) is a South-Mesopotamian Bedouin-type and a *galat*²-type dialect of Arabic spoken in the province of Khuzestan, among many other dialects and languages (Leitner, 2022). Although the Arabization of Khuzestan began with the immigration of Arab tribes to this province, Arab settlement in southern and western Iran (i.e., Khuzestan and Fars) dates back to the Sasanian times (226-651) (Leitner, 2022). Until very recently, there existed very few descriptions of KhA. The most recent dialectological contribution to KhA is Leitner (2022). KhA is a minority language and has witnessed considerable influence from Persian especially on its lexicon, but also on its phonology and syntax (see Shabibi, 2006; Matras & Shabibi, 2007). As we will discuss later, KhA has a relatively flexible word order with SVO as the most commonly used order. It also has gender (masculine and feminine), number (singular, dual, and plural), and person (first, second, and third) distinctions. Furthermore, KhA has verbal predicates (i.e., those with a main lexical verb other than *fan* ‘was’) and verbless predicates (i.e., those with the present tense form of the verb *fan* ‘was’). Adjectives follow nouns and adverbs appear freely in most positions in the sentence.

Figure 1.4 Depicts the concentration of population of Arab villages in Khuzestan.

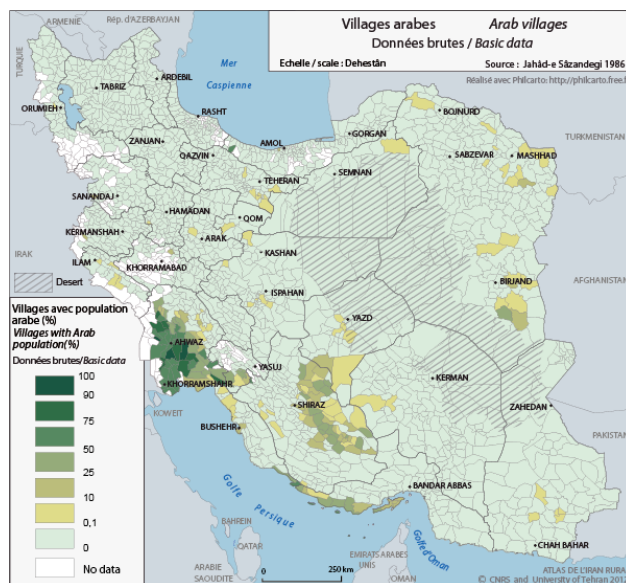


Figure 1.4. Arab Villages (basic data) in Khuzestan

(<http://www.irancarto.cnrs.fr/record.php?q=AR-040537&f=local&t=document&l=en>)

² *galat* is the first person singular perfective meaning ‘I said’.

1.4.3 Sadat Tawaher

Sadat Tawaher is a small village with a population of about 450 people and is located in the suburbs of Ahvaz, the capital city of Khuzestan. The only spoken language is KhA; however, most inhabitants know Persian to different degrees of proficiency as a second language and often use it in written communication. The majority of the inhabitants engage in agriculture, hunting, fishing, and livestock breeding. An important source of income for many of the inhabitants is a vast marshland that almost surrounds the village. There is one elementary school in Sadat Tawaher and most people at least have elementary education and quite a few have a high school diploma.



Figure 1.5. Village of Sadat Tawaher



Figure 1.6. Farming in Sadat Tawaher

1.4.4 Sadat Tawaher Sign Language (STSL)

The sign language system that is under scrutiny in this dissertation emerged naturally and spontaneously in Sadat Tawaher approximately sixty years ago after a man named *Hanash* lost his hearing. This man's family had no history of genetic deafness and there were no other deaf people in Sadat Tawaher. Allegedly, Hanash fell ill for a few days in his early twenties and then lost his hearing completely and became the first deaf person in that town. He was most likely single at the time and living with his own parents and siblings but got married soon after he became deaf and was married for at least 25 years until his wife passed away 31 years ago (at the time of writing this dissertation). Since Hanash's oldest child who is almost 55 does not remember when his father became deaf, we infer that his deafness must have occurred at least 55 years ago. Hanash had not gone to school and thus had no reading or writing literacy; however, since by the time he lost his hearing he had already fully mastered his native language (KhA), he could speak normally.

Therefore, Hanash never signed but used speech to communicate with others³. His vocabulary remained unchanged over the years, however, and differed from that of the younger generation. For instance, he referred to the ‘room’ and ‘bowl’ as *hiræh* and *gafyiæh*, respectively, while the new generation uses *dar* and *tʰasæh*. We will refer to the sign language as Sadat Tawaher Sign Language (STSL).

Today, everyone in Sadat Tawaher is hearing and, to the best of my knowledge, no other sign language system has ever been in use in this town than STSL. There was no deaf school near Sadat Tawaher when Hanash lost his hearing nor did his family and friends have any access to deaf education. Thus, his family had to creatively devise a sign system of their own to communicate with him. Note that, because of Hanash’s illiteracy, writing to him was not an option. This sign system was later learned by his extended family, neighbors, and friends. Later, this sign system was acquired by Hanash’s own children as their native language. His children also learned KhA simultaneously as their spoken mother tongue⁴. More recently, STSL has been acquired by a third generation, i.e., Hanash’s own grandchildren. Overall, STSL has been acquired by the following groups of people: (i) by Hanash’s own family and friends as soon as he became deaf (i.e., his siblings, uncles, friends) as a second language, (ii) by his children as one of their native languages, and (iii) by his grandchildren also as one of their first languages. Pictures of each generation are included below.

³ Indeed, most people who met Hanash for the first time mistakenly thought he was hearing!

⁴ This type of bilingualism in which an individual is competent in a spoken language and a signed language is called bimodal bilingualism.



Figure 1.7. Hanash



Figure 1.8. Hanash's Close Friends



Figure 1.9. Hanash's Grandson, Son, Neighbor, Friends & Hanash (Left to Right)



Figure 1.10. Hanash's Daughter-in-Law & His Grandchildren

1.5 Sign Language Typology

Sign languages come in different types and a basic knowledge of this field is helpful in determining where to situate STSL in the typology of sign languages. In this section, I will present a brief description of the different types of sign languages that have been described in the literature. My rationale for including this section is two-fold: (i) to familiarize the reader with the typology of sign languages and (ii) to argue that STSL does not in fact fit into any of the types of sign languages described thus far; rather, it might be more accurate to regard STSL as forming its own category. A complete picture of the category that STSL would fit into is beyond the scope of this dissertation. Thus, I will not go into the details as a brief description will suffice here.

Sign language typology lies at the intersection of sign language linguistics and linguistic typology and aims to uncover and account for the patterns of similarities and differences that are found among different sign languages (Zeshan & Palfreyman, 2017). Thus, this field has two orientations: (a) to expand the field of linguistic typology to include sign languages and (b) to investigate the linguistic diversity within sign languages (p. 178). Since linguistic typologists take an interest in all human languages, regardless of the modality in which they are expressed, sign language typology is argued to bring the two subfields of spoken and signed language typology closer with the aim of developing a theory that accounts for the variation among human languages in general. In this regard, the current study falls within cross-modal typology, a field which is characterized by the systematic inclusion of both spoken and signed language data (Zeshan & Palfreyman, 2017).

The field of sign language typology is, however, plagued by a few quandaries: (i) The number of documented sign languages, especially those fully so, compared to that of spoken languages is very small; (ii) As a result of the first issue, the sign languages used in sign language typology research have been unintentionally heavily biased in favor of specific geographical areas, mostly Western Europe; (iii) The issue of genetic relationship between sign languages has not been applied nor well understood in sign language typology research.

According to Bauer (2012), the seminal work of scholars like Stokoe (2005 [1960]) in the United States of America in the 1960s is considered the pioneering work on American Sign Language (ASL) that began the field of sign language research. Bauer describes four types of sign language. The first type is homesigns (also called home signs) which refer to “[...] idiosyncratic sign languages or gestural behavior that is developed when deaf individuals are

isolated from other deaf people and need to communicate with the hearing people around them” (Frishberg, 1987, p. 128). A more recent definition of homesigns is “systems of gestural communication, typically limited to a single family household and the few other communication partners of a single deaf individual” (Coppola & Senghas, 2010, p. 546). Two features that result in the development of a homesign system are: (i) the child’s inability to learn the spoken language and (ii) his/her lack of exposure to sign language (Goldin-Meadow, 2012).

The second type of sign language discussed by Bauer (2012) is hybrid-sign systems. These are sign systems that represent the spoken language and that are developed for educational purposes. Examples of these include Manually-Coded English (MCE) in the United States and Lautsprachbegleitende Gebärden (LBG, speech supported signs) in Germany. According to Bauer, hybrid systems are not considered natural languages but artificial systems because (a) they manually represent the spoken language, (b) they are not used for communicative purposes outside the education environment, and (c) they did not evolve naturally (p. 12).

Bauer (2012) then discusses four types of sign languages which are as follows. The first group is what she, following Padden (2011), refers to as Deaf community sign languages (DCSLs). These include American Sign Language (ASL) Australian Sign Language (Auslan), and Russian Sign Languages (RSL), among others that have also been referred to by such terms as *national* sign language, *urban* sign languages, *established* sign languages, *macro-community* sign languages (Bauer’s italics). DCSLs are normally found in large, urban areas, are formed in different locations like schools by deaf people from different backgrounds and countries, and are learned by only a small number of the hearing community. DCSLs are characterized by “the presence of the formal separate institutions of or for deaf people, including special schools, deaf associations and other deaf organizations (such as deaf theaters, dance groups, clubs, etc.)” (p. 13). The second group of sign languages discussed by Bauer is *emerging sign languages* which are different from DCSLs by one important feature: young age. This feature allows researchers to trace and follow their development over time. Emerging sign languages are the product of contact between two or more existing sign languages or homesigns and can be compared to spoken pidgins and creoles. Examples include Nicaraguan Sign Language (NSL), the Israeli Sign Language (ISL) and the Mauritian Sign Language (MSL). The third group of sign languages is *village sign languages*, also referred to as ‘indigenous’ or ‘shared sign languages’, are distinguished from DCSLs on various grounds: (i) socioeconomic and demographic settings, (ii) social homogeneity,

(iii) (socio)linguistic context and (iv) degree of endangerment (Bauer, 2012, p. 17). Regarding these differences, village sign languages develop in rural communities which are small, isolated, and close-knit and which share a social and cultural environment. In such sign languages, there is high incidence of genetic deafness (Pfau, 2012) and deaf people form an integral part of the village, not an isolated group with hearing people being fluent in the sign language. Furthermore, village sign languages are in danger as a result of contact with a larger sign language and death of deaf people. Examples of village sign languages include Martha's Vineyard Sign Language (MVSL), Kata Kolok, Al-Sayyid Bedouin, and Adamorobe. The last group is *alternate sign languages*, also referred to as 'secondary' or 'auxiliary' sign languages. These sign languages arise when hearing individuals, who have normal and complete access to spoken language, wish to find alternative ways of communication besides speech because of various circumstances such as noisy environments, hunting, religious purposes, etc. As a result, alternate sign languages are heavily influenced by spoken languages and differ from each other with respect their origin, linguistic structure, and function (p. 34). Examples of such sign languages include Sawmill Sign Language developed by hearing sawmill workers in British Columbia (Canada) and monastic sign languages developed in different monasteries by monks to observe silence as a religious practice.

In the next section, I will briefly introduce word order prior to discussing negation since a basic understanding of word order is necessary to gain insights into the position of negation in the broader clausal structure of STSL.

1.6 Typology of Sadat Tawaher Sign Language (STSL)

In this section, I will briefly attempt to situate STSL within the classification of sign languages. First, STSL is not a homesign since the deaf person was not born deaf but was deafened later in life, long after he had fully acquired his native language and used it for many years. Furthermore, he did not sign to communicate with others as he could speak his native language, KhA, fluently and with no difficulty whatsoever. Thus, he did not play a role, or at least not a key one, in creating STSL although it was created for his sake; rather, other hearing members had to come up with the signs that were needed for communication with him. Thus, it was not the deaf person but other people such as his family and friends who came up with this sign language to communicate with him. Note that in a homesign, it is the deaf child who uses gestures to communicate with his/her hearing parents who then adopt those gestures to communicate with

him/her. Second, STSL is not a village sign language either as the deaf individual in this study was the only deaf person in his hometown, Sadat Tawaher, while village sign languages are used in communities with high incidence of deafness (Pfau, 2012). In a village sign language, a group of deaf signers get together to create a sign system to communicate with one another. Lastly, STSL is not a community sign language since it was the only sign system that was used in Sadat Tawaher, and not a combination of different sign languages. This sign language does not seem to fall into any of the categories of sign language typology outlined in the literature.

From the description above of various types of sign languages, it becomes clear that STSL is not a homesign, a village sign language, or a community sign language, raising the question as to what type of sign language STSL is if it is none of the ones mentioned above. Therefore, we suggest that STSL needs to be in its own category of emerging sign languages, one that, to the best of my knowledge, I have not seen in the literature. This is another issue worth further investigation in the future as it might show that the visual modality alone might be enough to turn a gesturally based system of communication into a natural sign language. Also worth examining is whether STSL is a ‘secondary sign language’ (Pfau, 2012) which is used by hearing people to communicate with each other when use of speech is not possible due to, for instance, immense noise as in saw mills. This is a complicated issue because STSL was developed and used by *hearing* people not for communication among themselves but for communication with a *deaf* person.

One could further argue STSL might be a manually-coded communication system like Manually-Coded English (MCE) in the United States and Nederlands met Gebaren (Sign-supported Dutch) in the Netherlands which can show great variation among themselves from representing the copula verb *be* and bound morphemes like *-ing* to being closer to some ‘true’ sign languages by preserving such features as non-manual marking while still following the grammar of the spoken language (Pfau, 2012). STSL does not seem to be a manually-coded communication system either since it does not exhibit the structural properties of KhA in its grammatical system, as we will see in the case of negation. Finally, STSL could simply be a new type of sign language, one of the further types that Zeshan (2008) hypothesized would yet to be discovered.

1.7 Objectives and Significance

The main objective of this dissertation is to carry out a comparison of a spoken language, i.e., Khuzestani Arabic (KhA), and a sign language that we will refer to as Sadat Tawaher Sign

Language (STSL). Our purpose is to uncover the underlying structural similarities and differences between KhA and STSL by examining the way these two linguistic systems, which use different modalities, express negation. In some sense, thus, the current work falls within the category of language typology, i.e., “the systematic study of cross-linguistic patterns and cross-linguistic variation, in other words, of similarities and differences among languages” (Dahl, 2010, p. 9). More specifically, this study falls within the field of cross-modal typology in which a systematic use is made of data from both spoken and signed languages (Zeshan & Palfreyman, 2017). Here, I will cover negation in general including the negation of constituents, sentences of various types, and negative responses.

However, I will give special attention to *standard negation*, that is, “the basic way(s) a language has for negating declarative verbal main clauses” (Miestamo, 2005, p. 1). That is because declarative sentences are the types of sentences frequently used to identify the basic word order of a language.⁵ However, I am not concerned by the lack of a definition of ‘standard’ in this dissertation. Instead, I will follow Payne (1985) in including non-verbal predicates such as those with present tense *be* verb in Khuzestani Arabic (KhA) and weather predicates that can have zero valency, e.g., *rain*. In a nutshell, for the sake of this dissertation, I define standard negation as the negation of a sentence (i.e., clausal and sentential negation), whether with a verbal predicate or non-verbal predicate, with or without an agentive subject. I will also use ‘negative marker’ and ‘negator’ interchangeably for convenience in referring to the negative particles in KhA and the negative signs in STSL.

The current study of STSL will contribute to the field of linguistic typology in two ways. First, it will add to the repertoire of sign languages documented so far although STSL as we explained in section 1.6. Secondly, STSL is used in an area where no sign language had ever been previously documented or reported. This second fact is important here since STSL is no longer used by anyone due to the deaf person’s passing away about two years ago. Thus, STSL will soon die and this dissertation serves as a way to document it.

⁵ As noted by Dahl (2010), the term ‘standard’ in standard negation has not been explicitly defined and there is no clear reason why the imperative ‘*Go!*’ and present tense verbless sentences in some languages like Arabic should be considered non-standard compared to the declarative ‘*I went to the store*’ and their verbal counterparts in other languages like English, respectively.

CHAPTER 2. REVIEW OF LITERATURE

In this chapter, I will review negation which is the focus of the current study. Since a basic understanding of the word order of declarative sentences serves a baseline to understanding the clause structure of a language, we start off with word order to familiarize the reader with this area in the service of negation.

2.1 Word Order

Word order is a central issue in syntax as it deals with the linear order of the units of a linguistic expression. For a long time, linguists have recognized that word order is a critical concept in any given language as it was taken to mean as grounded in human beings' faculty of language (Frey, 2015). The search for word order universals was initiated by Greenberg (1963) who attempted to find overall regularities for word order among various languages. Frey (2015) cautions that this search presupposed that it was possible to identify subjects, objects, etc., across all language and that all languages have a basic word order which might be true of languages with restrictive word orders. This latter presumption is challenged once one considers languages with 'free' word order (e.g., Latin or Polish), languages with varying word orders (e.g., German), or languages with discontinuous constituents (e.g., Latin).

Nonetheless, formulating a basic word order for a language is done by considering the position of the subject, object, and main verb in a declarative clause as well as the position of other constituents such as nouns, adjectives, etc (Frey, 2015). In this regard, most languages are SOV (e.g., Japanese, Mongolian, Turkish,) or SVO (e.g., English, Chinese, French, Igbo). A smaller number of languages have a VSO order such as Arabic and Welsh. As Frey (2015) states, in searching for the basic word order of a language, one must consider clauses with full NPs, not weak pronominals. For example, French, a Romance language, has an SVO order when the clause has a subject and object that are full noun phrases (Jean a acheté une pomme 'John bought an apple') but an SOV order when the object is a pronominal (Jean l'a mangé 'John ate it'). Word order refers to "the order of the words in some phrase or construction, most often a sentence" (Munro, 2013, p. 118). For instance, English is considered a rigidly Subject-Verb-Object (SVO) language:

(2.1) The man fixed the car.

Word order plays an important role in the linguistic analysis of any language since it relates to the phrase structure of that language (De Quadros & Lillo-Martin, 2010). Most, if not all, languages, spoken and signed, seem to have a dominant *basic* word order, although such a word order can be, and in fact is, altered for a variety of reasons that have to do with such syntactic processes as object shift, focus, and topicalization (De Quadros & Lillo-Martin, 2010). Thus, an important notion in word order research is that of the ‘basic’ or ‘canonical’ word order which is based on the “unmarked simple clauses” used in pragmatically neutral settings.

Therefore, the above syntactic operations are applied to the underlying word order to produce the basic word order in the surface structure. While there are six possibilities to combine the subject (S), object (O), and verb (V): SOV, SVO, VSO, VOS, OVS, and OSV, languages seem to show a strong preference for certain word orders while disfavoring other orders. According to the World Atlas of Language Structures Online (WALS, <https://wals.info/>), the first two word orders, i.e., SOV and SVO, are used in 564 and 488 languages respectively, thus accounting for 76% of the 1376 spoken languages considered. On the other hand, the last two word orders, OVS and OSV, account for only 15 languages. WALS also reports that 189 languages lack a dominant word order. Napoli and Sutton-Spence (2014) believe that languages, both spoken and signed, prefer to keep V and O adjacent and therefore favor SOV and SVO.

Furthermore, they mention SOV as being grammatical in all sign languages. Therefore, the subject is immediately identifiable through word order because, everything else being equal, it almost always precedes the verb and object as noted by Greenberg (1963). Research has shown that SVO and SOV are the most common word orders across the world’s sign languages (Quer, Cecchetto, Donati, Geraci, Kelepir, Pfau, & Steinbach, 2017).

The basic word order refers to the position of the nominal subject, nominal object, and verb in affirmative main transitive declarative sentences. If in such sentences a language allows more than one word order, the word order that speakers find the least marked (i.e., most neutral) and use most frequently is considered the basic word order (Munro, 2013).

- (2.2) a. Who fixed the car and the bike? was it the man?
b. No, the man fixed the car. The bike the boy fixed.

The second sentence in (2.2b), i.e., *The bike the boy fixed*, is an example of a sentence that has an OSV order to imply a specific pragmatic effect, in this case, topic fronting. Therefore, sentences with fronted topics are not used to identify the basic word order of a language. The reason that the subject and object arguments need to be nominal to examine the basic word order in a language is that many languages (e.g., French, the Chickasaw language of Oklahoma, Munro, 2013, Farsi, personal knowledge) have different word orders for pronominal subjects and objects as opposed to nominal subjects and objects. For instance, Farsi follows an SOV order in sentences with nominal subject and object (3a) but this word changes to SVO if the object is pronominal (2.3b):

- (2.3) a. pesære ketab ro xund Farsi
 boy.DEF book OBJ read.3S.PRF
 ‘The boy read the book.’
- b. pesare xunde-ŋ Farsi
 boy read.3S.PRF-it
 ‘The boy read it.’

There are other types of factors and grammatical structures that can impact the basic word order in a language. One factor is voice, i.e., whether the sentence is active or passive. For instance, in English passive voice sentences, the object, which appears post-verbally in active voice transitive sentences, appears in subject position. Thus, although both (2.1) and (2.4) have their subject positions filled, the former, i.e., the active sentence, has a ‘doer’ subject in that position while the latter, i.e., the passive sentence, has a non-agentive subject in the same position. Thus, passive sentences are not considered neutral:

- (2.4) The car was fixed by the man.

Other factors that impact the basic word order include aspectual features of verbs (e.g., active vs. stative verbs in Middle Egyptian), negation (e.g., the Bafut language of Cameroon), type of clause (e.g., main vs. subordinate in German), type of object (e.g., nominal vs. pronominal objects in Spanish or French), and the type of sentence (e.g., declaratives vs. questions in English). Typological research has also focused on sorting sign languages by the basic syntactic order of subject, verb, and object. In this regard, it has been argued that sign languages appear restricted to SOV and SVO, likely due to their use of visual modality and signing space (Napoli & Sutton-Spence, 2014). In this regard, young languages offer a potential opportunity for investigating these hypotheses (Senghas, Coppola, Newport, & Supalla, 1997). Typological research on word order started with Greenberg's (1963) seminal work in which he proposed 45 linguistic universals from a sample of 30 languages. His major finding was that there was a correlation between the basic order of the major constituents of subject, verb, and object, and that of minor elements relative to each other such as noun and adjective, noun and genitive, adposition and noun.

Despite the striking similarities between spoken and signed languages, there are constraints on sign language constituent orders due to their use of the visual-gestural channel for communication (Fischer, 2014). Thus, a considerable amount of sign language research has emphasized the role and effect of the visual modality on constituent order. For instance, in a survey study of sign order in 42 sign languages, Napoli and Sutton-Spence (2014) conclude that pressures from the visual modality make sign languages tend to have SVO or SOV orders in declaratives since "the patterns for sign order in sign languages are determined by what makes sense visually" (p. 1). They attribute the variety in sign order to such factors as the data collection methods used, the theoretical approach adopted, stylistic and grammatical issues, or even to how a clause or sentence is defined.

Research shows that even sign languages that allow a variety of word orders in fact have a basic, unmarked, dominant word order. For instance, De Quadros and Lillo-Martin (2010) argue that although both American Sign Language (ASL) and Brazilian Sign Language (LSB) have SVO as their basic word order, other word orders are also possible as a result of different syntactic operations. They also speculate that the type of verb, i.e., whether it is an agreeing verb or a plain verb, influences the syntactic behavior. An agreeing verb is one which "displays person or location inflection" (De Quadros & Lillo-Martin, 2010, p. 227) for instance through eye gaze and shift in body orientation while plain verbs lack such agreement. Various types of agreeing verbs have been

mentioned in the literature: transitive verbs such as HELP in Russian Sign Language (RSL) which involve a movement from the subject to the object, ditransitive verbs such as SEND in which the movement is from the subject to the indirect object, and ‘backward agreeing verbs’ like INVITE where there is movement from the direct object to the subject (Kimmelman, 2022, p. 25). Thus, it is important to note that although sign languages often conform to a basic word order, the aforementioned factors might result in a different order. For instance, while ASL is argued to be an SVO language (Fischer, 2014), the verb might end up in sentence-final position because the object is topicalized as in (2.5):

(2.5) GIRL IX_a BOY IX_b bPUSH_a

‘The boy pushed the girl.’

(Example 1 from Fischer, 2014, p. 6)

In (2.5), PUSH is an agreeing verb and the indices *a* and *b* on the verb show the object and subject arguments, respectively. Thus, switching the indices results in a different order of arguments: ‘The girl pushed the boy’.

The following criteria have been mentioned in the literature for identifying the basic word order in a language: (a) the most frequent, (b) the least pragmatically marked, and (c) less morphologically marked (Quer, et al., 2017, p. 376). As for sentence type, simple (as opposed to complex), active (as opposed to passive) declarative sentences are used as a criterion for determining the basic word order of a language since, in comparison to other types of sentences, they are the most common and the most pragmatically neutral sentence type in any language (Kimmelman, 2012; Napoli & Sutton-Spence, 2014; Quer, et al., 2017).

Two semantic factors believed to influence the word order in many SLs are reversibility and animacy (Kimmelman, 2012). Reversibility deals with the ordering of arguments in the positions of the agent (subject) and theme (object). With reversible arguments, both the agent and theme can perform the action on each other, e.g., *The teacher met Sally*. In such cases, our only clue to know which argument is the agent and which one the theme is the word order (Quer, et al., 2017). It has been found that, in ambiguous sentences, sign languages prefer SVO with reversible predicates (Kimmelman, 2012), and SOV with non-reversible ones. It is then argued that the SVO order is preferred in reversible clauses to avoid ambiguity / confusability by inserting the verb between and thus separating the subject and object (Milković, Bradarić-Jončić, & Wilbur, 2006).

Since an ordering of NP1 NP2 V in reversible sentences can be interpreted as OSV or SOV, that is, the first NP can be interpreted as either the subject or object, sign languages might have at their disposal strategies to disambiguate such sentences. For instance, in Russian Sign Language, OSV is a marked word order because of the prosodic cues that mark O in this order which indicate topicalization.

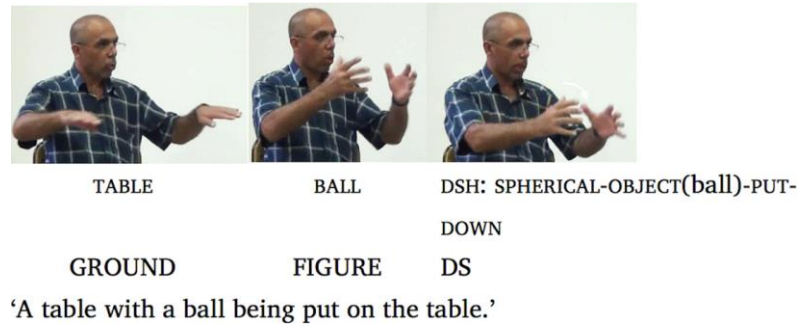
Preference for SVO order in reversible sentences with plain verbs is also confirmed by Napoli and Sutton-Spence (2014) which they believe to be a generalization true of sign languages. In a recent study on word order in Serbian Sign Language (SZJ), Bajić, Vermeerbergen, Schembri, and Herreweghe (2021) analyzed 810, mostly single, clauses consisting of reversible, non-reversible, and locative clauses. They found that in transitive non-reversible and reversible clauses, SVO is the most frequent word order, followed by SOV. In SOV clauses, the first constituent was found to be topicalized. The authors conclude that SVO should be the neutral word order in SZJ for transitive clauses as it disfavors topicalized constituents. The most common order in intransitive clauses was found to be SV. They also found verb-only clauses, i.e., clauses in which the two arguments of subject and object are not expressed overtly as in (2.6). The authors note that such clauses always follow another clause in which both of those arguments are overtly expressed and thus implied and represented through in the verb-only clause.

(2.6) GRANDMA LOVE CHILDREN // HUG
 S V O V
 ‘Grandma loves children. (She) hugs (them).’

(Example 35 from Bajić, et al., 2021, p. 18; modified)

Furthermore, in non-locative clauses, they found verb type, reversibility, and topicality to be three significant factors that influence word order in SZJ. In locatives, the signers used 15 different patterns with first-GROUND-then-FIGURE being the most frequently used order followed by a first-FIGURE-then-GROUND order. Locatives also exhibited the most simultaneous constructions mostly in clause final position. The authors found that simultaneous constructions were the most preferred used way of expressing a locative relation (2.7) followed by a preposition as the next strategy (2.8).

(2.7)



(Example 39 from Bajić, et al., 2021, p. 21)

(2.8)



(Example 40 from Bajić, et al., 2021, p. 21)

In another study, Coppola and Newport (2005) examined argument structure among three isolated Nicaraguan homesigners by focusing on the grammatical category of subject. Their objective was to investigate whether the homesigners would exhibit different word orders when dealing with agentive subjects (e.g., ‘Woman runs’) as opposed to non-agentive subjects, i.e., subjects of different thematic roles such as experiencer (e.g., ‘Woman sees man’). They found that the homesigners used the clause-initial position to encode both agentive and non-agentive subjects, rather than reserve this position for agents only. Thus, the thematic role of the subject of the sentence did not play a role in its appearance in the sentence, hence the evidence for the existence of the grammatical category of subject as a universal property of human language. Supalla (1995, cited in Senghas, et al., (1997) argues that the rich morphology that sign languages develop manifests itself in their incorporation of directional and spatial markers into the verb to express grammatical relations.

As Leeson and Saeed (2012) show in a study of word orders of different well-studied sign languages, overall, sign languages seem to show strong preference for two main word orders, SVO and SOV, in ordering their constituents / words. Examples of SVO sign languages include ASL, LSB (Brazilian Sign Language), Hong Kong Sign Language (HKSL), and Taiwan Sign Language (TSL). Examples of SOV sign languages include DGS (German Sign Language), JSL (Japanese Sign Language) and LIS (Italian Sign Language) (Fischer, 2014). It has been argued that sign language argument structure is heavily influenced by modality effects, such as iconicity, simultaneity, and use of signing space (Kimmelman, 2022).

In the next section, I will review some of the research that has been carried out on negation in both spoken and sign languages.

2.2 Negation

Negation is a unique and universal feature of all natural languages (Horn, 2001). Both spoken and signed languages have negation available at their disposal (Gökgöz, 2021). Thus, negative constructions constitute a central domain of the grammar of all languages, both spoken and signed, since speakers of any language need to negate sentences (Zeshan, 2006b). These constructions have been extensively studied cross-linguistically, providing a valuable opportunity to conduct language typology research. Cross-linguistic variation with regard to negation includes the form, number, and position of negative elements (Zeijlstra, 2013). Negation is the opposite of affirmation. To negate a sentence means “to reverse the truth value of the propositional content of a sentence” (Zeijlstra, 2013, p. 793).

In terms of markedness, negation is considered marked and affirmative unmarked. For instance, it is a well-known fact that while negation is always overtly morphologically marked, affirmative sentences have no such marking (Greenberg, 1966b). Negation has received considerable attention in scholarly literature and has been, to quote Dahl (2010, p. 9), ‘a low-hanging fruit’ in typological research for such features as its straightforward semantics across languages. Zeijlstra (2013) notes that in all languages studied so far negative sentences are always marked in comparison to affirmative ones. In other words, in no languages are affirmative sentences marked and negative ones unmarked. The markedness of negative sentences manifests itself on other levels and thus, as Miestamo (2005) notes, negation can be examined from different perspectives. From a cognitive viewpoint, negative sentences require longer processing time than

their affirmative counterparts. Pragmatically, in contexts in which negative sentences are used, the corresponding affirmative sentences are present as the background information. Finally, semantically, negation results in a reorganization of semantic domains.

2.2.1 Constituent vs. Sentential Negation

In examining negation, an important distinction is often made between constituent negation and sentential negation. Both express negation; however, constituent negation involves the negation of a single constituent while in sentential negation the whole sentence is negated. Consider the following example from Zeijlstra (2013, p. 794) which the author dates to Jackendoff (1972):

- (2.9) a. with no clothes is Sue attractive
b. with no clothes Sue is attractive

Examples (2.9a) and (2.9b) produce completely different readings. In (2.9a), Sue is not attractive no matter what she wears while (2.9b) does not deny Sue's attractiveness although under certain circumstances. (2.9a) and (2.9b) are also syntactically different: (2.9a) involves V-to-C movement while (2.9b) does not. This is because the preposition phrase (PP) *with no clothes* negates the whole sentence in (2.9a) while in (2.9b) only the PP *with no clothes* itself is negative and *Sue is attractive* remains affirmative. Thus, (2.9a) and (2.9b) involve sentential and constituent negation, respectively. Citing Klima (1964), Zeijlstra (2013) also provides diagnostics for sentential negation such as the fact sentences involving sentential negation are grammatical with positive question tags and *either* while those with constituent negation are grammatical with negative question tags and *too*. A detailed analysis of these tests and of the criticisms leveled at the mutual exclusivity of sentential and constituent negation is beyond the scope of this dissertation and will not be carried out here. In this regard, Zanuttini (2001) views understanding sentential negation as key to understanding grammar due to the interaction of negative markers with each other and with other grammatical information such as tense and mood as well as their sensitivity to other elements in the clause.

2.3 Negation in Spoken Language

Numerous attempts have been made to analyze the negation systems of the world's languages from a typological perspective including Klima (1964), Dahl (1979), Payne (1985), Zanuttini (2001), Dahl (2010), Miestamo (2005), among others. In a state-of-the-art crosslinguistic overview of negative elements, Zanuttini (2001), citing Payne (1985), mentions four major classes of such elements through which sentential negation is realized in the languages of the world: (i) negative verbs, (ii) finite auxiliaries, (iii) negative particles, and (iv) negative markers in the form of derivational morphemes. These negative elements are described below.

The first class, negative verbs, includes negative markers that behave like main verbs that take sentential complements but are also different from main verbs in their sensitivity to the aspect marking of the clause. The brackets indicate clausal boundary.

- (2.10) Na'e 'ikai [ke 'alu 'a Siale] Tongan
AspNegAsp-go AbsoluteCharlie
'Charlie didn't go.' (Example 1 from Zanuttini, 2001, p. 513)

The second class consists of negative markers with the properties of finite auxiliaries in that they carry such information as tense, aspect, number, gender, etc., and a non-finite participial form of a lexical verb as the complement.

- (2.11) Bi ə-ə-w dukuwən-ma duku-ra. Evenki
I-NegPast1Sg letter-Obj write-Part
'I didn't write a letter.' (Example 2 Zanuttini, 2001, p. 513)

In the third strategy, negative particles, a negative marker is used that has the following features: (a) it can be sensitive or insensitive to aspect, mood, or tense, (b) it might immediately precede the verb, and (c) (if occurring preverbally), it might be 'reinforced' by an element in postverbal position as in the well-known example of the French *ne* and *pas* or Welsh *nid* and *ddim*:

- (2.12) *Nid* yw'r bachgen (*ddim*) yn hoffi coffi. Welsh
Neg is-the boy Neg in like coffee

‘The boy does not like coffee.’

(Example 3 from Zanuttini, 2001, p. 513)

The last strategy consists of derivational morphemes (e.g., prefixes, suffixes, or infixes). An example is Turkish *-me-* which precedes tense and mood, but follows affixes that express reciprocals and reflexives, amongst other affixes.

Similarly, Dahl (2010) discusses three types of negation identified in three classifications of negation, namely Dahl (1979), Payne (1985), and Dryer (2005a). These types are (i) morphological or affixal negation, (ii) negative particles, and (iii) negative verbs. Morphological or affixal negation expresses negation through an affix that attaches to the verb or auxiliary. In Turkish, for instance, the suffix *-mV* is the standard negator (with the vowel harmonizing with other vowels). Since the function of this suffix is to negate the whole sentence, it is distinguished from lexical negation (e.g., English prefix *un-* in *unhappy*) which negates a lexical item only.

(2.13) a. Oku-yor-um

read-PROG-1SG

‘I am reading.’

b. Oku-mu-yor-um

read-NEG-PROG-1SG

‘I am not reading.’

(Dahl, 2010, p. 14)

Dryer (1988) explains that the cross-linguistic position of negation in the clause is theoretically important since it can help choose from among the various word order generalizations and correlations as put forward by Greenberg (1963). In this regard, in a pioneering study, Dryer (1988) examined the position of negation in 345 languages in various word order possibilities. In SVO order, for instance, there are four possibilities for negation as follows: sentence-initial (NegSVO), between subject and verb (SNegVO), between verb and object (SVNegO), and sentence-final (SVONeg). Dryer found that all these four possibilities existed in the SVO languages that he examined (p. 95):

(2.14) a. NegSVO: Iquito (Andean)

káá	kanamáákatúúwa	paahiáárikana	naki-híná
NEG	our, ancestors	able, PAST	jungle-in

ámayaakíini	núúkiika	kááyá
go. about	one	man

‘Our ancestors were not able to go about in the jungle alone.’

(Eastman & Eastman, 1963, p. 158)

b. SNegVO: Mokilese (Austronesian)

ih	joah	poaloahdi	suhkoah-u
he	NEG	chop	tree-that

‘He didn’t chop down that tree.’

(Harrison, 1976, p. 187)

c. SVNegO: Igbo (Kwa)

àda ebū-ghi	ìtè
Ada carry-NEG	pot

‘Ada is not carrying a pot.’

(Emenanjo, 1978, p. 173)

d. SVONeg: Tera (Chadic)

ɲá ndola	[ciɓtə karda]	ba
I like	wash trousers	NEG

‘I don’t like washing trousers.’

(Newman, 1970, p. 65)

Dryer’s data showed that the above four possibilities for the position of negation exist in SOV, VSO, and VOS languages. Overall, Dryer obtained the following results:

Table 2.1. Position of Negatives in SVO Languages (p. 94)

Word order	Number of languages	Number of families
NegSVO	4	3
SNegVO	47	13
SVNegO	3	1
SVONeg	13	4
Total	67	15

Table 2.2. Position of Negatives in SOV Languages (p. 96)

Word order	Number of languages	Number of families
NegSOV	8	5
SNegOV	6	3
SONegV	39	15
SOVNeg	64	18
Total	117	23

Table 2.3. Position of Negatives in Verb-initial Languages (VSO & VOS) (Dryer, 1988, p. 97)

Word order	Number of languages	Number of families	Word order	Number of languages	Number of families
NegVSO	34	10	NegVOS	8	6
VNegSO	0	0	VNegOS	0	0
VSNegO	0	0	VONegS	0	0
VSONeg	0	0	VOSNeg	0	0
Total	34	10	Total	8	6

Table 2.1 shows that the most common position for negation in SVO order is between the subject and verb (SNegVO) which, according to Dryer, is significant at the .01 level. In SOV languages (Table 2.2), the preverbal and sentence-final positions (SONegV and SOVNeg, respectively) are the most common. Nevertheless, unlike SVO and SOV languages that allowed all four positions for negation, in verb-initial languages (Table 2.3) only one position, sentence-initial position (NegVSO and NegVOS), was attested. Although there were 11 languages which Dryer was not able to classify for various reasons, he found sentence-initial negation to be true of 52 out of 53 verb-initial languages in his sample, concluding the universal that the negative precedes the verb in verb-initial languages. Finally, in languages with double negation such as

French *ne ... pas* with one optional and one obligatory negator, although Dryer includes both negators in his analysis, he considers the obligatory negator to be the basic position for negation.

Dryer offers some explanations for the position of negation in these orders as well. He argues that in SVO languages, the negative combines with and has scope over the semantic domain of the clause (i.e., the VP) and that the subject is outside this domain, hence the SNegVO being the most common position for negation followed by SVONeg. He refers to the fact whereby the negative combines with the VO unit in such orders as the Negative-Plus-VO Principle. This contrasts with other orders in which the negative cannot combine with its unit of semantic domain because either it is separated from the VO as in NegSVO or because it interrupts this unit as in SVNegO. According to Dryer, the Negative-Plus-VO Principle explains why orders like NegSVO and SVNegO are rare. However, this reasoning runs into problems once we consider SONegV which is fairly common in Dryer's sample although negation intervenes between the object and verb and thus should be infrequent. Dryer explains that SVO languages are more rigid and 'configurational' and do not normally allow anything to intervene and break the VO unit, hence the rarity of SVNegO while SOV are less so and more flexible.

Dryer also mentions that a second factor that might account for the frequency of the SONegV order is that the negation precedes the verb in this order (but follows the verb in SVNegO) which is a common tendency of negation. Indeed, 227 out of the 325 languages that Dryer investigated had negation preceding the verb. Dryer refers to this tendency as the Negative-Before-Verb Principle. In addition to classification of negation systems, one of the most central issues surrounding discussions of negation is the cross-linguistic encoding of sentential negation. Such discussions were sparked by the work of Kayne (1989) and Pollock (1989) who argued for the existence of a Negation Phrase (NegP) which subsequently led to the following issues regarding negation: (i) the existence of a NegP in all languages, (ii) the existence of a unique projection dedicated to encoding sentential negation, (iii) the position of such a unique projection in the clause, (iv) the possibility of having more than one NegP, and (v) the stability of sentential negation over time (Haegeman & Lohndal, 2013, p. 355). It is observed that different languages have different numbers of negative particles which also behave differently. For instance, while Italian *non* and English *not* express sentential negation on their own, French *ne* is incapable of doing so by itself and needs to be accompanied by *pas* or other forms such as *rien* or *personne*:

(2.15) Jean ne lit pas Le Monde. French
 Jean ne reads not Le Monde
 ‘Jean doesn’t read Le Monde.’

(2.16) Gianni non legge La Repubblica. Italian
 Gianni non reads La Repubblica
 ‘Gianni does not read La Repubblica.’

(2.17) John doesn’t read The Guardian. English
 (Haegeman & Lohndal, 2013, p. 354-355)

Furthermore, while in French and Italian negative particles may appear with such negative adverbs as *jamais* “never” and *mai* “never” without impacting the negative force of the sentence, English *not* cannot do so; instead, in English *never* expresses negation on its own and the presence of *not* renders the sentence affirmative:

(2.18) Jean ne lit jamais Le Monde. French
 Jean ne reads never Le Monde
 ‘Jean never reads Le Monde.’

(2.19) Gianni non legge mai La Repubblica. Italian
 Gianni non reads never La Repubblica
 ‘Gianni never reads La Repubblica.’

(2.20) John has never (#not) read The Guardian. English
 (Haegeman & Lohndal, 2013, p. 355-356)

The same pattern is seen with other negative elements like negative arguments, e.g., *nothing*: Italian *niente* and French *rien* can co-occur with the sentential negative particles while in English *nothing* expresses negation on its own and the presence of *not* or *never* results in an affirmative reading. Thus, in Standard American English two negative expressions cancel each other out and engender an affirmative sentence. This phenomenon is known as ‘double negation’.

However, in French and Italian, two negative expressions, e.g., *jamais* and *rien* or *mai* and *niente*, along with the sentential negation marker, can co-occur with each contributing to the expression of the negative force while resulting a single negation. This latter phenomenon is known as ‘negative concord’ (Haegeman & Lohndal, 2013).

2.4 Negation in Sign Language

In sign languages, just as in spoken languages, a substantial amount of scholarly literature exists on negation that has examined such constructions cross-linguistically. This research has addressed different aspects of negative constructions such as different types of markers (i.e., manual vs. non-manual) and syntactic patterns used in negative sentences. Manual markers refer to the use of hands in signing while non-manual markers (NMMs) are facial expressions and body movements (Zeshan, 2006b). NMMs serve various functions including phonological, morphological, syntactic, and prosodic (Pfau, 2016).

Another important notion is that of spread or scope defined as “the extent of a string of manual signs co-occurring with the non-manual marker” (Zeshan, 2006b, p. 39). The minimum scope is a single sign and the maximum scope is the whole clause. In all sign languages studied so far, it has been found that sentential negation is done using two basic components: manual signs and non-manual markers (e.g., Bembridge, 2016; Quer, 2012; Pfau, 2016). Manual signs indicate negative meanings which range from basic negative operators to specific ones. The non-manual negator is mostly a headshake. Research has shown that there is extensive cross-linguistic variation in how these two markers (i.e., manual signs and non-manual markers) interact.

Zeshan (2006) is considered the first comprehensive and comparative study of negative forms in sign languages. Zeshan examined negative structures (and interrogatives) in 37 sign languages although the corpus was heavily biased and skewed towards European sign languages. In sign language research on negation, two issues have received considerable attention: (i) ‘the position of negation in the clause structure’, and (ii) ‘the role and distribution of the non-manual markers’ (Gökgöz, 2021, p. 267). An important distinction between sign languages in terms of negation is the dichotomy first formulated by Ulrike Zeshan (2004, 2006a). Zeshan divides sign languages into two types in terms of negation: (i) manual-dominant and (ii) non-manual dominant. In the first type, negation is done using both manual and non-manual markers but is possible through NMMs alone, too. These are known as non-manual dominant sign languages which

include American Sign Language (ASL), German Sign Language (DGS), Catalan Sign Language (LSC), and Swedish Sign Language (Svenska Teckenspråket), amongst others.

On the other hand, in the second type, only the manual negator is required and the non-manual negator is optional. These are known as manual-dominant sign languages. Italian Sign Language (LIS), Jordanian Sign Language (LIU), Turkish Sign Language (Türk İşaret Dili, TİD), Kata Kolok (Bali), and Japanese Sign Language (JSL) fall into this category. Research has also indicated that there are differences among sign languages with respect to (a) the morphological nature of the negative marker and (b) its position with regard to the verb (Pfau, 2016). For instance, while negation is postverbal in DGS and TİD, it is preverbal in LIS and Berber Sign Language. However, in general, as Zeshan (2006b) notes, negative signs tend to occur sentence finally and are usually accompanied by the following NMMs: frowning, squinted eyes, nose wrinkling, lips spread, pursed or with the corners down, puffed cheeks, air puff, tongue protruding, etc. (Quer, 2012).

In this section, I will provide an overview of negation in a variety of sign languages. In American Sign Language (ASL), the basic syntactic negator, NOT, can occupy either a preverbal position or a sentence-final position if the rest of the sentence is topicalized (Fischer, 2006). NOT is almost always accompanied by a headshake (hs) as the negative non-manual marker.



Figure 2.1. NOT in ASL (Fischer, 2006, p. 187)

(2.21) $\frac{\text{neg}}{\text{(INDEX}_1\text{) UNDERSTAND}}$

‘I don’t understand.’

(Fischer, 2006, p. 186)

According to Fischer (2006), if the NMM spreads over the subject in (2.21), the meaning becomes more like ‘It is not the case that I understand’ (p. 186). Therefore, in ASL, a negative headshake can negate a sentence on its own or reinforce the negation of a negative sentence (Fischer, 2006). In addition to a side-to-side headshake, the basic negator in ASL is accompanied by frown, brow squint, wrinkling of the nose and a raised upper lip (Aarons, 1994).

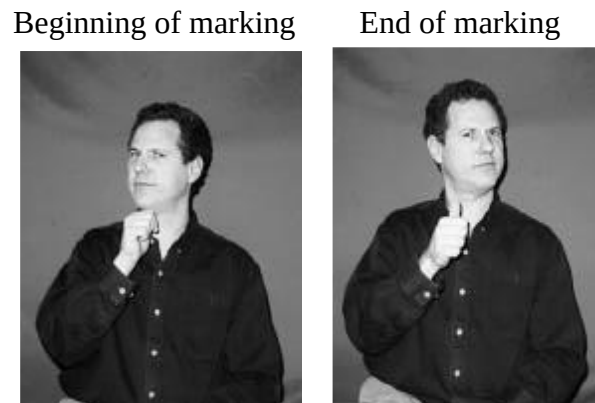


Figure 2.2. Negation in ASL (Aarons, 1994, p. 72)

In ASL, the negative headshake can be used along with the lexical negator, glossed as ‘NOT’. In this case, the headshake does not have to spread over the entire verb phrase as can be seen from (2.22) below:

hs

(2.22) JOHN NOT BUY HOUSE

However, in the absence of the manual negator NOT, the headshake can be used on its own to negate a sentence in which case it must extend over the entire verbal domain:

hs

(2.23) JOHN BUY HOUSE

(Pfau & Quer, 2002, p. 75)

Pfau and Quer (2002) argue that the verb never raises to Neg⁰ in ASL. In the presence of NOT, the realization of [+neg], i.e., headshake, associates with NOT and the sentence is grammatical. However, the absence of NOT would mean that there is no manual material that NOT

can be articulated with and therefore the headshake would be forced to spread over its entire c-command. This argument is represented schematically below in Figure 2.3.

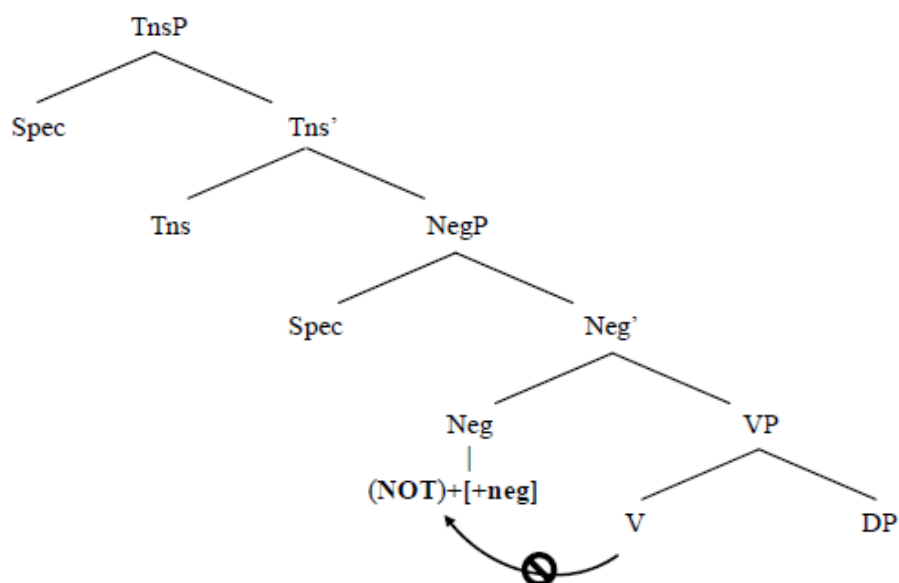


Figure 2.3. Negation in ASL (Pfau & Quer, 2002, p. 77)

The aforementioned argument explains why the following sentence is ungrammatical in ASL: the lexical negator NOT is absent, leaving the non-manual negator hanging with no (negative) lexical material for it to be associated with.

hs

(2.24) *JOHN BUY HOUSE

Antzakas (2006) investigated negation through different types of head movements in Greek Sign Language (GSL) via data gathered from free conversations, interviews and storytelling. He found the following head movements in negation: backward head tilt, side-to-side headshake and one-side head turn. He found that amplitude, temporal length, and duration with which head tilt, headshake, and head turn were signed indicated the weakness or strength of negation. He further reports that the informants in his study combined head turn with headshake or head tilt and head tilt with headshake for emphasis. The author also reported the following NMMs as accompanying

head movements: lowered and raised brows, closed eyes, corners of the mouth pulled down, upper lip raised and lower lip protruded. In GSL, head tilt and headshake can take scope over the whole clause:

TILT⁶

(2.25) INDEX1 AGAIN GO NOT-B

SHAKE

(2.26) INDEX1 AGAIN GO NOT-G

‘I won’t go (there) again.’

(Example 1 from Antzakas, 2006, 265)

Antzakas, however, reports that such co-occurrence (i.e., head movement across the whole clause) was rare and was done mainly for emphasis. Instead, head movements accompanied the negative signs only (2.27) or the negative sign and the verb (2.28):

TILT

(2.27) MORNING EAT INDEX1 CANNOT | ONLY COFFEE

‘I can’t eat anything in the morning, I only drink a coffee.’

(Example 5 from Antzakas, 2006, 26)

TURN

(2.28) STORY JUST ₁SPEAK₂ INDEX1 FORGET NEVER

‘I will never forget the story I have just told you.’ (Example 7 from Antzakas, 2006, 266)

In addition to GSL, another sign language in which backward head tilt (glossed below as ‘neg-tilt’) is found is Turkish Sign Language (TİD) which uses a basic clause negator DEĞİL ‘NOT’ (capital letters original to Zeshan, 2006c). This sign usually appears in post-predicate

⁶ Capital letters are original to the glossing from Antzakas (2006).

position which is also clause-final position (because TİD is SOV), thus making it difficult to verify whether the negator is actually in post-predicate or clause-final position (Zeshan, 2006c).

- (2.29) BEN KONUŞMAK $\overline{\text{neg-tilt DEĞİL}}$
INDEX1 SPEAK NOT
‘I am not a speaking (person).’ (Example 27 from Zeshan, 2006c, p. 148)

NOT also often criticizes to a preceding host sign and is not usually followed by another sign. As for NMMs, NOT is accompanied by a backward head tilt and brow raise which usually scope over the preceding sign as well.

- (2.30) WATER SEA INDEX1 NONE $\overline{\text{neg-tilt GO } \wedge \text{NOT}}$
‘I never went to the seaside at all.’ (Zeshan, 2006c, p. 150)

Another facial expression reported in TİD includes puffed cheeks and release of air. TİD is believed to have a manual-primary negation strategy and the non-manual negators, i.e., negative side-to-side headshake and backward head tilt, glossed as ‘neg-shake’ and ‘neg-tilt’, respectively, tightly bound to the manual negator. However, while non-manual negators also do have grammatical functions, they cannot negate a clause on their own:

- (2.31) (a): $\overline{\text{neg-tilt KNOW NOT}}$ (b) * $\overline{\text{neg-tilt KNOW}}$ (Zeshan, 2006c, p. 163)

Thus, based on the above information, i.e., scope on the manual negators and inability to negate the sentence on their own, TİD is manual-dominant. (Zeshan, 2006c) also reports that in TİD the sentence remains negative even though there might be more than one negative sign in the same sentence as in (2.32). This finding indicates that TİD is a negative concord (NC) language.

- (2.32) BEN HİÇ ŞANS $\overline{\text{neg-tilt OLMAK } \wedge \text{DEĞİL}}$
INDEX1 NONE(1) LUCK BECOME $\wedge \text{NOT}$
‘I’ve had no luck at all.’

Other NMMs that can optionally mark negative headshake are narrowed eyes and lowered brows. In the case of backward head tilt, this additional non-manual marker is brow raise, similar to GSL.

In a more recent study on TİD, Gökgöz (2011) examined the syntactic behavior of negation from a generative perspective. He found that the non-manual negative marker backward head tilt occurs in 48% of the negative sentences, either on the negative marker DEĞİL ‘not’ or on another single negative sign which sometimes spread beyond the predicate. Two other NMMs in negation were headshake, observed in 50% of the negative sentences, and a single head-turn, seen in only 11%. Gökgöz (2011), however, unlike previous research, found two other NMMs, namely, brow-lowering (‘bl’) and brow-raising (‘br’), that he terms ‘non-neutral brow position’ (nbp). Gökgöz considered these two to be the major non-manuals marking negation in TİD and based his conclusion on the spreading domain of brow-lowering and brow-raising (which is the whole clause) and their high frequency of occurrence.

A more recent study by De Barros and Siebörger (2016) of sentential negation in South African Sign Language (SALS) showed that the manual negative particle, NOT, appears clause-finally and is marked by a side-to-side headshake as the NNM (Figure 2.4).



Figure 2.4. Manual Negation NOT in SALS (De Barros & Siebörger, 2016, p. 8)

Their study also indicated that the manual negator is optional and the headshake, being obligatory, is the main negator in SALS. Therefore, the headshake plays a primary role while the role of the manual negator is of secondary importance in SALS, indicating that SALS is non-manual dominant. Depending on the type of negation, this headshake might span over the manual negator as the following example shows:

headshake

(2.33) MOTHER SHOCK SPEAK NOT

‘My mother was shocked that I could not speak.’ (De Barros & Siebörger, 2016, p. 10)

Furthermore, in terms of facial expressions, De Barros and Siebörger (2016) found frown, i.e., lowered eyebrows, pursed lips, and a wrinkled nose to be the most common NMMs of negation in SALS with varying degrees of scope over different constituents. Nonetheless, a headshake can appear on its own with no VP as a negative response in question-answer pairs. For instance, a recent study by Huddleston (2017) on polar question-answer clauses in SALS showed that the majority of the negative sentences did not have a manual negator but were formed using a headshake only which occurred mainly after, and sometimes on, the last signed element in the sentence. Huddleston’s (2017) findings show that the scope of the negative non-manual may not necessarily include the verb matrix. For instance, in (58), the headshake (hs), which is the answer part of the question-answer clause, does not spread on the VP. Note that re and le stand for raised eyebrows and lowered eyebrows, respectively. Huddleston (2017) found that while the ‘question’ part of question-answer clauses was marked by raised eyebrows (re), the ‘answer’ part is accompanied by headshake (hs) and lowered eyebrows (le).

hs
re le

(2.34) k-i-t-a INDEX3 EAT SUGAR

‘Kita does not eat sugar.’ (Huddleston, 2017, p. 100)

Finally, Hendriks (2007) examined negation in Jordanian Sign Language (LIU) and found that the manual negator, which occurs clause-finally and which takes the following form when signed neutrally (i.e., not emphatically), is the most common way of negating a clause.



Figure 2.5. Neutral Clause Negator NEG in LIU (Hendriks, 2007, p. 107)

Hendriks also found the following NMMs in negation in LIU: backward head-tilt, sideward head-turn, pursed lips, corners of the mouth down, and forward head-tilt. In LIU, the head tilt does not serve a grammatical purpose since it cannot negate a clause on its own and seems to be used by deaf people the way the hearing population uses it.

This short review of literature on sign language negation shows that negation is done in different sign languages by both manual and non-manual negators with different sign languages preferring either the manual negator or the non-manual negator as the primary clause negator although both negators might be present in a negative sentence. The manual basic clause negator can appear in different positions in the clause, e.g., preverbally, postverbally, or clause-finally. Furthermore, the sentence can be negated by using a main non-manual negator such as a side-to-side headshake or backward head tilt. Such a non-manual marker is usually accompanied by other NMMs such as brow raise, lowered brows, pursed lips, wrinkled nose, and head-turn. Leaving out the manual negator can cause variation in the scope of the non-manual negator. NMMs can occur either on the manual negator only or spread across their entire c-command domain which can prompt different sentence interpretations. Sign languages also display varying degrees in their number of negative particles such as negative pronouns and negative imperfective aspect. Finally, configurationally, negators might behave differently in different sign languages. For instance, they can display cliticization or affixation in some languages.

This review shows that most studies on negation have focused either on spoken language or sign language but not both. In this dissertation, we compare negation in a spoken language,

Khuzestani Arabic (KhA), and a sign language, Sadat Tawaher Sign Language (STSL). We present evidence to support that claim that KhA and STSL have different negation systems, which is rather unexpected since STSL emerged and grew adjacent with KhA and would logically conform in its grammar to that of KhA. A comparative study of KhA and STSL is especially important since STSL signers speak when they sign, providing a unique opportunity to compare speech and sign simultaneously. If the signers' signing turns out to conform to the word order of KhA, then the inevitable conclusion would be that STSL is simply a manual communication system copied from KhA. In other words, STSL would be classified as signed KhA. However, if STSL turns out to have a different grammar than that of KhA, then it is an independent grammatical system in which case, it should be treated as a sign language and compared on a par with other sign languages in terms of non-manual activities, position of negation, handshape configuration of negative signs, the spreading of non-manual markers, and whether the manual and non-manual markers are optional or obligatory.

CHAPTER 3. METHODOLOGY

In this chapter, I will describe the participants, types of data, data collection and analysis.

3.1 Participants

In this section, I provide general information about the participants which is independent of any of the data collection techniques described below. Any specific information about the participants as pertaining to a specific data collection technique will be provided in its own relevant section. Overall, eleven signers participated in the study, all of whom were fluent in signing STSL and all but one were native signers. They consisted of seven males and four females and ranged in age from 18 to 54. Since not every STSL signer used this sign language since birth, we did not emphasize being a native signer as a requirement for participation in the study. However, we used fluency in both KhA and STSL as an important criterion to be eligible to be recruited as a participant. All our participants speak KhA as their mother tongue and some are fluent in Farsi (Persian) and know some English from high school or college. Speaking KhA natively thus was not an issue. However, reading knowledge of Farsi became important as will be explained later. Except for one participant, all other participants lived in Sadat Tawaher where Hanash lived his whole life and interacted with him frequently. Table 3.1 provides more information about the participants.

Table 3.1. Participants' Profiles

Participant	Sex	Age	Education	Proficiency in STSL	Relation to Hanash	Relation to researcher	Proficiency in other languages
P #1	M	54	none	native	son	brother	some Farsi
P #2	F	51	none	non-native	daughter-in-law	sister-in-law	none
P #3	F	51	none	native	daughter	sister	none
P #4	F	42	none	native	daughter	sister	none
P #5	M	40	elementary	native	son	brother	some Farsi
P #6	M	36	elementary school	native	son	brother	fluent in Farsi
P #7	M	34	PhD student	native	son	self	fluent in Farsi & English
P #8	M	29	BA in English	native	grandson	nephew	fluent in Farsi & English
P #9	M	27	elementary school	native	grandson	nephew	fluent in Farsi
P #10	M	24	high school	native	grandson	nephew	fluent in Farsi, some English
P #11	F	18	high school	native	granddaughter	niece	fluent in Farsi, some English

3.2 Data

I gathered data regarding declaratives, interrogatives, and negatives using the following three techniques: (a) signed productions, (b) stories, and (c) grammaticality judgments. For various reasons we did not have access to all the participants for all three data collection techniques and therefore we used different techniques with different participants. Word order and interrogative data were gathered using signed productions while all three techniques were employed in collecting negation data. Therefore, while we did gather all three types of data (declaratives, interrogatives, and negatives), we did so using different methods and from different participants as shown in Table 3.2.

Table 3.2. Type of Data & Data Collection Technique for Each Participant

Participant	Type of data collected	Data collection technique used		
		signed productions	stories	grammaticality judgments
P #1	declaratives	√		
	negatives	√	√	√
P #2	declaratives	√		
	negatives	√	√	
P #8	declaratives	√		
	negatives	√		
P #4	declaratives	√		
	negatives	√	√	√
P #5	declaratives	√		
	negatives	√		
P #3	declaratives	√		
P #9	declaratives	√		
	negatives		√	
P #11	declaratives	√		
P #6	declaratives	√		
P #10	declaratives	√		
P#7	declaratives	√		

Table 3.3. Number of Participants, Data Type & Data Collection Method

Type of data	Data collection technique		
	signed productions	stories	grammaticality judgments
declaratives	11 participants	-	-
negatives	5 participants	4 participants	2 participants

3.2.1 Word Order Data

(1) Signed Productions

We gathered the data for declaratives by using signed productions in the form of stimulus sentences. To do so, we prepared 36 stimulus sentences, all being main (i.e., non-embedded) active voice declarative sentences. We designed these stimuli in Khuzestani Arabic, the participants' native spoken language, but we also translated them into Persian (Farsi) for three reasons. First, the participants had varying degrees of literacy skills in either language. Second, since Persian is the dominant and official language in the country and since Khuzestani Arabic is mostly spoken and less often written, the participants might not be totally familiar with it in written form. Third, most speakers of KhA use Persian when they write one another (e.g., text over the phone) and thus are more familiar with Persian in written form. Thus, we anticipated that including Persian translation of stimuli would facilitate the comprehension of the stimuli. In designing the stimuli, we included intransitive, transitive and ditransitive verbs (e.g., *fall*, *push*, and *give*, respectively) in both reversible and non-reversible situations (e.g., 'The teacher saw the doctor' and 'The people pushed the car', respectively). Furthermore, almost all subject and object arguments were nominal. The (di)transitive verbs included plain verbs (e.g., *love*, *eat*), agreeing verbs (e.g., *give*, *push*), body-anchored locatives (e.g., *poke someone in the eye*, *pull someone's hair*), and classifier constructions (e.g., *put*, *throw*). All the transitive verbs require two arguments, a subject, mostly of the agent type, and an object, mostly theme or patient. Intransitive verbs are those that require only one argument, subject. The stimuli consisted of two sets of intransitive verbs. The first set were verbal (predicational) sentences, i.e., sentences with main lexical verbs, and included different types of intransitive verbs such as ergative verbs (e.g., *move*), unaccusative verbs (e.g., *fall*), and weather verbs (e.g., *rain*). Note that weather verbs can optionally have a subject in Khuzestani Arabic (e.g., lit. 'The world/sky is raining'). The second set consisted of verbless (non-predicational) sentences, i.e., those without a main lexical verb because the verb is the present tense *to be* which is not overtly phonetically realized in Khuzestani Arabic and in most sign languages. These were locative sentences, e.g., 'The ball is behind the wall'. After the stimuli were prepared, we shared them with the participants via WhatsApp in two formats: (i) written format (in both Khuzestani Arabic and Persian) with those participants who had excellent reading skills and (ii) audio-recorded in Khuzestani Arabic format with those participants who lacked reading literacy. The word order stimuli are in Appendix A.

3.2.2 Negation Data

We used three methods to collect negation data: (i) signed productions, (ii) stories, and (iii) grammaticality judgments. These are described below.

(1) Signed productions

We used the declarative stimuli to elicit negatives by using the negated counterpart of each corresponding declarative sentence of all 36 original sentences. As will be described later, Khuzestani Arabic has three negative particles: *ma*, *la*, and *mu(f)*. We negated the sentences in the signed productions task with *ma* only as this negative particle is the most neutral and most frequent negator. *mu(f)* and *la* are used in other marked negative contexts. For instance, *la* is used with imperatives: *La truḥ* ‘Don’t go!’. Once again and just like the declarative data, we provided both the written form and audio-recorded version of the negative stimuli for the participants.

(2) Stories

Inclusion of stories was a way to provide the participants enough freedom to sign more naturally since in the signed productions task, the participants were only asked to sign fixed sentences without changing anything else. We included two types of stories as described below: (i) Everyday activities and (ii) Bedtime stories.

(2a) Stories Type I: Everyday Activities

In this task, we simply asked the participants to describe any everyday activities that they were involved in such as doing dishes, harvesting season, visiting a friend, etc., or anything that has recently happened to them, e.g., a car accident. Since every person engages in some type of activity on a daily basis, including such activities as a source of data was certain to engender at least some desired outcomes that we were seeking. We further encouraged the participants to think of not only the things they did do but also of things they did not do or of things that did not happen so that we would have negative sentences produced as a result. For instance, when narrating a visit to a friend’s house, the participants were asked to report on what they talked about / ate / watched, /etc. as well as what they did not do / talk about / etc. Another example is harvesting season in which we asked the participants to describe whether or not the crops had grown well the year before.

(2b) Stories Type II: Bedtime Stories

Since the participants were likely to narrate everyday activities without providing detailed descriptions which would have resulted in few negative sentences, we decided to include a second type of stories, ‘bedtime stories’. These are stories that have been frequently narrated in Arabic and mostly by female individuals in Khuzestan and likewise in Hanash’s family over the years. Collectively known as *سوالف الشام* *suwalif ash-sham* (‘Stories of the Levant’), these stories are told at bedtime and are meant to be amusing but also teach a moral principle. Some popular bedtime stories are *The Poor Man and The Fish*, *The Horse Rider and The Witch*, and *The Annoying Mother-in-Law*. Thus, bedtime stories provided a solid template for the participants to follow. We asked the participants to narrate as many bedtime stories as they could remember and as accurately as possible. These stories are not as well-known or remembered as in the past, hence only a few participants were able to narrate some of them.. Like everyday activities, bedtime stories provided a natural context for the participants to sign without the pressure to stick to a specific set of sentences. As expected, these stories provided a large number of negative constructions of various types (e.g., negative main sentences, negative constituents, and negative responses).

(3) Grammaticality Judgments

Grammaticality judgments were the last data collection task we used after we had collected the data through signed productions and story-telling. We used grammaticality judgments for negation only. Our rationale for including grammaticality judgments was our prediction that the participants would sign what they believe to be grammatical and thus naturally avoid ungrammatical sentences altogether. Thus, by the principle that absence of evidence is not evidence of absence, we included grammaticality judgments to find out whether or not the participants would find alternative ways of signing negative sentences grammatical. To conduct grammaticality judgments, we had to take into account a few issues. The first was the choice of word order in which the negative sentences for grammaticality judgments were to be signed and presented to the participants. We chose SVO and SOV because, as will be discussed in Chapter 4, these two orders are the most commonly used orders in STSL although other word orders are also acceptable.

Next, we had to wait until the results from the signed production and story-telling data for negation were available to find out how the participants had signed negative sentences, that is, what position in the sentence the negators were, what constituent(s) the nonmanual marker, if any, had spread over, etc. For instance, the results from signed productions and story-telling for negatives showed that in almost all the cases the participants placed the negative markers in sentence-final position. Based on this finding, I signed sentences in which the negator appeared in other positions such as preverbally and sentence-initially and then sought grammaticality judgments. In addition to changing the position of the negator, other variations we applied to the negative sentences in this task included altering the spreading domain of the nonmanual marker (over the VP, whole sentence, etc.) and negating the clause with only the manual negator, only the nonmanual negator, or both manual and nonmanual negator.

The third issue was the choice of negative markers. As I will explain later in Chapter 4, STSL has several negative markers with different handshapes and nonmanual marking. Of these negators, we chose what we have labeled NEG_{basic} and NEG_{proh} for grammaticality judgments for three reasons. First, NEG_{basic} and NEG_{proh} seem to be the most frequently used negators, especially NEG_{basic}, which as we will later argue is the basic clause negator. Second, NEG_{basic} and NEG_{proh} have backward head-tilt and side-to-side headshake, respectively, as their associated nonmanual markers. It seems to be physiologically far easier for a signer to spread these nonmanual markers beyond the manual negators that they are associated with than, say, NEG_{poss} whose nonmanual marker is air blown out of the mouth: Try blowing air out of the mouth while gesturing the sentence ‘The boy did not give the book to the girl’. The third reason for selecting NEG_{basic} and NEG_{proh} was simply time constraint. As we had several sentences signed with NEG_{basic} and NEG_{proh} for which we needed to elicit grammaticality judgments, increasing the number of negators would have increased the number of sentences for grammaticality judgments and would have consequently made this task far more exhausting for the participants and would thus have affected the reliability of the results. Therefore, I avoided adding any more negators. Furthermore, as for the word order of the sentences that were sent to the participants for grammaticality judgment, since SVO is the most common word order of STSL, followed by SOV, we decided to have both orders with both sentences to examine whether word order affects the participants’ judgment. Thus, overall, there were two main affirmative sentences in two different orders, namely SVO and SOV

which were negated with two negators, namely NEG_{basic} and NEG_{proh}, signed in several different ways as shown in Tables 3.4 and 3.5.

The procedure was as follows. First, I, as a native signer of STSL, was recorded signing two affirmative sentences and then their corresponding negative counterparts which I signed several times in two ways: (i) the same way the signers had signed them in the last two tasks, namely, signed productions and stories and (ii) differently than the way the participants had signed them in the last two tasks. We elicited grammaticality judgments from two participants. The recordings were sent via the messaging software WhatsApp to one participant who did not have a reliable Internet connection for a live session. She was asked to rate the sentences as *Correct*, *Incorrect*, or *Undecided*. We avoided the terms *Grammatical* and *Ungrammatical* since the participant was not familiar with such technical terms. Furthermore, we asked her to provide the grammatical counterpart of any sentence that she deemed to have been signed incorrectly. For another participant who had access to reliable Internet connection, we carried out and recorded this task live on WhatsApp. For the grammaticality judgment stimuli, see Appendix B.

Two example sentences that were used for grammaticality judgment are as follows:

(i) Grammaticality judgments with NEG_{basic}:

(a) Declarative sentence #1: FRIEND BUY CAR (SVO) / FRIEND CAR BUY (SOV)

Table 3.4. Grammaticality Judgments for NEG_{basic} / SVO Order

Test
(a) Sentence-final neg + bht: FRIEND BUY CAR <u>NEG_{basic}</u> <u>bht</u>
(b) Sentence-final neg without bht: FRIEND BUY CAR NEG _{basic}
(c) Sentence-final neg + bht spreading over whole clause: FRIEND BUY CAR <u>NEG_{basic}</u> <u>bht</u>
(d) Sentence-final neg + bht spreading over VP: FRIEND BUY <u>CAR NEG_{basic}</u> <u>bht</u>
(e) Only bht; no spread; no manual neg: FRIEND BUY CAR <u>bht</u>
(f) Spreading hs; no manual neg: FRIEND <u>BUY CAR</u> <u>bht</u>
(g) Sentence-initial neg with bht: <u>NEG_{basic}</u> FRIEND BUY CAR <u>bht</u>
(h) Sentence-initial neg without bht: NEG _{basic} FRIEND BUY CAR
(i) Sentence-initial neg + spreading bht: <u>NEG_{basic}</u> FRIEND BUY CAR <u>bht</u>
(j) Preverbal neg with bht: FRIEND <u>NEG_{basic}</u> BUY CAR <u>bht</u>
(k) Preverbal neg without bht: FRIEND NEG _{basic} BUY CAR
(l) Preverbal neg + spreading bht: FRIEND <u>NEG_{basic}</u> EAT MEAT <u>bht</u>

Note: bht = backward head tilt

(ii) Grammaticality judgments with NEG_{proh}:

- Declarative sentence #2: FRIEND EAT MEAT (SVO) / FRIEND MEAT EAT (SOV)
‘My friend eats meat.’

Table 3.5. Grammaticality Judgments for NEG_{proh} / SVO Order

Test
(a) Sentence-final neg + hs: FRIEND EAT MEAT <u>NEG_{proh} hs</u>
(b) Sentence-final neg without hs: FRIEND EAT MEAT NEG _{proh}
(c) Sentence-final neg + hs spreading over whole clause: <u>FRIEND EAT MEAT NEG_{proh} hs</u>
(d) Sentence-final neg + hs spreading over VP: FRIEND <u>EAT MEAT NEG_{proh} hs</u>
(e) Only hs; no spread; no manual neg: FRIEND EAT MEAT <u>hs</u>
(f) Spreading hs; no manual neg: FRIEND EAT MEAT <u>hs</u>
(g) Sentence-initial neg with hs: <u>NEG_{proh} hs</u> FRIEND EAT MEAT
(h) Sentence-initial neg without hs: NEG _{proh} FRIEND EAT MEAT
(i) Sentence-initial neg + spreading hs: <u>NEG_{proh} hs</u> FRIEND EAT MEAT
(j) Preverbal neg with hs: FRIEND <u>NEG_{proh} hs</u> EAT MEAT
(k) Preverbal neg without hs: FRIEND NEG _{proh} EAT MEAT
(l) Preverbal neg + spreading hs: FRIEND <u>NEG_{proh} hs</u> EAT MEAT

Note: hs = headshake

3.3 Procedure

Prior to collecting the data, we explained the whole study to the participants and asked for their consent to participate in the study. Specifically, we explained what the study was about, what it involved, what was needed from the participants, what their participation included, and how the data was going to be collected, stored, and used. We made sure the participants fully understood everything before we collected the data. The data was collected online due to the travel restrictions that I was faced with. After I received the data, I downloaded and stored them on my laptop and then transferred and stored them in a specific folder on the computer at the Sign Language Data Analysis Lab at Purdue University. Besides my advisor and me who were able to access the data using our login credentials, no one else had access. As for my laptop, it is secured with a password that is not shared with anyone. Furthermore, all the folders on my laptop in which the data was stored were password protected.

3.4 Data Analysis

I analyzed the data in the current study using distributional linguistic analysis in which I tried to find and isolate recurring patterns regarding word order, interrogatives, and negation. The analysis also involved descriptive statistics (e.g., raw frequencies and percentages). Furthermore, I imported the data into ELAN for annotation and transcription and used the non-manual template that was created by colleagues in the Sign Language Data Analysis Lab at Purdue University.

CHAPTER 4. RESULTS AND DISCUSSION

This chapter addresses three grammatical areas in KhA and STSL: word order (declaratives), interrogatives (question forms), and negatives. The first two areas will be examined briefly and the last area extensively as negation is the focus of this dissertation. In the next section, I present a general descriptive sketch of the syntax of KhA and STSL. My hope is that this section will help the reader better understand the rest of this dissertation since no literature exists on STSL and very scarce literature on KhA.

4.1 Overview of the Syntax of KhA and STSL

KhA is a head-initial, pro-drop language. Complements follow their heads as reflected in the preposition-NP order in preposition phrases (4.1a). The verb most commonly precedes its complement (as in SVO order, 4.1b) and in cases in which it follows the object (e.g., SOV, OSV, etc.), the verb most often also takes a resumptive pronoun (4.1c). Furthermore, although KhA allows *wh*-in situ, *wh*-movement is the common strategy to form *wh*-questions (4.1d). The perfect aspect marker and negative particles also appear to the left of the verb (4.1e & 4.1f).

(4.1) a. lə-l-mædrəsæh

to-the-school

‘to school’

b. əl-bət ʃaf-æt əl-wælæd

the-girl see-3SF.PST the-boy

‘The girl saw the boy.’

c. əl-bət əl-wælæd ʃaf-ət-æh

the-girl the-boy see-3SF.PRF-him

‘The boy, the girl saw him.’

d. wyiəæn ʃəf-ət sʰaħb-æk
 where see-2SM.PRF friend-2SM.GEN
 ‘Where did you meet with your friend?’

e. əl-wælæd xællæsʰ əgrayæh
 the-boy finish.3SM.PRF reading.PRTC
 ‘The boy is done reading.’

f. əs-sæyyaræh ma-tə-ʃtəyəl
 the-car NEG-3SF-work.IMPRF
 ‘The car won’t work.’

STSL, on the other hand, seems to be head-final. First, although SVO is the most common word order, SOV is also perfectly acceptable (4.2a and 4.2b). Additionally, other pieces of evidence point to a head-final direction. Various functional markers such as *wh*-signs (4.2c), the perfective aspect marker *FINISH* (4.2d), and negation markers (4.2e) all appear on the right of the verb in sentence-final position.

(4.2) a. BOY EAT APPLE

b. BOY APPLE EAT
 ‘The boy is eating an apple.’

c. BOY EAT WHAT
 ‘What is the boy eating?’

d. BOY EAT APPLE FINISH
 ‘The boy has eaten the apple.’

e. BOY EAT APPLE NEG
 ‘The boy is not eating an apple.’

4.2 Findings

I present the findings for negation in the section 4.2.2. Prior to doing so, I will present an overview of the findings related to word order (of declarative sentences) in KhA and STSL.

4.2.1 Word Order

Since basic familiarity with word order in declaratives is necessary and helps the reader better understand negation, in this section, I will describe various word order possibilities in Khuzestani Arabic (KhA) and Sadat Tawaher Sign Language (STSL).

4.2.1.1 Word Order in Khuzestani Arabic (KhA)

KhA has a rather flexible word order and allows all the word order possibilities: SVO, SOV, OSV, OVS, VSO, and VOS. Word order in KhA, however, is not free. Rather, not only are these orders used under certain pragmatic circumstances and with varying frequency but also favored to different degrees. SVO is the order most commonly used and seems to be the basic word order in KhA. For instance, in starting a narrative with no context previously provided, SVO is used (4.3a) whereas a different word order, especially those with the object fronted, would result in very odd readings, if not ungrammaticality, in such contexts (4.3b).

- (4.3) a. wælæd kælæ tæffahæh (SVO)
 boy.INDEF eat.3SM.PRF apple.INDEF
 ‘A boy ate an apple.’
- b. ? tæffahæh wælæd kælæ (OSV)
 apple.INDEF boy.INDEF eat.3SM.PRF

Nevertheless, in KhA, to shift emphasis on the object, it is frequently emphasized and thus topicalized through OSV and OVS orders. In such cases, if the object is a definite NP, a resumptive pronoun which agrees in gender and number with the object is required (4.4a) but not if the object is an indefinite NP (4.4b) or a generic / common definite NP (4.4c):

- (4.4) a. $\text{\ae t-t\ae ffah\ae h}$ $\text{l-w\ae l\ae d}$ $\text{k\ae la-h\ae h}$ (OSV + resumptive pronoun)
 the-apple_i the-boy eat.3SM.PRF-it_i
 ‘The boy ate an apple.’
- b. $\text{t\ae ffah\ae h}$ $\text{l-w\ae l\ae d}$ $\text{k\ae la}$ (OSV ; no resumptive pronoun)
 apple.INDEF the-boy eat.3SM.PRF
 ‘The boy ate an apple.’
- b. $\text{t\ae ffah}$ $\text{l-w\ae l\ae d}$ $\text{k\ae la}$ (OSV ; no resumptive pronoun)
 apple.DEF.PL the-boy eat.3SM.PRF
 ‘The boy ate apples.’

Examples of SOV and OVS are provided below. Once again, if the object is a definite noun, a resumptive is required (3a); if it is an indefinite or generic / common noun, no resumptive pronoun is required (4.4b & 4.4c). In SOV, the emphasis is on the subject and object whereas the OVS shifts the emphasis on the object and the verb (i.e., the action that occurred to the object):

- (4.4) a. $\text{\ae l-w\ae l\ae d}$ $\text{t-t\ae ffah\ae h}_i$ $\text{k\ae la-h\ae h}_i$ (SOV + resumptive pronoun)
 the-boy the-apple ate.3SM.PRF-it
 ‘The boy ate the apple.’
- b. $\text{\ae l-w\ae l\ae d}$ $\text{t\ae ffah\ae h}$ $\text{k\ae l\ae}$ (SOV; no resumptive pronoun)
 the-boy apple.INDEF ate.3SM.PRF
 ‘The boy ate an apple.’
- c. $\text{\ae l-w\ae l\ae d}$ $\text{t\ae ffah}$ $\text{k\ae l\ae}$ (SOV; no resumptive pronoun)
 the-boy apple.PL ate.3SM.PRF
 ‘The boy ate apples.’
- d. $\text{\ae t-t\ae ffah\ae h}_i$ $\text{k\ae la-h\ae h}_i$ $\text{\ae l-w\ae l\ae d}$ (OVS + resumptive pronoun)
 the-apple eat.3SM.PRF-it the-boy
 ‘The apple, the boy ate.’

Verb-initial orders (i.e., VSO and VOS) are used to emphasize the action but are generally used less frequently. If used, they tend to occur with an animate subject and an inanimate object (4.5a). In other words, these orders are favored in irreversible predicates. However, since KhA does not have case which can result in ambiguity in semantic interpretation especially if the subject and object are both animate, verb-initial orders are used far less frequently in reversible predicates (4.5b). Resumptive pronouns are not allowed in these orders:

- (4.5) a. kælæ əl-wælæd ət-təffahæh (VSO)
 eat.3SM.PRF the-boy the-apple
 ‘The boy ate the apple.’
- b. ʃaf əl-wælæd əl-moʃælləm (VOS/VSO)
 see.3SM.PRF the-boy the-teacher
 ‘The boy saw the teacher.’ (or possible reading: ‘The teacher saw the boy.’)

4.2.1.2 Word Order in Sadat Tawaher Sign Language (STSL)

For word order, 11 participants were recorded signing 36 stimulus sentences. Thus, word order data consist of signed productions only and stories were not analyzed for this purpose since word order is not the focus of this dissertation. While 36 stimulus sentences and 11 participants should have provided us with a total of 396 sentences, we considered 298 sentences for the final analysis. The rest either the participants failed to sign or signed the wrong sentences. Since we were interested in the ordering possibilities of the subject, verb, and object, we did not include intransitive sentences, i.e., those with a subject and intransitive verb such as ‘The tree fell’, which only generated SV orders. As we explained in Chapter 3, the stimulus sentences consisted of plain verbs, agreeing verbs, body-anchored verbs, classifier constructions, and locatives. Before we present the findings for word order, we provide examples of each of these construction types for clarification below.

(4.6) Examples of construction types from the word order stimuli

(a) Plain verb: DRINK



FRIEND

DRINK

TEA

‘My friend drank tea.’

(b) Agreeing verb: SHOOT



SOLDIER

SHOOT

MAN

IX-3

‘The soldier shot the man.’

(c) Body-anchored verb:



BOY

PULL

GIRL

HAIR

‘The boy pulled the girl’s hair.’

(d) Classifier construction: PUT



(e) Locative: ON TOP OF



The findings regarding word order are as summarized in Table 4.1.

Table 4.1. Word Orders in STSL (Excluding Intransitive Sentences)

Word order	SVO	SOV	SVOV	OSV	SOVOV	SOVO
Construction type						
Plain	35	5	1	0	0	0
Agreeing	44	18	3	1	1	0
Body-anchored locatives	37	11	4	2	0	0
Classifier constructions	29	29	1	1	0	1
Locatives	45	15	10	4	1	0
Total: 298	190	78	19	8	2	1
Percentage	63.7%	26.2%	6.4%	2.7%	0.7%	0.3%

As Table 4.1 shows, SVO is the most common word order in STSL with 190 out of 298 sentences (63.7%) in this order. SVO is used almost equally with all verb types and with reversible and irreversible predicates. Thus, verb type and reversibility of predicate did not impact choice of word order. SVO is then followed by SOV which comprises 78 instances (26.2%). Therefore, SVO and SOV are used in the vast majority of the sentences (89.9%). Furthermore, while both SVO and SOV are used in all types of constructions, SOV is especially used with agreeing verbs (e.g., SEE, PUSH) and classifier constructions (e.g., PUT).

The remaining four orders, SVOV, OSV, SOVOV, and SOVO, all seem to be variations on these two orders with the verb or object or both repeated. Another issue is the use of SVO and SOV with locatives. These were sentences in which a relation (Re) is established between a figure (Fig) and ground (Gr), for instance, ‘**The man** [Figure] is **next to** [Relation] **the wall** [Ground]’. In such sentences, there is no lexical verb and the relation is taken to be equivalent to the verb whereas figure and ground correspond to the subject and object, respectively. Thus, an SVO order in locatives means the following order: Figure – Relation – Ground. SVOV was used especially with locatives and body-anchored locative constructions (e.g., POKE SOMEONE IN THE EYE). The repetition of the verb in this order causes the object to be sandwiched between the two repetitions. It is likely that the repeated verb was used to establish the relation between the object and the location of the action in the case of body-anchored locatives (e.g., POKE SOMEONE IN THE EYE) and between the figure and the ground in the case of locatives (e.g., MAN NEXT-TO WALL). Of the word orders in Table 4.1, SVO and SOV are shared with KhA.

The next section examines negation in Khuzestani Arabic (KhA) and Sadat Tawaher Sign Language (STSL).

4.2.2 Negation

In this section, I will describe negation in Khuzestani Arabic (KhA) and Sadat Tawaher Sign Language (STSL).

4.2.2.1 Negation in Khuzestani Arabic (KhA)

There are three negative markers of sentential negation in KhA: *ma*, *mu(f)*, and *la*. All three negative markers can only appear in immediately preverbal position and are the outermost

morphemes in the word. Furthermore, all three negators take over the stress from the following verb and form a prosodic unit with it similarly reported for Persian and Czech (Dahl, 2010). The standard negator, however, is *ma* which negates both perfective (4.7a) and imperfective (4.7c) verbs, conditionals (4.7d), existential *ʔæku* ‘there is/are’ (4.7e), and possessive *ʕad-* ‘have / has’ (4.7f):

- (4.7) a. əl-wælæd ma-kælæ ət-təffahæh
 the-boy NEG-eat.3SM.PRF the-apple
 ‘The boy ate the apple.’
- b. ma-rah lə-l-mædrəsæh æməs
 NEG-go-3SM.PRF to-the-school yesterday
 ‘He did not go to school yesterday.’
- c. ma-y-ruh lə-l-mædrəsæh baʔfər
 NEG-3SM-go.IMPRF to-the-school tomorrow
 ‘He will not go to school tomorrow.’
- d. lo tə-mtʕər ma-nə-lʕæb
 if 3SF-rain.IMPRF NEG-1PL-play.IMPRF
 ‘If it rains, we won’t play.’
- e. ma-ku mætʕʕæm əhnək
 NEG-exist restaurant there
 ‘There isn’t a restaurant there.’
- f. ma-ʕədd-æh səyyaræh
 NEG-with-him car
 ‘He doesn’t have a car.’

In the so-called verbless sentences with nominal / adjectival / prepositional predicates (19), *ma* must combine with a subject pronominal clitic which agrees in number and gender with the subject as also reported for Egyptian, Kuwaiti, and Moroccan Arabic. Failing to do so results in ungrammaticality.

- (4.8) ʔæħmæd mə-hu moʕælləm / tʕəwil / b-əl-mætʕæm
 Ahmad NEG-him teacher / tall / at-the-restaurant
 ‘Ahmad is not a teacher / tall / at the restaurant.’

Ma also appears frequently in negative discourse expressions (4.9a) and serves as a negative quantifier with indefinite singular NPs (4.9b).

- (4.9) a. mə-ʕælyiæ-k
 NEG-on-you
 ‘Don’t worry about it!’
- b. mə wælæd raħ
 NEG boy.INDEF go.3SM.PRF
 ‘No boy left.’

Furthermore, a shared feature of all three negative markers is that they can express negative imperatives. However, they imply pragmatically different functions in that *la* implies prohibition, *mu(f)* communicates warning, and *ma* conveys that presupposed information has already been shared with the addressee.

- (4.10) a. la t-ruħ le:yad
 NEG 2SM-go.IMPRF there
 ‘Don’t go there!’
- b. mu(f) t-ruħ le:yad
 NEG 2SM-go.IMPRF there
 ‘You’re not supposed to go there!’

c. *ma t-ruḥ le:yad*
 NEG 2SM-go.IMPRF there
 ‘You’re not going there!’

Of these, only *la* can be used as a negative response:

(4.11) a. *ʃan ʃədd-æk sʰæff əlyom?*
 was with-you class today
 ‘Did you have a class today?’

b. *la*
 ‘No’

La is also used in negating a series of items and thus communicating negative contrasts (4.12a), as an emphatic negative quantifier with *wæ-* in verbal and verbless sentences (4.12b & 4.12c, respectively), and for invoking emotions (4.12d). When preceded by *wæ-*, *la* can host subject clitics (4.12e) in verbless sentences, indicating that *la*, like *ma*, also heads its own projection.

(4.12) a. *la ʔanæ w la əntæ*
 NEG I and NEG you
 ‘neither I nor you’

b. *wæ-la wælæd bə-l-mædrəsəh*
wæ-NEG boy.INDEF at-the-school
 ‘Not a single boy is at school.’

c. *wæ-la ʃədd-əh bæxæt*
wæ-NEG with-him conscience
 ‘He has no conscience whatsoever.’

d. ællæh la-y-rəzdʒ-æh
 God NEG-3SM-bless-him
 ‘May Allah not bless him!’

e. wæ-la-ni mɒhændəs
 wæ-NEG-me engineer
 ‘I’m not an engineer at all’ / ‘I am no engineer!’

f. wæ-la wælæd rəh
 wæ-NEG boy go.3SM.PRF
 ‘No boy left.’

g. * wæ-la wælæd mɑ-rəh
 wæ-NEG boy NEG-go.3SM.PRF
 Intended meaning: ‘No boy left.’

When *wæ-la* occurs in preverbal position, negation is not allowed (4.12g above) unless a Double Negation (DN) reading is intended (4.13a). However, when *wæ-la* is in postverbal position, the presence of negation is required (4.13b). Thus, when in lower position than verbal Neg, *wæ-la* is a Negative Concord (NC) item and its behavior patterns well with similar items in languages like Georgian (4.13c). We thus classify KhA as a non-strict NC language.

(4.13) a. wæ-la wælæd mɑ-rəh
 wæ-NEG boy NEG-go.3SM.PRF
 Intended meaning: ‘No boy didn’t leave’ (i.e., ‘Every boy left.’)

b. mɑ-rəh wæ-la wælæd
 NEG-go.3SM.PRF wæ-NEG boy
 ‘No boy left.’

c. shen-i megobar-i arasodes **ar** sv-am-s lud-s
 your-NOM friend-NOM never NEG drink-TH-3SBJ beer-DAT
 ‘Your friend never drinks beer.’

(Example 7b from Pfau, Makharoblidze, & Zeijlstra, 2022)

Finally, *mu* is a proclitic that can optionally combine with enclitic *-f* to form *muf*. Unlike some other dialects like Egyptian Arabic, in KhA the two morphemes are always continuous (i.e., never separated by another constituent). *Muf* occurs in verbless predicates and its scope of negation can be sentential or constituent. It also occurs in past tense (perfective) context (4.14). (Note that the following example has rising intonation).

(4.14) *Mu(f)* ʃənət b-əl-mædrəsəh?
 NEG be.2SM.PRF at-the-school
 ‘Weren’t you at school?’

In line with Aoun, Benmamoun, and Choueiri, (2010), we analyze *mu(f)* as an interrogative with a null expletive subject and thus an independent clause that takes another clause as its complement. Two pieces of supporting evidence for this claim are as follows:

(i) The fact that the verbal clause can be negated shows that *mu(f)* does not immediately precede the verb as the following example illustrates (Note that the following example also has rising intonation):

(4.15) *Mu(f)* ma-ʃənət b-əl-mædrəsəh?
 NEG NEG-be.2SM.PRF at-the-school
 ‘Isn’t it the case that you were not at school?’

(ii) Unlike *ma* and *la*, *mu(f)* can be moved, for instance, to the right periphery of the clause to form a tag question. Taken together, these results explain that the embedded clause verb does not raise to T to merge with Neg *mu(f)*. Note that there is a prosodic break between *mu(f)* and the rest of the sentence and obligatory rising intonation on sentence-final *mu(f)* since it forms an interrogative in KhA in this position.

- (4.16) mɑ-ʈənət b-əl-mædrəsæh, muʃ
 NEG-be.2SM.PRF at-the-school NEG
 ‘You were not at school, weren’t you?’

4.2.2.2 Negation in Sadat Tawaher Sign Language (STSL)

In this section, I will report the findings for negation in STSL. As stated earlier in Chapter 3, we used three data collection methods to examine negation in STSL, namely signed productions, stories, and grammaticality judgments. Below, I will report the findings for each data collection method, i.e., signed productions, stories, and grammaticality judgments, separately.

4.2.2.2.1 Negation Markers

STSL has, at least and as far as I am aware as a signer, six different negators which I have labelled as follows: NEG_{basic}, NEG_{ref}, NEG_{proh}, NEG_{emph}, NEG_{poss}, and NEG_{exist}. For each negator, I have incorporated the main semantic function as the subscript in its given title although some negators have more than one semantic function. These subscripts stand for the following: *basic* for basic, *ref* for refusive, *proh* for prohibitive, *emph* for emphatic, *poss* for possessive, and *exist* for existential. For instance, NEG_{poss} is used mainly to negate possession (e.g., ‘X does not have Y’).

Furthermore, note that two negators, namely NEG_{ref} and NEG_{emph}, were not attested in the data set and thus will not be included in the results reported. By the main semantic function, I mean the most frequent meaning for which that negator is used. However, as will become clear, the meanings that these negators convey overlap are not clear-cut. Thus, some negators can be used to convey multiple meanings. In addition to serving as clause negators, all these negators can be used as negative responses although, as we will see, only two were actually used for this function. Based on my own knowledge, each negative marker consists of a manual element and one or more non-manual elements associated with the manual element. In reporting the data, I will gloss the manual negator as *NEG* and the non-manual element(s) as abbreviations (e.g., *hs* for headshake, *bht* for backward head tilt, etc.). The line extending above the manual negator indicates the scope that the non-manual marker takes. STSL negators are briefly described below and a summary of them is provided in Table 4.3.

(1) NEG_{basic}:

Configurally, NEG_{basic} consists of a single upward movement of the dominant hand or of both hands from the wrist and a backward head-tilt (bht) simultaneously. I will present evidence in the section 4.2.4 that NEG_{basic} is the basic clause negator in STSL.



Figure 4.1. NEG_{basic}

bht

(4.17) IX-1 SMOKE NEG_{basic}

‘I don’t smoke.’

(2) NEG_{ref}:

NEG_{ref} consists of a simultaneous single rightward movement of the dominant hand and a single leftward head movement (lhm). In other words, the hand and head move in opposite directions. NEG_{ref} expresses refusal to do an action. It might be accompanied by pouted lips and furrowed brows. Since this negator was not attested in the data, it will not be included in the reporting of the results.



Figure 4.2. NEG_{ref}

- lhm
- (4.18) CHILD EAT BREAKFAST NEG_{ref}
 ‘The child refused to eat breakfast.’

(3) NEG_{proh}:

NEG_{proh} comprises repeated side-to-side movement of both hands or only the dominant hand and can be accompanied by a simultaneous side-to-side headshake (hs) and brow raise (br). Semantically, it expresses prohibition, warning, or simply emphatic negation.



Figure 4.3. NEG_{proh}

- hs
- (4.19) FRIEND EAT MEAT NEG_{proh}
 ‘My friend never eats meat / doesn’t eat meat at all.’

(4) NEG_{emph}:

NEG_{emph} is similar to NEG_{proh} in its pragmatic function (i.e., emphasis) and type and direction of the movement of the manual and non-manual elements (i.e., side-to-side headshake) except for the handshape which is composed of the repeated, fast wagging of the index finger of the dominant hand and brow raise. As a signer, I do not use NEG_{emph} as a clause negator. Like NEG_{ref}, this negator was also not attested in the data and therefore will be excluded from the results.



Figure 4.4 NEG_{emph}

If NEG_{emph} turns out to be capable of negating clauses like other negators, then it would look the following example:

hs

(4.20) MONEY IX-1 STEAL NEG_{emph}
 ‘I did not steal the money at all.’

(5) NEG_{poss}:

NEG_{poss} consists of the tips of all fingers of the dominant hand (or of both hands) swiping down the tip of the nose with air blown out (abo) of the mouth simultaneously. This negator is used to indicate lack of possession (‘not-have’) and non-existence.



Figure 4.5. NEG_{poss}

abo

(4.21) MAN IX-3 CAR NEG_{poss}

‘That man doesn’t have a car.’

(6) NEG_{exist}:

NEG_{exist} consists of the hands alternately up and down at least twice and side-to-side headshake. It communicates non-existence and lack of possession.



Figure 4.6. NEG_{exist}

hs

(4.22) CHILD YARD NEG_{exist}

‘There are no kids in the yard.’

NEG_{exist} is the negative existential marker and thus suggests primarily non-existence but also lack of possession. NEG_{exist} is interchangeable with NEG_{basic} but seems to have a different pragmatic function: NEG_{exist} implies a sense of unexpectedness while NEG_{basic} simply negates the existence of something, another reason to argue that NEG_{basic} is the basic negator. An interesting feature of NEG_{exist} is that it uses the same sign that is used for marking completive aspect in STSL, i.e., *FINISH*:

(4.23) CHILD READ BOOK FINISH

‘The child finished reading the book.’

The use of completive aspect markers to communicate a negative existential meaning has also been reported for some emerging sign languages, e.g., Kata Kolok (Perniss & Zeshan, 2008).

Table 4.2 provides a summary of STSL negators. In the next section, I will report the findings from the signed productions, as pertains to sentential negation.

Table 4.2. STSL Negators

Negator Features	NEG _{basic}	NEG _{ref}	NEG _{proh}	NEG _{emph}	NEG _{poss}	NEG _{exist}
Manual negator handshape	upward hand movement	rightward hand movement	side-to-side hand movement	index finger wagging	fingertips swiping down nose	both hands swiping each other up & down
Manual negator: optional or obligatory?	obligatory	obligatory	obligatory	obligatory	obligatory	obligatory
Non-manual negator handshape	backward head tilt	leftward head movement, furrowed brows, pouted lips	side-to-side headshake, brow raise	side-to-side headshake, brow raise	air blow out of mouth	side-to-side headshake
Non-manual negator: optional or obligatory?	optional	optional	optional	optional	optional	optional
Spreading domain of NMM	manual negator	manual negator	manual negator	manual negator	manual negator	manual negator
Spreading of non-manual negator on other signs / constituents	disallowed	disallowed	disallowed	disallowed	disallowed	disallowed
Clause negator?	√	√	√	√	√	√
Used as a negative response particle?	Yes	Yes	Yes	Yes	Yes	Yes
Negative response marker(s)	both	√	√	√	√	√
	MM only	√	√	√	√	√
	NMM only	√	√	√	X	X

Table 4.2 continued

Pragmatic function	standard negation, non-existence, lack of possession	refusal	prohibition, warning, emphasis	emphasis	lack of possession, non-existence	non-existence
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4.2.2.2.2 Findings from Signed Productions

In this section, I present the results of the signed productions for negation. As we mentioned before in Chapter 3: Methodology, we provided the participants with 36 negative stimulus sentences in KhA, either in written or audio form and asked them to sign them. The results are presented in Table 4.3.

Table 4.3. Signed Productions for Negation

Negation Markers				Other																	
MM		NMM		NMMs																	
		(bht)		Word order & Position of NEG																	
Overt	Non-overt	Overt	Non-overt	Brow raise	Eye gaze	SVO-Neg	SOV-Neg	SV-Neg	SOVO-Neg	SVOV-Neg	SVOVO-Neg	SdOVIO-Neg	SdOiOV-Neg	SVdOiO-Neg	SVO-Neg-VO-Neg	S-Neg-VO-Neg	SO-Neg	V-Neg	S-Neg	SV-Neg-O	SdOV-Neg-iO
158	9	43	124	52	5	47	45	41	1	3	3	2	1	1	1	1	6	2	1	2	1

Note: dO: direct object, iO: indirect object, NMM: non-manual marker, O: object, S: subject, V: Verb

Five signers participated in this task and, since there were 36 original stimulus sentences, we were expecting 180 sentences to be produced; however, the participants either failed to sign some sentences or signed some wrong ones which we had to discard. Thus, the final total number of sentences was lower than we should have received. Overall, we considered 167 negative sentences for the final analysis. Before we present these findings in detail, it is noteworthy that the only negation marker that the participants used in all the negative sentences from the signed productions task was NEG_{basic}.

As can be seen from Table 4.4, of the 167 negative sentences, the manual marker of negation is overtly signed in 158 cases (94.6%) but is non-overt in only 9 cases (5.4%). In the 9 sentences in which the manual negator was not overt, the non-manual negator was not overt, either. Thus, there was no signed negator, manual or non-manual, in these nine cases. However, we still considered these sentences to be negative since the signers spoke negative Arabic sentences although they failed to sign a negative marker. On the other hand, the non-manual marker of negation (i.e., backward head-tilt) was overt in 43 sentences (25.7%) and non-overt in 124 sentences (74.3%). In the 43 sentences in which the non-manual negator was present, backward head tilt (bht) was either only slightly or very slightly visible and only occasionally noticeably visible. As for other non-manual activities, two non-manual markers were attested in the data: 52 cases of brow raise (br) (32.9%) and 5 cases of eye gaze (3.1%). Both of these non-manual negators occurred simultaneously on the manual negator but did not spread on any other adjacent signs.

As for its position in the sentence, the negation marker appears in sentence-final position in 155 sentences (98.1%) including two sentences in which the negator is repeated in preverbal and sentence-final positions. In 3 sentences (1.9%), the negator appears in non-sentence-final positions, in one case between V and O (SV-Neg-O) and in two cases between V and the indirect object (SdOV-Neg-iO). In 6 cases, no verb was signed (SO-Neg) and thus the position of the verb, and as a result the position of Neg is unclear. Furthermore, in 2 cases only a verb was signed followed by NEG (V-Neg) with no arguments. In both sentences the verb was of the weather type (e.g., rain) and perhaps not unexpected although all other sentences with weather verbs had overt subject arguments (which was equal to the subject argument in KhA for weather verbs, i.e., ‘the sky’, ‘the world’, or ‘the clouds’).

An interesting finding was that, since the signers verbally spoke these negative sentences while signing them, the negative element appeared simultaneously in two positions: in preverbal position in KhA and in sentence-final position in STSL (4.24a & 4.24b). The negative marker appeared in pre-verbal position in KhA regardless of the word order of the sentence when spoken. For instance, while (4.24a) and (4.24b) have SVO and SOV orders, respectively, the negative marker still appears preverbally (i.e., S-Neg-VO and SO-Neg-V).

(4.24) Position of negative marker in STSL (sentence-final) & KhA (preverbal)



KhA: əl-wadəm mɑ-ħəs^ɕd-æt ʃələb
the-people NEG-harvest-3SF.PRF rice

			bht

STSL:	PEOPLE	HARVEST	RICE
			NEG_{basic}

‘The people did not harvest the rice.’



KhA: əs-sərbəz əz-zəlmæh mɑ-rəmə-h
the-soldier the-man NEG-shoot.3SF.PRF-him

			bht + br

STSL:	SOLDIER	MAN	SHOOT
			NEG_{basic}

‘The soldier did not shoot the man.’

4.2.2.2.3 Findings from Story-telling

As we mentioned in Chapter 3: Methodology, our rationale for using stories as a data collection method was to provide the participants with enough freedom to sign as naturally and as comfortably as possible without worrying about diverging from a set of fixed sentences as in signed productions. We gathered two types of stories, namely bedtime stories and everyday activities, from four participants. Overall, we collected 32 bedtime stories (length: 01:17:38) and 208 everyday activities (length: 03:55:50). The total number and length of both types of stories equaled 208 stories / activities and 05:13:28 in length.

Table 4.4. Story-telling Data

	Stories I: Bedtime Stories			Stories II: Everyday Activities		
	# of videos	# of stories	Length	# of videos	# of stories	Length
P #1	30	3	00:21:36	25	25	00:13:05
P #4	33	18	00:50:37	268	151	03:34:33
P #2	2	2	00:00:30	21	21	00:04:06
P #9	9	9	00:04:55	11	11	00:04:36
Total	74	32	01:17:38	325	208	03:55:50
Total (stories I & II)	399 videos	240 stories / activities	05:13:28			

As Table 4.4 shows, the everyday activity videos exceeded the bedtime stories videos both in number and length significantly (almost 3:1). There are three reasons were for this discrepancy. First, there are far more everyday life activities that one can talk about than bedtime stories s/he can narrate. Second, most participants did not remember some of the bedtime stories well and probably decided to provide everyday activities instead. Third, one of the participants, P #4, provided many more videos of everyday activities than the three other participants combined (03:34:33 vs. 00:21:47). This is clearly one of the limitations of the current study which cannot be remedied at this point.

In this section, I present the results for negation from the stories. In Table 4.5, the results of single clausal negation are presented, that is, sentences in which a single negative marker (as opposed to, e.g., two or more negators) negated a clause (as opposed to, e.g., constituent negation).

Table 4.5. Single Clausal Neg – Stories I: Bedtime Stories & Stories II: Everyday Activities

Stories		NEG _{basic}	NEG _{proh}	NEG _{poss}	NEG _{exist}	Other	No overt Neg
I		93	2	10	16	27	22
	Total	148					22
II		653	35	49	29	91	69
	Total	857					69
I + II		746	37	59	45	118	91
	Total	1005					91

As can be seen from Table 4.5, of the six negators that we expected to find in STSL, four were used in the stories as clausal negators: NEG_{basic}, NEG_{proh}, NEG_{poss}, and NEG_{exist}. Altogether, we found 1005 instances of these clausal negators including a negative marker that we categorized separately as ‘Other’ which occurred 118 times. Excluding the negator ‘Other’, the most frequently used negator was NEG_{basic} (746 occurrences) followed by NEG_{poss} (59), NEG_{exist} (45), and NEG_{proh} (37). The following are examples of all five negators as used in the bedtime stories:

(4.25) Single clausal negation: NEG_{basic}



SOW

WATER

DIRECT

WATER

GRAVE



NEAR

FLOW

NEG_{basic}

‘They sowed the ground, watered it and directed the water towards the grave [but] the water did not go near the grave.’

(4.26) Single clausal negation: NEG_{proh}



GIRL

DIE

NEG_{proh}

‘The girl didn’t die.’

(4.27) Single clausal negation: NEG_{poss}



PERSON

SEVEN

GIRL



BOY

NEG_{poss}

‘A man had seven daughters, but he didn’t have a son [any sons].’

(4.28) Single clausal negation: NEG_{exist}



CLOCK

NEG_{exist}

EAT

SLEEP

‘We did not have a clock. We would eat and then go to bed.’

(4.29) Single clausal negation: Other [NOT-KNOW]



WITCH

HORSE

EAT

MAN

NOT-KNOW [Other]

‘The witch was eating the horse and the man did not know that.’

The following is an example of a bedtime story which narrates the account of a man who takes his elderly father to a remote area to leave him there because the son is tired of looking after his father. The moral of the story is karma, i.e., how one treats one’s parents is how one’s children will treat one:

(4.30) SAY MAN FATHER CARRY TAKE GO LAND BIG HOUSE OTHER **NEG_{exist}**. HOUSE PEOPLE CAR **NEG_{exist}**. GROUND **NEG_{proh}**. TREE ONE BIG. FATHER PUT TREE NEXT-TO, LEAVE. GO LITTLE. FATHER LAUGH. SON LOOK-BACK. SAY LAUGH WHY. SAY LAUGH PAST ME FATHER PUT HERE. NOW YOU ME PUT HERE. FUTURE SON YOU PUT HERE. FEEL SORRY. COME CARRY GO. LEAVE TREE UNDER **NEG_{basic}**.

‘It is said that a man carried his father and took him to a big area where there were no houses around. There were no houses, people, or cars. There was nothing in that area. There was one big tree. The man put his father down next to the tree and left. He walked a little. The father laughed. The man looked back and asked why he was laughing. The father said, “I’m laughing because in the past I also left my father here. Now you left me here. In the future, your son will leave you here.” The man felt sorry. He went back [to his father], carried him back and left. He didn’t leave him under the tree.’

In the above story, the negation markers appear in sentence-final position unaccompanied by any associated NMMs. In cases where the non-manual negator accompanied the manual negator, it never spread over any adjacent signs.

4.2.2.2.3.1 *Negative Marker ‘Other’*

In addition to the manual negators that neatly fit in one of the categories above (i.e., NEG_{basic}, NEG_{proh}, NEG_{exist}, and NEG_{poss}), we found 118 instances of manual negators that did not resemble any of these negators. These negators were used mostly and almost exclusively as negative irregular verbs or negative modals such as ‘NOT-WANT’, ‘NOT-POSSIBLE’, ‘NOT-LIKE’, ‘NOT-KNOW’, etc. The following are examples of the category ‘Other’:

(4.31) Negative marker ‘Other’



a. NOT-POSSIBLE

‘That’s not possible. [No way!]



b. GIRL SAY NOT-WANT

‘The girl said she didn’t want / like [her suitor].’

One relatively frequent non-manual marker of the negative category ‘Other’ was the use of both hands in a push-away motion away from the body as if the signer is indicating that s/he is pushing something or someone away to indicate lack of interest, willingness, and similar emotions. This was usually accompanied by a head turn in the opposite direction of that of the hands and looking away as if to avoid eye contact. These movements are in line with the semantics of this category of negator since they iconically indicate these emotions as shown in (4.32).

(4.32)



a. WOMAN SAY NOT-WANT

‘His wife said, ‘Let it go’ [don’t go near it!]

Given the above features, it seems that negative sentences of the category of ‘Other’ in STSL are examples of incorporated negation. According to Woodward (1974, p. 22), in the case of ASL, negative incorporation occurs in “verbs that may be negated by a bound outward twisting movement of the moving hand(s) from the place where the sign is made”. Antzakas (2006) reports instances of incorporated negation in GSL which, according to him, was used mostly for verbs of experience and sensation such as ‘have-not’, ‘like-not’, ‘want-not’, ‘know-not’, ‘disagree’, ‘believe-not’, ‘should-not’ (p. 261). It is possible that this category forms a group of suppletive forms similar to the ones described for TĪD. Zeshan (2006c) mentions that TĪD has different forms for EXIST and NOT-EXIST, that is, the forms are formationally different for the positive and for the negative form.

4.2.2.2.3.2 *Non-overt Negators in STSL*

One rather surprising finding of the current study is sentences with a non-over negator in the story-telling data Overall, of the 1005 NEG occurrences, there were 91 (9%) cases of non-overt

NEG which is somewhat high. These were cases in which although the spoken (Arabic) sentence did contain a negator, the signed one did not. Therefore, these are cases that we are certain that the sentence was in fact negative since the spoken counterpart had an overt negative marker. In this section, I will describe these sentences, provide examples, and present an explanation for their occurrence. Sentences with a non-overt negator occurred in both types of stories but far more frequently in the everyday activity type which is not surprising as there was a far larger number of such stories than the bedtime type. Analyzing the sentences with a non-overt negative marker and the contexts in which they occurred shows that there are three main reasons for the occurrence of these sentences. First, most non-overt negators occurred in sentences with a sequence of verbs, normally three verbs but at least two, in which the first verb, and occasionally the last one as well, was negated but not the other verbs. Thus, in such cases, it seems that the first instance of negation takes scope over the rest of the discourse which needs to be negated. The following example illustrates one such occurrence:

(4.33)



SEE COME HERE CRY NOT-GOOD SLEEP NEG_{basic}



EAT TALK LAUGH NOT-GOOD

‘I saw [them] and came back home and cried. I did not feel good. I didn’t sleep, [didn’t] eat, [didn’t] talk, [didn’t] laugh. I did not feel good.’

In (4.33) above, the participant describes her visit to her hometown when it was flooded and how sad she felt about the situation. Then, she states that she could not fall asleep as her mind was preoccupied with what was going on in her hometown. While mentioning this, she uses an overt negative marker, in this case NEG_{basic}, for the verb SLEEP but does not do so nor does she use any other overt negator(s) for the other verbs EAT, TALK, and LAUGH. Therefore, it seems that the participant intended the negator of the first verb, SLEEP, to take scope over the other verbs. We are not sure if she intended to do so consciously but this sounds like a plausible conclusion. Similarly, in the next example, only the verb EAT is negated overtly while the verb SHOWER has no overt negator although both verbs are overtly negated in their spoken form.

(4.34)



‘The girl climbed up the bed and fell off and broke her arm. [Now] she can’t eat or shower. She just holds her arm [like this].’

At other times, the same verb was used again not long after the first one which was overtly negated. In such cases, we are not dealing with a sequence of different verbs but the same verb repeated. However, the second one was not overtly negated. In such cases, it is very likely that the participant thought of negating the first mention of the verb as enough to be extended to the second mention of the same verb. In the following example, which illustrates this occurrence, the participant negates the verb LIFT overtly but does not do so when he uses the same verb again. Note that once again in the spoken sentence, both occurrences of the verb LIFT are overtly negated.

(4.35)



$$\frac{\text{bht} + \text{br}}{\text{NEG}_{\text{basic}}}$$
 LIFT LIFT
 ‘(They) couldn’t lift (it). They [couldn’t] lift (it).’

The second reason for the occurrence of sentences with non-overt negative markers could be due to contextual information. In the following example, the participant mentions the pain and swelling that she has frequently had recently in her finger and then goes on to mention that, due to this pain, she cannot knead the dough or bake the bread. Although the Arabic sentences were overtly negated, the signed ones had no overt negators, leading us to argue that it is possible that the participant thought of contextual information as sufficient for interpreting the sentence as negative since a physically unpleasant experience, in this case a swollen finger, makes it hard to get things done.

(4.36)



‘My finger has swollen. I [can’t] knead the dough and [can’t] bake the bread.’

The third and last reason for sentences with a non-overt negator could be distraction. Since all the STSL signers speak when they sign, it is very well possible that the participants get distracted at times by their spoken sentences and fail to sign everything they speak. This could be because they assume that they have in fact signed everything since they heard themselves say it.

4.2.2.2.3.3 *Manual and Non-manual Markers of Negation in STSL*

In this section, I present the results regarding the use of manual and non-manual markers of sentential negation as used in the story-telling data. Table 4.6 shows the frequency and percentage with which the manual marker and non-manual marker of each of the STSL negators were used. I have not included the category ‘Other’ since there is not a single specific manual or non-manual marker associated with this category which was addressed separately in section 4.2.2.2.3.1.

Table 4.6. Occurrence of Manual & Non-manual Markers of Sentential Negation in Stories (Excluding ‘Other’)

Marker Negator	Manual marker		Non-manual marker		
	Overt	Non-overt	Non-manual marker	Overt	Non-overt
NEG _{basic}	746 (100%)	0 (0%)	backward head tilt (bht)	83 (11.1%)	663 (88.9%)
NEG _{proh}	37 (100%)	0 (0%)	headshake (hs)	23 (62.2%)	14 (37.8%)
NEG _{poss}	59 (100%)	0 (0%)	air blown out (abo)	0 (0%)	59 (100%)
NEG _{exist}	45 (100%)	0 (0%)	headshake (hs)	1 (2.2%)	44 (97.8%)
Total	887 (100%)	0 (0%)	Total	107 (12.1%)	780 (87.9%)

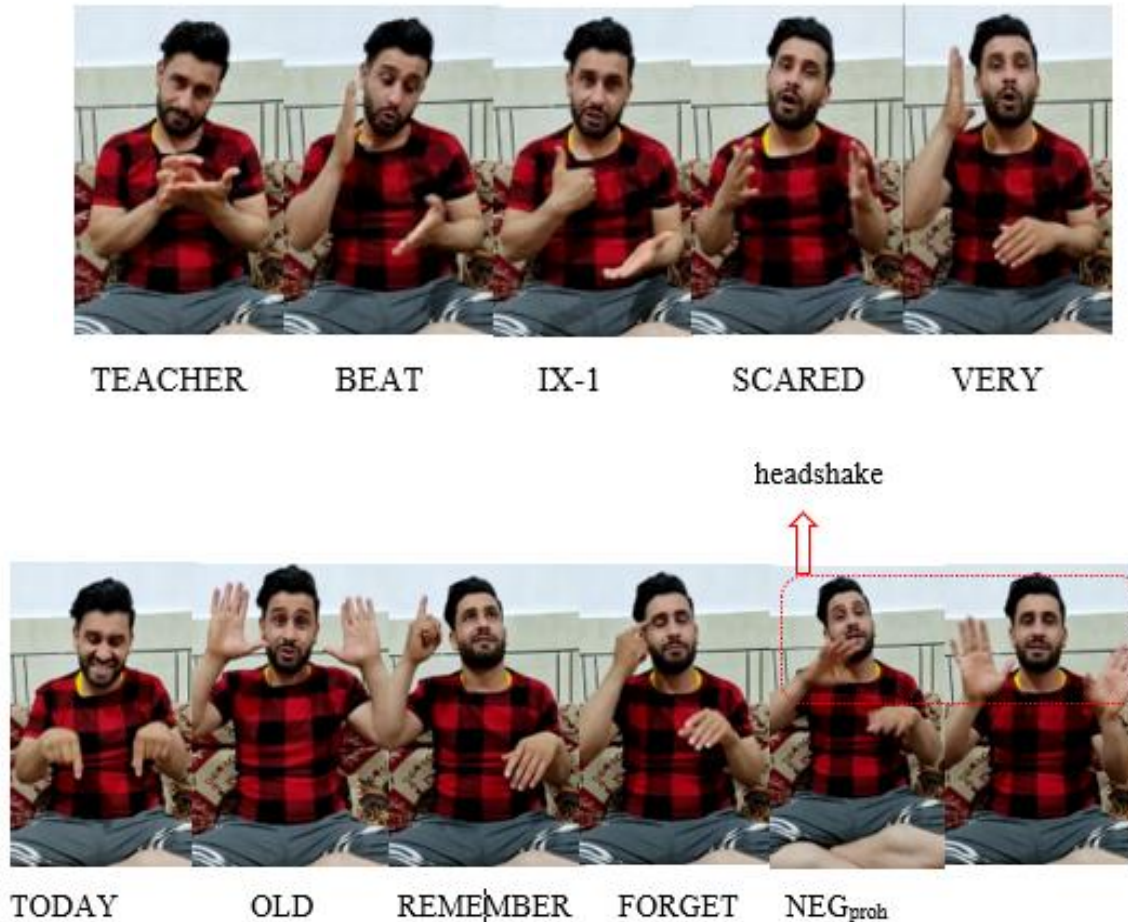
As can be seen from Table 4.6, for the negators that were in fact used in the stories (i.e., NEG_{basic}, NEG_{proh}, NEG_{poss}, and NEG_{exist}), the manual negator was present in all the negative sentences. In this regard, NEG_{basic} was used far more frequently (746 times) than the other three negators which, combined, were used 141 times.

The non-manual negators, on the other hand, were present with three manual negators, namely, NEG_{basic}, NEG_{proh}, and NEG_{exist}, but not with NEG_{poss}. Overall, out of the total number of 887 negative sentences, only 107 ones (12.1%) had a non-manual negator in addition to a manual

negator while 780 sentences (87.9%) had no main non-manual negator. This means that while all the negative sentences had a manual negator, only one fifth of them had a non-manual negator. Thus, there was not a single case of a sentence negated by a non-manual negator alone while the vast majority of the sentences (87.9%) had a manual negator only and 12.1% had both a manual and a non-manual negator. In this regard, headshake, the main non-manual marker of NEG_{exist}, was used only once but absent in almost 98% of the cases in which NEG_{exist} was used and is thus negligible.

Therefore, one can say that only two manual negators were accompanied by their associated non-manual markers: NEG_{basic} and NEG_{proh}. Of these two negators, headshake was used quite frequently with NEG_{proh} (62.2%) while backward head tilt, the non-manual marker of NEG_{basic} was used only 83 times (11.1%) of the cases. Since the proportion of use of NEG_{basic} to NEG_{proh} is 20 to 1 (as there are 20 times more cases of NEG_{basic} than NEG_{proh}), this 11.1% of backward head tilt use becomes even more insignificant than the headshake of NEG_{proh} which was used almost six times (i.e., 62.2%) more than the backward head tilt. What is noteworthy about these non-manual markers, i.e., headshake and backward head tilt, is that they were mostly slightly visible and only occasionally well noticeable. The following are examples of visible and slightly visible cases of both headshake and backward head tilt.

(4.37) Visible headshake with NEG_{proh}



‘The teacher punished me, and I was very scared. To this day, that I’m old now, I still remember it. I have not forgotten at all.’

Both headshake and backward head tilt appeared simultaneously with the manual negators and never spread over any other constituents or even adjacent signs.

The main non-manual markers were not the only ones that were used with the manual negators. There were four other non-manuals that occasionally accompanied manual negators: brow raise, head turn, backward head tilt, and eye gaze. Once again as in the main non-manual negators, the shared feature of all of these non-manual negators is that none of them spread over any other constituents or adjacent signs. Table 4.7 presents the frequency with which these non-manual markers were used.

Table 4.7. Occurrence of Other Non-manual Markers of Sentential Negation in Stories

Marker	Non-manual marker			
	brow raise	head turn	backward head tilt	eye gaze
NEG _{basic}	132 (17.7%)	6 (0.8%)	N/A	7 (0.9%)
NEG _{proh}	5 (13.5%)	34 (91.9%)	1 (2.7%)	0 (0%)
NEG _{poss}	8 (13.5%)	0 (0%)	7 (11.8%%)	0 (0%)
NEG _{exist}	3 (6.6%)	2 (4.4%)	0 (0%)	0 (0%)
Total	148	42	8	7

Table 4.7 shows that brow raise was the most frequent non-manual marker in terms of both the frequency with which it occurred (148 times) and the number of manual negators that it appeared with, namely four negators, which are as follows: NEG_{basic}, NEG_{proh}, NEG_{poss}, and NEG_{exist}. However, it is seen that brow raise was most frequently used with NEG_{basic} (132 out of 148 times) while it was used only 16 times with the other three negators. This is a ratio of 8 to 1. Clearly, this is not quite surprising since NEG_{basic} was also used 746 times and the other negators 141 times which is a ratio of 5 to 1. Thus, of the 746 sentences that were negated with NEG_{basic}, 132 (17.7%) were also marked by brow raise, i.e., approximately one fifth of the total number. The next manual negators, NEG_{proh} and NEG_{poss} had lower percentage of brow raise use (13.5%) and even lower for NEG_{exist} (6.6%).

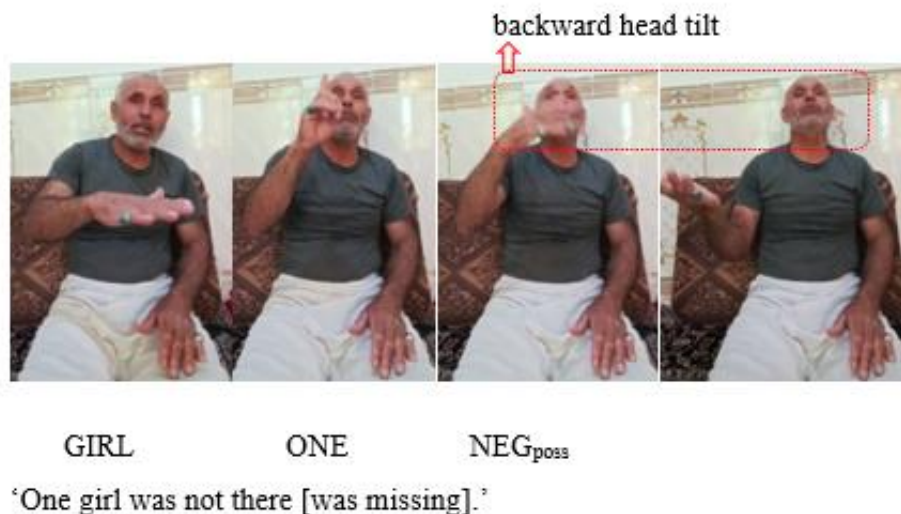
The next non-manual marker was head turn which was used very frequently with NEG_{proh} (34 out of 37) but far less so with NEG_{exist} (4.4%) and less than one percent with NEG_{basic}. The extensive use of head turn with NEG_{proh} warrants some discussion. An interesting finding in the case of head turn is that, out of the 42 cases of its occurrence, it occurred with NEG_{proh} in 34 cases (80.9%). In such cases, the head turn is then followed by headshake which in most cases is slight. This correlation between head turn and headshake seems to be more than just a coincidence and warrants separate analysis. The following example illustrates the use of head turn with NEG_{proh}.

(4.38) Head turn with NEG_{proh}



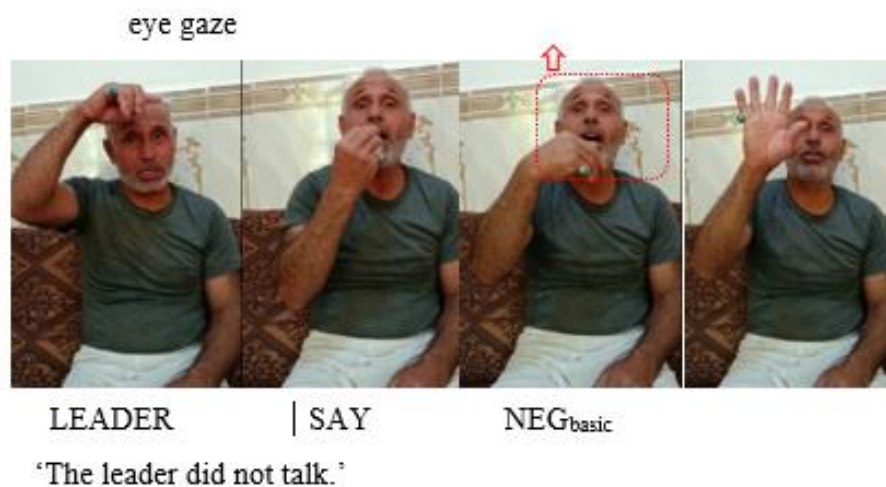
The third non-manual marker was backward head tilt which occurred only once with NEG_{proh} but 7 times with NEG_{poss}. Considering the downward movement of the dominant hand or of both hands which is the manual negator in NEG_{poss}, it is likely that backward head tilt automatically occurred as a result of the physiological alignment of the movement of the hand(s) and head in opposite directions (i.e., downward and upward, respectively) but along the same axis (i.e., Y axis). The following example illustrates this non-manual marker as it occurred with NEG_{poss}:

(4.39) Backward head tilt with NEG_{poss}



The last non-manual marker which occurred in less than 1% of the cases and only with NEG_{basic} was eye gaze. The insignificant frequency of occurrence of eye gaze might suggest that it does not serve any syntactic or phonological purposes but occurred automatically due to the occurrence of brow raise. This is a plausible hypothesis especially because all the seven cases of eye gaze were sentences that had brow raise on NEG_{basic}. The following is an example of eye gaze with NEG_{basic}.

(4.40) Eye gaze with NEG_{basic}



4.2.2.2.4 Findings from Grammaticality Judgments

For grammaticality judgments, we chose two negators, namely NEG_{basic} and NEG_{proh}, to test our hypothesis that STSL is manual-dominant. We used both SVO and SOV orders to control for the possible effect of word order on acceptability. Our choice of these negators is justified on the following facts:

- (i) NEG_{basic} and NEG_{proh} are both very frequently used;
- (ii) They both have various semantic functions and can be used in almost any context (e.g., existence, possession);
- (iii) The NMMs associated with NEG_{basic} and NEG_{proh} (backward head-tilt and headshake, respectively) can commonly spread in many sign languages; and finally,
- (iv) NEG_{poss} and NEG_{exist} seem reserved for more specialized contexts as the results from the stories showed (e.g., possession or existence) and thus not very frequently used. Additionally,

it is hard, if not impossible, to imagine how the non-manual marker of NEG_{poss} which is air blown out of the mouth would physiologically spread. Try blowing air out of your mouth over the entire VP / sentence for NEG_{poss}! The two sentences that we used for grammaticality judgments, the tests for examining the manual vs. nonmanual dominance, and the results are shown in Tables 4.8 - 4.11.

(i) Grammaticality judgments with NEG_{basic}:

(a) Declarative sentence #1: FRIEND BUY CAR (SVO)

‘My friend bought a car.’

Table 4.8. Grammaticality Judgments for NEG_{basic} / SVO Order

Test	Result
(a) Sentence-final neg + bht: FRIEND BUY CAR <u>NEG_{basic}</u> <u>bht</u>	grammatical
(b) Sentence-final neg without bht: FRIEND BUY CAR NEG _{basic}	grammatical
(c) Sentence-final neg + bht spreading over whole clause: FRIEND BUY CAR <u>NEG_{basic}</u> <u>bht</u>	ungrammatical
(d) Sentence-final neg + bht spreading over VP: FRIEND <u>BUY CAR</u> <u>NEG_{basic}</u> <u>bht</u>	ungrammatical
(e) Only bht; no spread; no manual neg: FRIEND BUY CAR <u>bht</u>	ungrammatical
(f) Spreading bht; no manual neg: FRIEND BUY CAR <u>bht</u>	ungrammatical
(g) Sentence-initial neg with bht: <u>NEG_{basic}</u> FRIEND BUY CAR <u>bht</u>	ungrammatical
(h) Sentence-initial neg without bht: NEG _{basic} FRIEND BUY CAR	ungrammatical
(i) Sentence-initial neg + spreading bht: <u>NEG_{basic}</u> FRIEND BUY CAR <u>bht</u>	ungrammatical
(j) Preverbal neg with bht: FRIEND <u>NEG_{basic}</u> <u>BUY CAR</u> <u>bht</u>	ungrammatical
(k) Preverbal neg without bht: FRIEND NEG _{basic} BUY CAR	ungrammatical
(l) Preverbal neg + spreading bht: FRIEND <u>NEG_{basic}</u> <u>EAT MEAT</u> <u>bht</u>	ungrammatical

Note: bht = backward head tilt

(b) Declarative sentence #1: FRIEND CAR BUY (SOV)

‘My friend bought a car.’

Table 4.9. Grammaticality Judgments for NEG_{basic} / SOV Order

Test	Result
(a) Sentence-final neg + bht: FRIEND CAR BUY <u>NEG_{basic}</u> <u>bht</u>	grammatical
(b) Sentence-final neg without bht: FRIEND CAR BUY NEG _{basic}	grammatical
(c) Sentence-final neg + bht spreading over whole clause: FRIEND CAR BUY <u>NEG_{basic}</u> <u>bht</u>	ungrammatical
(d) Sentence-final neg + bht spreading over VP: FRIEND CAR BUY <u>NEG_{basic}</u> <u>bht</u>	ungrammatical
(e) Only bht; no spread; no manual neg: FRIEND CAR BUY <u>bht</u>	ungrammatical
(f) Spreading bht; no manual neg: FRIEND <u>CAR BUY</u> <u>bht</u>	ungrammatical
(g) Sentence-initial neg with bht: <u>NEG_{basic}</u> <u>FRIEND CAR BUY</u> <u>bht</u>	ungrammatical
(h) Sentence-initial neg without bht: NEG _{basic} FRIEND CAR BUY	ungrammatical
(i) Sentence-initial neg + spreading bht: <u>NEG_{basic}</u> <u>FRIEND CAR BUY</u> <u>bht</u>	ungrammatical
(j) Preverbal neg with bht: FRIEND <u>NEG_{basic}</u> <u>CAR BUY</u> <u>bht</u>	ungrammatical
(k) Preverbal neg without bht: FRIEND NEG _{basic} CAR BUY	ungrammatical
(l) Preverbal neg + spreading bht: FRIEND <u>NEG_{basic}</u> <u>CAR BUY</u> <u>bht</u>	ungrammatical

Note: bht = backward head tilt

(ii) Grammaticality judgments with NEG_{proh}:

(a) Declarative sentence #2: FRIEND EAT MEAT (SVO)

‘My friend eats meat.’

Table 4.10. Grammaticality Judgments for NEG_{proh} / SVO Order

Test	Result
(a) Sentence-final neg + hs: FRIEND EAT MEAT <u>NEG_{proh}</u> <u>hs</u>	grammatical
(b) Sentence-final neg without hs: FRIEND EAT MEAT NEG _{proh}	grammatical
(c) Sentence-final neg + hs spreading over whole clause: FRIEND EAT MEAT <u>NEG_{proh}</u> <u>hs</u>	ungrammatical
(d) Sentence-final neg + hs spreading over VP: FRIEND EAT <u>MEAT NEG_{proh}</u> <u>hs</u>	ungrammatical
(e) Only hs; no spread; no manual neg: FRIEND EAT MEAT <u>hs</u>	ungrammatical
(f) Spreading hs; no manual neg: FRIEND EAT <u>MEAT</u> <u>hs</u>	ungrammatical
(g) Sentence-initial neg with hs: <u>NEG_{proh}</u> <u>hs</u> FRIEND EAT MEAT	ungrammatical
(h) Sentence-initial neg without hs: NEG _{proh} FRIEND EAT MEAT	ungrammatical
(i) Sentence-initial neg + spreading hs: <u>NEG_{proh}</u> <u>hs</u> FRIEND EAT MEAT	ungrammatical
(j) Preverbal neg with hs: FRIEND <u>NEG_{proh}</u> <u>hs</u> EAT MEAT	ungrammatical
(k) Preverbal neg without hs: FRIEND NEG _{proh} EAT MEAT	ungrammatical
(l) Preverbal neg + spreading hs: FRIEND <u>NEG_{proh}</u> <u>hs</u> EAT MEAT	ungrammatical

Note: hs = headshake

(b) Declarative sentence #2: FRIEND MEAT EAT (SOV)

‘My friend eats meat.’

Table 4.11. Grammaticality Judgments for NEG_{proh} / SOV Order

Test	Result
(a) Sentence-final neg + hs: FRIEND MEAT EAT <u>NEG_{proh}</u> <u>hs</u>	grammatical
(b) Sentence-final neg without hs: FRIEND MEAT EAT NEG _{proh}	grammatical
(c) Sentence-final neg + hs spreading over whole clause: FRIEND MEAT EAT <u>NEG_{proh}</u> <u>hs</u>	ungrammatical
(d) Sentence-final neg + hs spreading over VP: FRIEND MEAT <u>EAT NEG_{proh}</u> <u>hs</u>	ungrammatical
(e) Only hs; no spread; no manual neg: FRIEND MEAT EAT <u>hs</u>	ungrammatical
(f) Spreading hs; no manual neg: FRIEND <u>MEAT EAT</u> <u>hs</u>	ungrammatical
(g) Sentence-initial neg with hs: <u>NEG_{proh}</u> <u>hs</u> FRIEND MEAT EAT	ungrammatical
(h) Sentence-initial neg without hs: NEG _{proh} FRIEND MEAT EAT	ungrammatical
(i) Sentence-initial neg + spreading hs: <u>NEG_{proh}</u> <u>hs</u> FRIEND MEAT EAT	ungrammatical
(j) Preverbal neg with hs: FRIEND <u>NEG_{proh}</u> <u>hs</u> MEAT EAT	ungrammatical
(k) Preverbal neg without hs: FRIEND NEG _{proh} MEAT EAT	ungrammatical
(l) Preverbal neg + spreading hs: FRIEND <u>NEG_{proh}</u> <u>hs</u> MEAT EAT	ungrammatical

Note: hs = headshake

As can be seen from Tables 4.8 - 4.11, the only sentences that the participants found grammatical shared the following features: (i) Either the manual negator or both the manual and non-manual negators were signed, and (ii) The negative marker, consisting of either the manual negator or both manual and non-manual negators, was in sentence-final position. As these tables clearly show, any other variations on these two features were found unacceptable including the following: (a) the negator in any other position (e.g., sentence-initial, preverbal, etc.) than the sentence-final position, (b) the spreading of the non-manual negator over the VP or whole clause, (c) the absence of the manual negator. The results also show that word order does not affect grammaticality and both SVO and SOV are acceptable so long as the above features are present in the sentence.

In addition to grammaticality judgments, we also elicited metalinguistic data by asking the participants directly about how negative sentences are to be signed. When asked how to sign a negative sentence, one participant said the following: “To say NO, use your hand or your hand and head, but not just your head”. When I allowed headshake or backward head tilt to spread over the entire clause or VP, the same participant reacted by saying: “Don’t keep shaking your head! Only shake your head when you sign NO with your hand”. We further asked where in the sentence NO should appear to which he responded: “Sign what you want to say, and then sign NO”. Finally, when we asked why he signed differently from how he spoke Arabic (because negation is preverbal in Arabic), he responded: “We don’t care about Arabic. Arabic is different!”. The other participant found the spreading headshake and backward head tilt funny and burst into laughter after watching me sign the sentence that way. She also responded by saying, “That’s really funny! You made me laugh. Why are you signing that way?” She explained that the signer should use his/her hand when s/he signs ‘NO’ to negate a sentence and further added that shaking one’s head or tilting it backwards is not important but that “you can choose to shake your head if you want. Up to you!”. Overall, these results indicate that STSL is strictly manual-dominant with the manual marker being the primary negator and the non-manual element as the secondary and very often optional marker. This means that the non-manual element of negation is simply part of the phonological make-up of the manual negative element incapable of expressing negation on its own on a sentential level. However, note that headshake, head turn, and head tilt, and even brow raise alone, but not air blown out of the mouth, can readily express negation as negative response particles.

4.2.3 Semantics of STSL Negators

In terms of their semantics, most, if not all, of the STSL negators seem to have two features: (i) they share different meanings, and (ii) they can be used interchangeably. For example, the signers consistently used NEG_{basic} and NEG_{proh} to negate an action verb and quite frequently used NEG_{exist} and NEG_{poss} interchangeably to convey non-existence and lack of possession (e.g., ‘I don’t have a car’ = ‘There is no car’). The following examples taken from the signed productions and stories show this usage.

- Negating an action verb:

(4.41) a. EMPTY-HANDED COME NEG_{basic}

‘I won’t come empty-handed.’

b. WOMAN OLD PREGNANT NEG_{basic}

‘The older woman doesn’t get pregnant.’

c. WATER FLOW NEG_{proh}

‘The water didn’t flow.’

- Negating existence:

(4.42) a. WATER NEG_{proh}

‘There is no water.’

b. PAST COMBINE NEG_{poss} RICE SICKLE

‘In the past, there were no combines. We harvested the rice with sickles.’

c. PAST ELECTRICITY NEG_{basic}

‘In the past, there was no electricity.’

- Negating possession:

(4.43) a. IX-1 DAUGHTER NEG_{proh}

‘We don’t have any daughters.’

b. DRESS RED IX-1 NEG_{poss}

‘I don’t have a red dress.’

c. PEOPLE TRACTOR NEG_{poss}

‘People didn’t have tractors.’

In several cases, the interchangeable use of these negators was done even in the same discourse. For instance, in the following story, the participant narrates that in the past there were no official jobs in his hometown and people relied on agriculture to make ends meet:

(4.44) PAST JOB NEG_{poss} MONEY NEG_{basic} MARSHLAND NEAR PEOPLE ALL GO CUT GRASS SELL

‘In the past, there were no jobs; there was no money; everyone would go to the marshland and cut grass and sell it.’

The meanings for which the STSL negators were used also seems to be participant-specific to some extent. The following participant, for instance, used only NEG_{poss} and NEG_{exist} exclusively to negate both possession and existence. It seems that lack of possession and lack of existence mean roughly the same to this participant.

(4.45)



PAST ELECTRICITY NEG_{exist}

‘In the past, we didn’t have electricity / there was no electricity.’

(4.46)



PAST DOCTOR NEG_{poss}

‘In the past, we didn’t have doctors / there were no doctors.’

Thus, most of these negation markers express the same meaning and function grammatically the same way which is why they can be used interchangeably. This finding is perhaps not surprising. Antzakas (2006), for instance, reports that head turn, headshake, and head turn did not differ from each other grammatically in terms of expressing negation in GSL; however, these head movements differed from each other phonologically in that they were synchronized with negative hand movements that they accompanied; i.e., head backward tilt accompanied the upward hand movement (Figure 4.7) while headshake was articulatorily harmonized with side-to-side hand movement (Figure 4.8).



Figure 4.7. Synchronization of Negative Manual & Non-manual Markers in GSL (Antzakas, 2006, p. 264)



Figure 4.8. Synchronization of Negative Manual & Non-manual Markers in GSL (Antzakas, 2006, p. 264)

4.2.4 Basic Clause Negator in STSL

I have reserved a detailed discussion of the basic clause negator in STSL until here for one main reason: To claim that STSL has a basic clause negator, I could not have simply relied on my intuition as a signer. Rather, I need to present different types of evidence from the data to support my claim which is why I needed to examine all the data we gathered and look for a pattern with regard to the basic negator. My claim is that NEG_{basic} is the basic clause negator which configurationally looks like the following figure:



Figure 4.9. Basic Clause Negator (NEG_{basic}) in STSL

NEG_{basic} has as its manual and non-manual elements a single upward movement of the dominant hand or of both hands from the wrist and a simultaneous backward head tilt (bht), respectively. As can be seen below, NEG_{basic} is identical in its manual and non-manual elements to the clause negators of GSL (Antzakas, 2006) and of TĪD (Zeshan, 2006c).



Figure 4.10. Negation in GSL (Antzakas, 2006, p. 264)



Figure 4.11. Manual Negator in TìD (Zeshan, 2006c, p. 150)

The manual and non-manual negator can also be used to say ‘no’ as an independent sign in STSL. When saying ‘no’ as a reply to a yes/no question, either element of the negator (manual or non-manual) or both elements can be used. In other words, neither negator is obligatory and either the manual negator or negative backward head tilt alone or both negators can be used as a negative response. This phenomenon bears some similarity to how TìD treats the manual and non-manual negator. The manual negator can be signed with both hands although the one-handed variant is much more common. We do not have any evidence that the duration of either or both elements of NEG_{basic} is any different when it is used as a negative response marker from when this same negator is used as a sentential negator. It has been noted that in TìD, the manual negator is far shorter in duration when used as an independent sign than when used as a clitic (Zeshan, 2006c). Furthermore, when used as an independent sign, NEG_{basic} is signed in neutral signing space; however, most often it is signed in the signing space in which the preceding sign is signed as the following example shows:

(4.47) NEG_{basic} in signing space



PREGNANT NEG_{basic}

‘She is not pregnant.’

This finding contradicts Antzakas’ (2006) finding in GSL about the scope of head tilt, along with that of headshake and head-turn, which was found to include either the negative sign or the whole sentence. Antzakas (2006), however, also notes that the spreading of the head movement across the whole sentence is strange and is mainly used for emphasis. Backward head tilt is also found in another sign language, i.e., TID, and is believed to be an unusual areal phenomenon found in some sign languages of the Eastern Mediterranean region (Zeshan, 2006c). The simultaneity of occurrence of the manual negator and backward head-tilt confirms Zeshan’s (2004) claim that the head tilt occurs with the upward or backward movement of the hand. Zeshan (2004) also contends that the similarity of manual negators in terms of the handshape in different, unrelated sign languages might be iconically motivated. My personal observation confirms that the gesture used by hearing people for negation in KhA in the village of residence of the deaf individual in the current study is identical to the one used in this family’s homesign. It can thus be said that the manual negator and the backward head tilt accompanying it have originated from the surrounding hearing culture. Such a claim is to some extent tenable as Fischer (2006) has also made a similar argument about ASL, stating that the negative headshake or frown used paralinguistically with negative sentences in hearing American culture are seen in spoken American English.

Although four negators were attested in STSL, one negator seems to serve as the *basic* clause negator: NEG_{basic}. My argument for this claim is based on the fact that NEG_{basic} is used to change the polarity of a pragmatically neutral sentence from affirmative to negative without adding any pragmatic or semantic flavor to the sentence. In other words, NEG_{basic} does not seem to have

an adverbial function and is used in pragmatically unmarked contexts while other negators such as NEG_{proh} must be used in marked contexts and add an adverbial meaning (‘never’, ‘not at all’, etc.). The first piece of evidence comes from the type of data collection method. Our claim here is straightforward: The data collection method impacted the participants’ choice of the negator in that all the five participants who signed the stimuli from the signed productions for negatives *only* used NEG_{basic} to negate the sentences. It is important to emphasize the data collection method here because signed productions likely provided the participants with sentences out of their natural context and thus were pragmatically neutral. As we will see later, the participants used almost all other negators in the stories which we believe is because sentences in stories are contextualized and pragmatically marked, providing the participants with an opportunity to use negators with specific semantics, e.g., NEG_{exist} for non-existence, NEG_{proh} for prohibition and warning, etc., while in the absence of a marked context, NEG_{basic} serves that purpose. The following examples support this claim:

(4.48) a. BOY LOVE FATHER NEG_{basic}

‘The boy doesn’t love his father.’

b. SOLDIER SHOOT MAN NEG_{basic}

‘The soldier didn’t shoot the man.’

c. MAN NEXT-TO WALL NEG_{basic}

‘The man is not next to the wall.’

d. MAN DIE NEG_{basic}

‘The man didn’t die.’

The second piece of evidence for NEG_{basic} as the basic negator is of the semantic type. In addition to negating clauses with main verbs, NEG_{basic} can also suggest non-existence and lack of possession. The results from the stories are especially telling as there quite a few examples of such usage of NEG_{basic}:

(4.49) a. IX-3 RICH NEG_{basic}

‘He’s not rich.’

b. IX-3 DRIVE NEG_{basic}

‘He doesn’t drive.’

c. HUSBAND FARM NEG_{basic}

‘My husband doesn’t have a farm.’

d. WATER EIGHT DAY BREAK WATER NEG_{basic}

‘The water has been shut off for eight days now. There is no water!’

The third piece of evidence is simply the frequency with which NEG_{basic} was used regardless of the context, semantics, and data collection method. As mentioned above, NEG_{basic} was the only negator used in signed productions and was used 158 times. In the stories, NEG_{basic} was used 746 times out of the total 1005 times (74.2%) that all the negators combined were used.

Finally, there is another piece of evidence for NEG_{basic} being the basic clause negator. However, this time the supporting evidence comes from negative responses, that is, simply saying ‘no’, that occurred in the stories, not from sentential negation. Although negative responses do not constitute clausal negation, they probably provide the most basic way the participants think of negation. One could imagine that, in creating a sign language, signers would have to first come up with a sign meaning ‘no’ and then employ that sign at the sentential level. In any case, the results from negative responses are as follows:

Table 4.12. Negative Response Markers from Stories

	NEG _{basic}	NEG _{proh}	NEG _{poss}	NEG _{exist}	Other
	42	3	0	0	2
Total	47				

As can be seen in Table 4.12, NEG_{basic} is used far more frequently (89.3%) than NEG_{proh} (6.3%) and Other (4.2%). Thus, NEG_{basic} is almost exclusively used as a negative response particle. When used as a negative response particle, like NEG_{proh} and Other, NEG_{basic} was signed in the neutral signing space.

4.2.5 Double Negation and Negative Concord in KhA and STSL

It is a well-known fact that negation occurs with quantification such as ‘nobody’ and ‘nothing’. One of the main criteria for classifying languages in this regard is whether a language requires or disallows the presence of the standard negator with negative quantifiers. Languages behave differently with negative quantifiers in that while some allow standard negation with negative quantifiers, others disallow it and yet for other languages it depends on whether the quantifier occurs in the subject or object position. For instance, while standard negation is obligatory with a negative quantifier in Russian, this is not the case in German:

(4.50) Nikto ne prišel Russian
 nobody NEG come.PST.M.SG
 ‘Nobody came’ (Dahl, 2010, p. 29)

(4.51) Niemand kam German
 nobody come.PST.3.SG
 ‘Nobody came’ (Dahl, 2010, p. 29)

KhA behaves like German in that negative quantifiers in subject position disallow the presence of standard negation and render the sentence ungrammatical:

(4.52) a. mæhhæd iyyæh
 nobody come.3SM.PRF
 ‘Nobody came.’
 b. *mæhhæd ma-yyæh
 nobody NEG-come.3SM.PRF
 ‘Intended meaning: ‘Nobody came.’’

Negative concord (NC), after Labov (1972), refers to “the phenomenon where two (or more) negative elements that are able to express negation in isolation yield only one semantic negation when combined” (Zeijlstra, 2013, p. 819). In other words, NC is the phenomenon where two negative elements have only one semantic negation as they contribute only one negative force (Zeijlstra, 2008).

NC is one of the most extensively researched areas in negation and is exhibited by a large number of languages including Afrikaans, West Flemish, Yiddish, Albanian and Greek (Zeijlstra, 2013), Persian (Kwak, 2010), and Iraqi Arabic (Albuarabi, 2021), among many others. Languages have different NC strategies at their disposal. Two major types of languages with regards to NC structures are strict NC languages and non-strict NC languages. In strict NC languages, all negative indefinites (n-words in Laka, 1990), regardless of their position and number, are accompanied by an obligatory negative marker (Zeijlstra, 2013). In (84) from Czech, the negative indefinites *nikoho* ‘n-body’ in postverbal object position in (4.53a) and *nikdo* ‘n-body’ in postverbal and preverbal subject position in (4.53b) and (4.53c) respectively, must be accompanied by the negative element *ne-*; otherwise, the sentence would be ungrammatical.

- (4.53) a. Milan **(ne-)vidi nikoho* Czech
 Milan NEG.saw n-body
 ‘Milan doesn’t see anybody’
- b. dnes **(ne-)volá nikdo*
 today NEG.calls n-body
 ‘today nobody calls’
- c. dnes *nikdo *(ne-)volá*
 today n-body NEG.calls
 ‘today nobody calls’ (Zeijlstra, 2013, p. 819)

In non-strict NC languages, an NC relationship “can only be established between n-words in postverbal position and one negative element in preverbal position, either an n-word or a negative marker” (Zeijlstra, 2013, p. 819). In the following examples from Italian, the NC structure

is between the preverbal negative marker *non* and the postverbal n-word *nessuno* ‘n-body’ regardless of the grammatical function of the n-word: subject in (4.54a) and (4.54b), while with the n-word in preverbal position, *non* is disallowed (4.54c).

(4.54) a. Gianni **(non)* ha telefonato a *nessuno* Italian

Gianni NEG has called to n-body
 ‘Gianni didn’t call anybody’

b. ieri **(non)* ha telefonato *nessuno*

yesterday NEG has called n-body
 ‘yesterday nobody called’

c. ieri *nessuno* **(non)* ha telefonato (a *nessuno*)

yesterday n-body NEG has called to n-body
 ‘yesterday nobody called (anybody)’

(Zeijlstra, 2013, p. 819)

STSL exhibits NC structures, too. In STSL, more than one manual negator can appear in the same simple sentence without rendering the sentence affirmative. This means that these negators do not cancel each other out since the sentence remains negative. Overall, we identified 86 cases of NC constructions from the stories, 14 of which occurred in the bedtime stories and 72 cases in the everyday activities. The following are examples of two negator and three negator NC constructions from the stories.

(4.55) Negative Concord in STSL

(a) $NEG_{basic} + NEG_{proh}$



MOTHER PAST PRAY FAST GIVE



MEAN NEG_{basic} $\frac{hs}{NEG_{proh}}$

‘His mother would pray and fast, and was generous. She wasn’t mean at all.’

(b) $NEG_{exist} + NEG_{basic}$



PAST AIR CONDITIONER FAN NEG_{exist} $\frac{bht}{NEG_{basic}}$

‘In the past, there were no air conditioners or fans.’

(c) $\text{NEG}_{\text{poss}} + \text{NEG}_{\text{exist}} + \text{NEG}_{\text{proh}}$



SAY PERSON MONEY NEG_{poss} $\text{NEG}_{\text{exist}}$ $\frac{\text{hs}}{\text{NEG}_{\text{proh}}}$
 'It is said that there was a man who did not have any money whatsoever!'

(d) $\text{NEG}_{\text{basic}} + \text{NEG}_{\text{exist}} + \text{NEG}_{\text{basic}}$



PAST PHONE $\frac{\text{bht} + \text{br}}{\text{NEG}_{\text{basic}}}$ $\text{NEG}_{\text{exist}}$ $\frac{\text{bht} + \text{br}}{\text{NEG}_{\text{basic}}}$



NOW PHONE MANY GOOD

'In the past, there were no phones. Now there are so many phones. It's convenient.'

Table 4.13 presents these negative concord combinations and whether they consisted of two or three negators. Furthermore, an analysis of these NC combinations showed that most of these constructions had NEG_{basic} as their first negator. Thus, in Table 4.13, we have also separated those NC constructions that have NEG_{basic} as their initial negator and those that have a negator other than NEG_{basic} as their initial negator.

Table 4.13. Negative Concord Combinations of Negators in STSL

Two-negator combinations												Three-negator combinations						
NEG _{basic} -Initial		Non-NEG _{basic} -Initial										NEG _{basic} -Initial						
NEG _{basic} NEG _{basic}	NEG _{basic} NEG _{proh}	NEG _{basic} NEG _{exist}	NEG _{exist} NEG _{proh}	NEG _{exist} NEG _{basic}	NEG _{exist} NEG _{basic}	NEG _{exist} NEG _{poss}	NEG _{proh} NEG _{exist}	NEG _{proh} NEG _{basic}	NEG _{poss} NEG _{other}	NEG _{poss} NEG _{exist}	NEG _{poss} NEG _{proh}	NEG _{poss} NEG _{basic}	NEG _{basic} NEG _{exist} NEG _{proh}	NEG _{basic} NEG _{proh} NEG _{basic}	NEG _{basic} NEG _{basic} NEG _{proh}	NEG _{basic} NEG _{basic} NEG _{basic}	NEG _{basic} NEG _{exist} NEG _{basic}	NEG _{poss} NEG _{exist} NEG _{proh}
10	33	2	20	1	2	1	3	1	1	3	2	1	1	1	1	1	1	1
86																		

Table 4.13 shows that 80 out of the 86 NC constructions (93%) are two negator combinations and thus constitute the vast majority of the examples. This indicates that STSL strongly favors two negators over three negators in NC constructions although three negators are clearly allowed and grammatical. Furthermore, we can see from Table 4.13 that 50 out of the total 86 NC combinations (58.1%) have NEG_{basic} as their initial negator and 36 combinations started with a different negator than NEG_{basic} (41.9%). Furthermore, of the 36 combinations that had a negator other than NEG_{basic} as their initial negator, 24 (66.6%) started with NEG_{exist}. Therefore, overall, STSL favors two negator combinations with NEG_{basic} as the first negator in NC constructions.

4.3 Discussion

In the next two sections, I will provide an analysis of sentential negation and negative concord in STSL.

4.3.1 A Formalization of Sentential Negation in STSL

All the evidence from the signed productions, stories, and grammaticality judgments strongly confirmed the above hypothesis. First, the above findings showed a clear pattern regarding negation in STSL: negation is almost always sentence-final and manually signed. The fact that the manual negator was absent in nine cases cannot be used as an argument against our hypothesis here because the non-manual negator (i.e., backward head-tilt) was absent, too. In other words, while either the manual negator or both the manual and non-manual negators were both present in all negative sentences, in no sentences was the non-manual negator the only negative marker. The nine sentences with neither a manual nor a non-manual negators were ruled out as mistakes on the part of the signers due to distraction or some other factor and were thus excluded from the data analysis. Thus, the signed productions clearly show that STSL does not allow non-manual negation only. What is more, when NEG appeared in sentence-final position, it occurred following not only the object argument(s) (i.e., direct and/or indirect objects) but also following adjuncts, if any. Therefore, sentence-final position for STSL in the case of negation includes the position following both arguments and adjuncts. Most of the brow raises occurred with backward head-tilt. It is likely that this is again an instance of physiological harmony in terms of movement between different parts of the body and face (i.e., the hand, head, and brows) all moving upward, perhaps unconsciously. Eye gaze might have been the result of brow raise since raising one's brows can cause one's eyes to gaze more visibly. In any case, the signed productions did not indicate that any major non-manual marker would be a likely candidate for marking the syntactic domain of negation. The backward head tilt was present in less than one third of the data (27.2%) and brow raise in one third of the cases (32.9%).

Turning to stories, we see a similar pattern. Overall, only 107 instances of the 887 negative manual signs were accompanied by some non-manual marker (12.1%). Backward head tilt was attested in a little over one tenth of the data set of NEG_{basic} (11.1%), headshake in only one case of the data set of NEG_{exist} (2.2%), and not attested at all in the case of NEG_{proh}. In NEG_{proh}, the

non-manual marker, i.e., headshake, is frequently attested: 62.2%. NEG_{proh} was also very frequently accompanied by head turn (91.9%). It is likely that head turn is a short form of headshake. However, given that in many cases the head turn was followed by headshake, considering head turn to be shortened headshake might not be totally accurate. Combining head turn with a different head movement has also been reported for other sign languages. For instance, Antzakas (2006) observed that, to express emphatic negation, Greek Sign Language (GSL) signers often combined head turn with headshake or both head tilt and head tilt with headshake. It seems that combining head turn with headshake also serves emphasis in STSL. Furthermore, bear in mind that the data is heavily skewed toward one participant who contributed far more data than any other participant. Most of the NEG_{proh} occurrences were found in this participant's data. Thus, the frequency of use of headshake with NEG_{proh} might very well be an idiosyncratic tendency of one participant. If this tendency turns out to be the case, then we can rule out headshake as a major NMM, too. However, it may well turn out that the headshake in NEG_{proh} plays a syntactic role. Obviously, this issue remains to be a limitation of the current study at this point and warrants further data from other participants to determine what role headshake and head turn play in negative sentences. Nevertheless, overall, I take these distributional observations of the non-manual markers (along with their lack of spreading when they are present as well as their inability to negate the sentence on their own) as evidence that they do not play any grammatical role in the syntax of STSL's negation. Rather, the non-manual markers constitute part of the phonological make-up of the negative manual signs. In other words, the manual negators are lexically specified for these non-manual markers. Thus, it makes more sense to call the non-manual markers lexical markers and not non-manual *negators* as they do not play a syntactic role. As Gökgöz (2011) similarly argues for TİD, a syntactically required NMM would occur much more often than in a small subset of the data. More crucially, we should at least have a handful of sentences negated by a non-manual marker. Neither of these two expectations are met in the case of negation in STSL. Table 4.14 below, following Gökgöz (2011), provides a summary of the distributional properties of the non-manual markers associated with the manual negators and hypotheses about their role.

Table 4.14. Hypotheses about the Function of Non-manual Markers of Negation in STSL

Non-manual marker	Distributional properties		Functions	
	Local	Spreading	Lexical	Grammatical
Backward head tilt (NEG _{basic})	√		√	
Backward head tilt (NEG _{poss})	√		√	
Headshake (NEG _{proh})	√			√ (?)
Headshake (NEG _{exist})	√		√	
Head turn (NEG _{proh})	√			√ (?)
Brow raise (NEG _{basic})	√		√	
Eye gaze (NEG _{basic})	√		√	

We hypothesized that STSL is manual-dominant. This hypothesis requires the following criteria to be met (e.g., Pfau, 2016; Pfau, et al., 2022; Zeshan, 2006): (a) the manual negator is required to negate the sentence (4.56) while the nonmanual negator is not only optional but also often left out (4.57); (b) the manual negator can negate a clause on its own (4.57) while the nonmanual negator cannot (4.58b); and (c) the nonmanual negator only accompanies the manual negator (4.56) and never spreads beyond the manual negator (4.58b) (bht = backward head tilt):

bht

(4.56) BROTHER DRINK TEA NEG_{basic}
 ‘My brother does not drink tea.’

(4.57) BROTHER DRINK TEA NEG_{basic}

bht

(4.58) (a) *BROTHER DRINK TEA

bht

(b) *BROTHER DRINK TEA

Therefore, we conclude that STSL is a strictly manual dominant sign language. At least in this regard, STSL's negation system is similar to other sign languages such as TİD (Gökgöz, 2011). However, note that Gökgöz (2011) argues that in TİD, it is not backward head tilt (bht) that is the major NMM of negation but what he calls 'non-neutral brow position' (nbp) as the following example shows:

- bht
nbp
-
- (4.59) IX-1 MUZ (ÖNE) ATMAK DEĞİL
 I banana (front) throw not
 'I did not throw the banana to the front.'
- (Example 21 from Gökgöz, 2011, p. 66)

Gökgöz (2011) found that this non-manual marker, unlike bht, was frequently attested in negative sentences in TİD (71%) and in 80% of the cases spread over the entire sentence and in 20% of the cases it only occurs on the manual negator DEĞİL 'not' Based on these facts, Gökgöz argues that nbp originates in the head position of NegP from which position it marks the c-command domain of negation as shown in (4.60):

- nbp
- (4.60) IX1 KELİME BİLMEK DEĞİL
 I word know not
 'I didn't know the words.'
- (Example 22a from Gökgöz, 2011, p. 69)

Gökgöz proposes the tree in Figure 4.12 to capture this c-command relation:

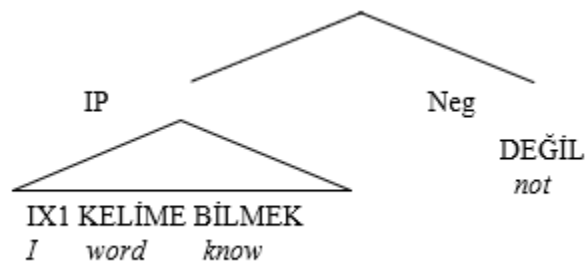


Figure 4.12. C-command Relation in TİD (Gökgöz, 2011, p. 69; modified)

However, the manual negative sign DEĞİL can be followed by the negative adverbial HİÇ which nbp also spreads on as in (4.61). Gökgöz realizes that c-command relation cannot account for this spreading of nbp because HİÇ falls outside the c-command domain of the nbp.

- (4.61) $\overline{\text{IX1 İŞARET BİLMEK DEĞİL HİÇ}}$ nbp
 I sign know not at all
 ‘I didn’t know (how to) sign at all.’ (Example 23a from Gökgöz, 2011, p. 69)

To explain this unexpected example, he proposes another syntactic relation, i.e., spec-head agreement, arguing that both c-command relation and spec-head relation are at play in such cases:

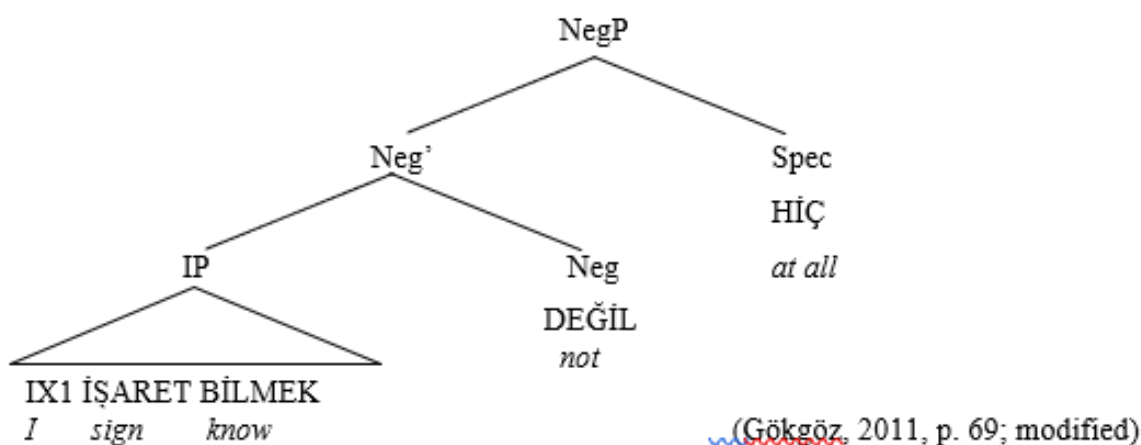


Figure 4.13. C-command and Spec-head Relation in TİD

We argue that since the non-manual negator is optional and is very frequently left out, the obligatory manual marker occupies the Neg^0 position of NegP . As we saw above, the nonmanual negator, e.g., backward head tilt, on the other hand, is optional and occurs in a small percentage of the data. It also requires that lexical material be associated with it for it to be phonologically realized. We argue that the non-manual marker is part of the lexical description of the negator. In other words, the non-manual marker constitutes an optional part of the phonological make-up of the negative manual sign. In STSL, the head position of NegP may not be void of phonological material. Rather, it must always be occupied by a phonologically overt negator which consists of

an obligatory manual element and an optional non-manual element. The specifier position of NegP, on the other hand, is occupied by a covert negative operator $Op \neg$ which then establishes a spec-head agreement with the manual negator in the head position. The lack of non-manual spreading in STSL negation is evidence that this relation is not of the c-command type. We further argue that the direction of the Neg head and the position of spec,NegP are both on the right. As for the hierarchical relationship between Neg and Tense, we do not have a definitive answer as of yet as to whether Neg dominates T or vice versa. STSL does not seem to have manually signed modals. Rather, modals might be expressed through non-manual markers. This issue, however, remains speculative at this point and needs to be verified by empirical evidence.

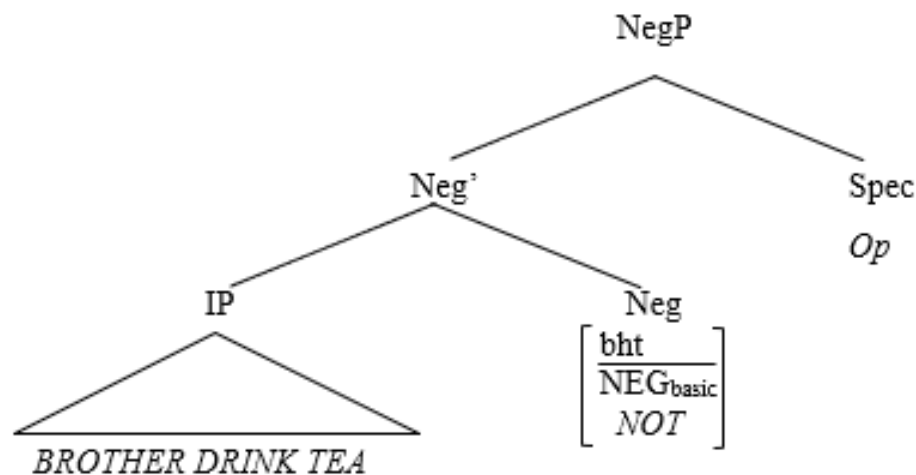


Figure 4.14. Projection of Negation and Spec-head Relation in STSL

At this point, we wish to add the facts and proposals of STSL to those of ASL, LSC, and DGS as reported by Pfau and Quer (2002) and TİD as reported by Gökgöz (2011, p. 67).

Table 4.15. Facts & Proposals on Negation in ASL, LSC, & DGS (Pfau & Quer, 2002), TİD (Gökgöz, 2011), & STSL

SL	Major manual negative marker	NMM observed with negation	Direction of the Neg head	The nature of negation	Neg head occupied by	Position of spec,NegP	spec,NegP occupied by	Hierarchy between T & Neg heads
ASL	NOT	‘hs’	left	syntactic	NOT and [+neg]	left	covert negative operator	T > Neg
LSC	NO	‘hs’	right	morphological	NO and [+neg] _{aff}	right	NO-RES	Neg > T
DGS	NICHT	‘hs’	right	morphological	[+neg] _{aff}	right	NICHT	Neg > T
TİD	DEĞİL	‘bht’ ‘hs’ ‘sht’ ‘nbp’	right	syntactic	DEĞİL	right	HİÇ, HAYIR, AVUÇ, HEP	Neg > T
STSL	NEG	‘bht’ ‘hs’ ‘ht’ ‘br’	right	syntactic	NEG	right	covert negative operator	Unclear

Note. bht = backward head tilt, br = brow raise, hs = headshake, ht = head turn, nbp = non-neutral brow position, sht = single head turn

4.3.2 A First Attempt to Analyze NC in STSL

A few points about NC constructions are in order here. First, all of the NC constructions negated simple sentences and thus the same verb, not more than one verb; Thus, semantically, these negators were used in NC constructions only for the purpose of emphasis, and not to add any other lexical meaning to the sentence. Second, just like non-NC sentences, all of these NC constructions were located in sentence-final position. Third, none of these NC constructions allowed any spreading of any the non-manual markers over any constituents or adjacent signs. In other words, each negative manual sign preserved its respective non-manual marker. A similar observation has been made in TİD in which negative manual signs that are marked by different non-manual markers keep their respective non-manual markers as in (4.62) although other possibilities also exist such as the spreading of the non-manual marker of one negative sign onto other negative signs:

	<u>neg-shake</u>	<u>neg-tilt</u>	
(4.62)	NONE(2)	GO^NOT	(Example 49 from Zeshan, 2006c, p. 159)

Fourth, none of these constructions rendered the sentence affirmative; in other words, none of the negators canceled each other out and all the sentences remained negative. Fifth, none of the non-manual elements of any of the negators spread over any other manual negators; that is, the non-manual negators, if present, remained solely on the manual negator that they were associated with. Finally, no signs can intervene between any of the negators in NC combinations. The above points indicate that NC is completely optional in STSL and serves emphatic purposes only. Since these constructions never result in an affirmative reading of the sentence, then the negators in such constructions do not cancel each other out and thus must modify the same negative head; that is, there is only one NegP modified by several negators (Haegeman, 1995).

An important point that needs to be made clear about NC in STSL is that the class of NC that STSL presents us with is different from the type presented in the spoken language examples provided above. The above examples (4.54) from spoken languages like Italian involve negative quantifiers whereas the examples from STSL, at least the ones identified in this study, do not involve any such quantifiers. Rather, they only involve negative verbs modified by other negative elements that seem to function as negative adverbials for emphatic purposes only. In fact, at this point, we are not certain if STSL has any negative quantifiers or negative indefinite expressions like ‘nobody’ (n-words, adopting Laka’s 1990 and Giannakidou’s, 2000, terminology) at its disposal to begin with. Thus, we ask for caution in interpreting these findings and leave this issue open to future research.

We argued above (section 4.2.4) that NEG_{basic} is the basic clause negator in STSL based on various types of evidence. Implied in that argument was the expectation that all NC combinations would have NEG_{basic} as their first negator since NEG_{basic} was hypothesized to be the basic clause negator. In other words, we hypothesized that in sentences that exhibit NC, the first negator negates the sentence while the second and third negator intensify that negation and thus serve as negative adverbials like *HİÇ* in TİD. As Table 4.13 shows, this is the case in nearly two thirds of all the NC combinations, but clearly not in all of them. This can be taken as evidence against our hypothesis that NEG_{basic} is in fact the main clause negator. However, although it is true that not all NC combinations start with NEG_{basic}, most of them do. Furthermore, if one considers altogether there were four negators that initiated the NC combinations and yet three of these negator combined accounted for slightly over two fifths of the total number of the NC constructions, then NEG_{basic} obtains great significance in NC sentences. The above explanation

serves only as a first attempt to examine NC in STSL and is by no means intended to be considered final at this stage. We leave a more robust analysis to future research.

CHAPTER 5. CONCLUSION

5.1 Introduction

The current dissertation was a study of spoken and sign language. The objective was to compare a sign language with the surrounding spoken language with the assumption that this sign language does have a grammatical system (in this case, negation) but that it is different from that of the spoken language. Thus, the main hypotheses put forward in this dissertation were: (i) STSL is not a copy of KhA, i.e., not *signed KhA*, rather an independent sign language with a grammar of its own at least in its negation system, and (ii) STSL aligns well with other well-documented sign languages in terms of negation. Various types of contextualized and decontextualized evidence, namely, stories, signed productions, and grammaticality judgments, robustly confirmed both hypotheses. The findings showed that STSL strongly and almost exclusively favors sentence-final negation while KhA allows negation only in preverbal position. Furthermore, while KhA has three negation markers which apply in some shared constructions and under certain circumstances but not in others (e.g., true imperatives with the negation marker *la* only), STSL has evolved even more negation markers all of which can negate any type of sentence although some are used far more extensively than others. In addition, each of STSL's negation markers is composed of a manual element which is obligatory and a non-manual element which is optional and most often omitted. STSL therefore falls into the manual-dominant category of sign languages. STSL also seems to have evolved some specialized negators such as NEG_{poss} to negate possession while still maintaining other negators like NEG_{basic} and NEG_{proh} for any semantic function. Furthermore, STSL has evolved what looks like irregular negative verbs such as NOT-LIKE, NOT-KNOW, NOT-WANT, etc., which formed their own category of negators, Other. STSL also allows up to three negators to negate the same verb for emphatic purposes which is another feature that is not shared with KhA. Thus, STSL is an optional negative concord (NC) language. KhA is an NC language, too, but under certain conditions (e.g., when *la* is prefixed by *wæ-* and is in postverbal position).

However, the fact that STSL exhibits a different grammar in its negation system (as well as in other grammatical constructions such as word order and interrogatives) from that of KhA is rather surprising. STSL was created and used by hearing people who knew and spoke KhA to

communicate with a deaf individual who also knew and spoke KhA. Thus, one would expect that the signers would logically follow the grammar of KhA while signing rather than create a new grammar which would complicate things. Creating a new grammar, for instance, would mean risking miscommunication with the deaf individual who, by virtue of his native language KhA, is used to only preverbal negation, not sentence-final negation. Yet, clearly the signers had opted to create a new grammar for STSL. We must, however, be careful with our choice of words when describing how STSL's new grammar came into existence. 'Opted' implies that the signers *consciously* chose a new grammar. We cannot confirm or reject this claim and our explanation of the possible reasons for these grammatical differences remain speculative at this point. As a signer myself, I do not have any recollection of any family members correcting any other family members' accuracy of signing although I do remember occasions on which corrections were made regarding the appropriateness of a sign. In one instance, for example, my older sister corrected my brother who used a chest sign to refer to a woman which she deemed as inappropriate and asked him to use a headscarf sign instead to refer to a woman.⁷ Thus, most corrections, if any, were concerned with content, not grammatical accuracy. Everyone else simply observed how others signed and learned how to sign, too, the same way they acquired their native spoken language, KhA.

Although it is not clear why STSL's grammar is different than that of KhA, one possibility is that STSL has a different grammar as a result of its use of a different modality than that of KhA, i.e., the gestural modality. As one of the participants in the grammaticality judgments pointed out, 'When you want to sign a negative sentence, sign what you want to say and then sign 'NO''. Thus, it seems that STSL signers think of negation as pertaining to the whole proposition and not just the verb phrase. A similar parallel can be drawn between spoken and sign languages here in that some spoken languages such as French have developed multiple elements to mark negation for which Dryer (1988) presents the following semantic explanation, "Negative morphemes carry a large communicative load in the sense that they carry an important part of the message. If a hearer fails to hear the negative morpheme in a sentence, they will have fundamentally misunderstood the sentence" (p. 102). Dryer (1988) believes that this explanation accounts for the very frequent tendency of negation to precede the verb since postponing negation until after the verb runs the risk of miscommunication as it can create a 'semantic garden path' (p. 102). Dryer refers to the tendency of negation to precede the verb as the Negative-Before-Verb Principle. In addition to

⁷ For context, women in Sadat Tawaher wear headscarves.

classification of negation systems, one of the most central issues surrounding discussions of negation is the cross-linguistic encoding of sentential negation. Although Dryer's reasoning might explain the position of the negative in KhA, the fact that the only attested position for negation in STSL is sentence-final (and thus following the verb, not preceding it) remains to be explained. Dryer is aware of this common tendency among languages and contends that the communicative burden is only one factor among many that results in negation preceding the verb.

We believe that the same idea can be viewed from a different perspective. In fact, it is likely that inserting negation in sentence-final position makes it easier for negation to take scope over the whole clause both syntactically and visually in terms of perception. In other words, if negation were inserted in a non-peripheral position in the sentence (e.g., preverbally in SVO orders), it would be easy to miss in case of memory failure if the deaf individual failed to see the negation marker whereas in sentence-peripheral positions (e.g., sentence-initial/final position) it would catch the addressee's attention (and vision) easily. Thus, the sentence-final position of negation might be plausibly accounted for in terms of the primacy-recency effect and serial position effect. Cognitive bias experiments in psychology, for instance, have shown that, when presented with a list of items to memorize, individuals tend to recall items in the middle the most poorly. This is known as the serial position effect. On the other hand, the primacy effect and recency effect refer to individuals' tendency to recall the first and last items, respectively, to be remembered the most easily (Corsini, 2001). It has been theorized that this tendency is due to the fact that the first and last items are stored in the long-term memory and working memory, respectively. Sentence-final position thus likely places the least cognitive burden on the addressee to recall negation. Nonetheless, modality seems to be the low-hanging fruit that is taken recourse to when it comes to finding a solution to the question of spoken / signed language differences. Thus, in our case, modality is by no means a definitive answer to the question of why STSL signers adhere to a different grammar than that of the surrounding spoken language.

STSL exhibits some important similarities to other fully-fledged sign languages, too. The fact that STSL's negation markers appear in sentence-final position is a tendency that has been reported for many sign languages (Zeshan, 2006b). Furthermore, all evidence robustly showed that STSL neatly fits into the manual-dominant category of sign languages in terms of negation. Thus, STSL is similar in this respect to such sign languages as Italian Sign Language (LIS), Jordanian Sign Language (LIU), Indo-Pakistani Sign Language (IPSL), Turkish Sign Language (Türk İşaret

Dili, TİD), Kata Kolok (Bali), and Japanese Sign Language (JSL), amongst others. Along the same lines, the particle negation / morphological / affixal negation distinction which has been applied to spoken languages can be used to ascertain the typology of negation of STSL. In this case, STSL has simple particle negation in that the negative (manual) particle may be lexically specified for a non-manual marker, e.g., backward head tilt (Pfau, 2012).

5.2 Implications for the Bilingual Mind

The findings of the current study might provide insights for our understanding of the bilingual mind. Since STSL signers speak when they sign, they are competent in two languages which use two different modalities: the vocal modality for KhA and the gestural modality for STSL. An important finding of this dissertation is the fact that while the STSL signers did speak KhA each time they signed a sentence, their signing followed a different grammar than their speech. That is, the signers did not conform to the same word order in their speech and sign, but followed two different grammars simultaneously, one via their vocal tracts and one via their hands and faces. In psycholinguistic research, this type of bilingualism is referred to as bimodal bilingualism, “a type of bilingualism that employs two different input-output channels, one involving spoken language and the other involving sign language” (Abutalebi & Clahsen, 2016, p. 221).

Thus, bimodal bilinguals are those competent in one spoken and one signed language. Unlike unimodal bilinguals who are competent in two spoken or two signed languages and thus the same modality (i.e., either spoken or signed, but not both spoken and signed), bimodal bilinguals have the extraordinary capability of expressing the same proposition in two different modalities, namely, speech and sign, by taking advantage of the fact that the physical constraints are removed. The finding regarding negation in KhA and STSL is extraordinary in that the signers simultaneously expressed negation in two different positions, i.e., preverbally in KhA and sentence-finally in STSL. Previous research has shown that bimodal bilinguals tend to simultaneously conform to the grammar of both the spoken language and the signed language if the two languages follow the same word order (e.g., Dutch & Sign Language of the Netherlands (NGT), Baker & Van den Bogaerde, 2008) but not if the languages are typologically different (Italian & Italian Sign Language (LIS), Branchini & Donati, 2016). For instance, Branchini and Donati’s (2016) study of bimodal bilinguals’ simultaneous production of negative and wh-question

strings in Italian, a head-initial language, and Italian Sign Language (LIS), a head-final language, showed that words and signs in these two languages mismatched in their order:

(5.1) Mismatching alignment of words and signs for negation in Italian / LIS:

Italian: Non ho capito
 not have.1SG understand.PTCP⁸

LIS: UNDERSTAND NEG
 NOT
 ‘I don’t understand.’

(Example 13 from Branchini & Donati, 2016, p. 11)

(5.2) Mismatching alignment of words and signs for wh-questions in Italian / LIS:

Italian: Cosa ha mangiato la rana?
 what have.3SG eat-PTCP the frog

LIS: FROG EAT WH
 WHAT
 ‘What did the frog eat?’

(Example 14 from Branchini & Donati, 2016, p. 11)

Similarly, the simultaneous production of negative strings⁹ in KhA and STSL exhibited one-hundred percent mismatch between in spoken and signed modalities:

(5.3) Mismatching alignment of words and signs for negation in KhA / STSL:

KhA: əf-ʃæləb mɑ- tʰəfær ən-næhær
 the-dog NEG-jump.3SM.PRF the-creek

STSL: DOG CREEK JUMP bht
 NEG_{basic}
 ‘The dog did not jump over the creek.’

⁸ PTCP = Participle

⁹ Such mismatching alignments were seen in declarative sentences and wh-questions, too.

Furthermore, the data suggest cross-modal influence from speech to sign and sign to speech and bimodal bilinguals' preference for code-blending over code-switching (e.g., English-ASL, Emmorey, Grabowski, McCullough, Damasio, Ponto, Hichwa, & Bellugi, 2003). Our results then have implications for the architecture of the bilingual mind and what it can achieve, confirming previous research (e.g., Donati, 2021) that these bilinguals avoid costly inhibitory control and additional processing costs by taking advantage of the suspension of articulatory constraints. Interestingly, the existence of a bimodal intentional communicative system in captive chimpanzees has been reported in some recent studies (Hopkins, Taglialatela, & Leavens, 2007). It has been argued that this system was preceded by communicative gestures in the common ancestor of humans and chimpanzees (Meguerditchian, Cochet, & Vauclair, 2011). In conclusion, the findings provide support for the historical connections between gesture and speech and for a single integrated communication system that oversees both vocal and gestural communication.

5.3 Limitations and Future Directions

There are a few issues that were not addressed in this dissertation for various reasons such as time and space constraints. First, the stimulus sentences designed to elicit signed productions consisted of only action verbs such as 'push', 'give', and 'shoot' but did not contain any stative or existential verbs such as 'have' or 'there are'. This was likely the reason why the participants only used NEG_{basic} to negate the sentences from this task. This issue needs to be addressed in future research, for instance, by including stimulus sentences with stative verbs to investigate the type of negative marker(s) that the participants would use to negate sentences like 'He doesn't have a car' and 'There aren't any cars in the parking lot'. Second, the stimulus sentences were sent in written form to some of the participants which likely affected their signed productions. Thus, future research should include alternative ways of gathering this type of data, for example, by including pictures or videos to elicit these sentences. Third, another issue is concerned with the semantics of the negative markers. At this point, we have not fully addressed the issue of the meaning(s) that each negative marker expresses. For instance, it seems that NEG_{basic} can be used with any meaning in mind such as lack of possession, lack of existence, etc., or that NEG_{poss} mostly expresses lack of possession. However, further research is needed to verify this claim and ascertain the precise semantics of each negative marker. Fourth, and relevant to the previous issue, we did not compare different generations or cohorts of signers with each other to obtain comparative data with regard

to the semantics of each negative marker. For instance, it is likely that one generation of signers uses NEG_{poss} to express lack of possession while another generation uses it to negate existence.

Fifth, unfortunately due to travel restrictions, I was not able to gather the data from the participants in person. On the other hand, some participants had frequent Internet connection issues and we were not able to gather as much data from them as we were hopeful to gather while another participant had access to good Internet connection and was able to submit so many videos that the data ended up skewed toward her. In future research, this issue needs to be addressed by collecting more data, whether online or in person, from those participants who were unable to submit as much data as desired to have a relatively equal amount of data from all or most participants. Finally, we are not aware as to what the status of the negation ‘gestures’ is in Sadat Tawaher among non-signers. For instance, although we know that at least some of the negation markers such as NEG_{basic} are certainly borrowed from the hearing community and are frequently used as a gesture to indicate negation by the non-signers, we are not sure if its use is systematic, if it is accompanied by any non-manual markers, etc. Thus, one possibly promising future direction is to examine the way non-signers use negation gestures. For instance, data can be collected from other hearing non-signers from Sadat Tawaher and the results can be compared with those from STSL signers to gain insights into the evolutionary past of STSL.

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APPENDIX A. STIMULUS SENTENCES FOR DECLARATIVES

Instructions:

لطفا هر جمله را یک یا دو بار برای با صدای رسا به زبان عربی برای شرکت کننده در این مطالعه بخوانید و سپس از شرکت کننده بخواهید که جمله را همانطور که برای پدر / پدر بزرگ ناشنوا خود توصیف میکند با حالتی کاملاً طبیعی و آرام توصیف کند. اگر شرکت میکنند می تواند هر جمله ای را به بیشتر از یک روش توصیف کند از او بخواهید این کار را انجام دهد. دوربین را روی یک سطح ثابت و بدون هیچ حرکتی نگه دارید. نور چراغ یا آفتاب و ... نباید از پشت سر شرکت کننده بتابد. تمام بدن شرکت کننده از جمله صورت و دست ها و بالاتنه باید در تصویر باشد. به شرکت کننده بگویید که استفاده از حرکات صورت مثل بالا بردن ابروها و ... در هنگام توصیف جمله ها کاملاً طبیعی است و هیچ اشکالی ندارد. لطفا تأکید کنید که هر جمله باید به حالتی کاملاً طبیعی توصیف شود. با تشکر

English:

Please read each sentence clearly once or twice to the participant in Arabic and then ask him/her to sign the sentence calmly and exactly the way he/she would sign it to his/her father/grandfather. If the participant wishes to or can sign any sentence in more than one way, ask him / her to do so. Place the camera steady in one spot without any movement. The light must not shine from behind the participant. The participant's whole body must be in the picture frame. Remind the participant that facial expressions like raising the eyebrows or shaking one's head is completely natural while signing. Please make sure you emphasize that each sentence must be signed as naturally as possible. Thank you.

(1) Transitive verbs:

(a) Plain verbs:

- | | | |
|-----------------------------|---------------|----------------|
| 1. əl-wæɫæd | y-ʕəzz | ʊbu-h |
| the-boy | 3SM-love.IMPF | father-GEN.3SM |
| 'The boy loves his father.' | | |

2. əs-sərbəz rəmæh əz-zəlmæh
 the-soldier shoot.3SM.PRF the-man
 ‘The soldier shot the man.’

3. sʰəhb-i ʃæræb tʃai
 friend-1S.GEN drink.3SM.PRF tea
 ‘My friend drank tea.’

4. əl-wadəm həsʰdæt ʃələb
 the-people harvest.3SM.PRF rice
 ‘The people harvested rice.’

(b) Agreeing verbs:

5. əl-wælæd zæŋŋ əl-bət
 the-boy push.3SM.PRF the-girl
 ‘The boy pushed the girl.’

6. əl-mʊʃælləm ʃaf əl-dæxtur
 the-teacher see.3SM.PRF the-doctor
 ‘The teacher saw the doctor.’

7. əz-zəlmæh tʰægg əl-γərbæh boksi
 the-man hit.3SM.PRF the-tree fist
 ‘The man punched the tree.’

8. əl-wadəm zæŋŋ-æʊ əl-məntəkær
 the-people push-3PLM.PRF the-car
 ‘The people pushed the car.’

9. əl-færæx ətʰtʰæh əl-əktab lə-l-bət
the-boy give.3SM.PRF the-book to-the-boy
‘The boy gave the book to the girl.’

10. sʰaħb-i ʃafa-ni æməs
friend-1S.GEN see.3SM.PRF-1S.ACC yesterday
‘My friend saw me yesterday.’

(c) Body-anchored locatives:

11. əz-zəlmæh tʰægg əl-wælæd əsʰtʰar
the-man hit.3SM.PRF the-boy slap
‘The man slapped the boy.’

12. əl-wælæd yærr əl-bət mən ʃæʃær ras-hæh
the-boy pull.3SM.PRF the-girl from hair head-3SF.GEN
‘The boy pulled the girl from the hair.’

13. əl-wælæd bæzz sʰaħb-æh əb-ʃe:n-æh
the-boy poke.3SM.PRF friend-3SM.GEN in-eye-3SM.GEN
‘The boy poked his friend in the eye.’

14. sʰaħb-i bæzza-ni əb-ʃe:n-i
friend-1S.GEN poke.3SM.PRF-1S.ACC in-eye-1S.GEN
‘My friend poked me in the eye.’

15. anæh bæzz-e:t sʰaħb-i əb-ʃe:n-æh
I poke.1S.PRF friend-1S.GEN in-eye-3SM.GEN
‘I poked my friend in the eye.’

16. huwwæh gæmməza-hæh
 he scare.3SM.PRF-3SF.ACC
 ‘He scared her.’

(d) Classifier constructions:

17. sʰəhb-i hætʰtʰ əz-zad bə-l-yæxtʃal
 friend-1S.GEN put.3SM.PRF the-food in-the-fridge
 ‘My friend put the food in the fridge.’

18. əl-wælæd hætʰtʰ əl-xədkar əb-wəsʰtʰ əl-əktaβ
 the-boy put.3SM.PRF the-pen in-middle the-book
 ‘The boy put the pen inside the book.’

19. əl-wælæd xətʰæf ət-tæbhæh wæræ ətʰ-tʰof
 the-boy throw.3SM.PRF the-ball behind the-wall
 ‘The boy threw the ball over the wall.’

20. əl-bət hætʰtʰ-æt əl-əglasʰ fog əl-myiaz
 the-girl put-3SF.PRF the-glass on the-table
 ‘The girl put the glass on the table.’

21. əʃ-ʃæləb tʰəfær ən-næhær
 the-dog jump.3SM.PRF the-creek
 ‘The dog jumped over the creek.’

(e) Locatives:

22. ət-tæbhæh wæræ ət^s-t^sof

the-ball behind the-wall

‘The ball is behind the wall.’

23. ət-tæbhæh fog əd-dar

the-ball on the-room

‘The ball is on top of the room.’

24. əz-zəlmæh əb-s^sæff əl-bab

the-man in-side the-door

‘The man is beside the door.’

25. ət^s-t^sas^sæh əb-s^sæf əl-bab

the-bowl in-side the-door

‘The bowl is next to the wall.’

26. ət-tærətər bə-l-gaŋ

the-tractor in-the-farm

‘The tractor is at the farm.’

27. əl-gufyiæh hædər əl-əktaḅ

the-phone under the-book

‘The phone is under the book.’

28. əl-xədkar əb-wæs^st^s əl-əktaḅ

the-pen in-middle the-book

‘The pen is inside the book.’

(2) Intransitive verbs:

29. sʰaḥb-i y-nam ʕælæ əl-gaʕ
friend-1S.GEN 3SM-sleep.IMPF on the-ground
‘My friend sleeps on the floor.’

30. əl-færæx dʒaʕəd yə-rkədʰ
the-boy sit.3SM.PART 3SM-run.IMPF
‘The boy is running.’

31. əl-γərbæh tʰaḥ-æt
the-tree fall-3SF.PRF
‘The tree fell.’

32. əz-zəlmæh təwæffæh
the-man die.3SM.PRF
‘The man died.’

33. əd-dənyæh dʒaʕd-æh tə-mtʰər
the-world sit-3SF.PART 3SF-rain.IMPF
‘It is raining.’

34. əd-dənyæh hɔwæh hæssæh
the-world windy now
‘It is windy right now.’

35. əl-ḥæyyæh dʒaʕd-æh tə-mʃi bə-l-maɪ bə-n-næhær
the-snake sit-3SF.PART 3SF-go.IMPF in-the-water in-the-creek
‘The snake is slithering in the creek.’

36. əl-məntəkær	dʒɑŋəd	yə-mʃi	ŋælæ	əs-səŋŋæh
the-car	sit.3SM.PART	3SM-go.IMPF	on	the-road
'The car is moving on the road.'				

APPENDIX B. STIMULUS SENTENCES FOR NEGATIVES

(1) Transitive verbs:

(a) Plain verbs:

1. əl-wælæd mə-y-ʕəzz ʊbu-h
 the-boy NEG-3SM-love.IMPF father-GEN.3SM
 ‘The boy does not love his father.’

2. əs-sərbaz mə-rəmæh əz-zəlmæh
 the-soldier NEG-shoot.3SM.PRF the-man
 ‘The soldier did not shoot the man.’

3. sʕaħb-i mə-ʃæræb ʃʕai
 friend-1S.GEN NEG-drink.3SM.PRF tea
 ‘My friend did not drink tea.’

4. əl-wadəm mə-ħəsʕdæt ʃələb
 the-people NEG-harvest.3SM.PRF rice
 ‘The people did not harvest rice.’

(b) Agreeing verbs:

5. əl-wælæd mə-zæʕʕ əl-bət
 the-boy NEG-push.3SM.PRF the-girl
 ‘The boy did not push the girl.’

6. əl-mʊʕəlləm mə-ʃaf əl-dæxtur
 the-teacher NEG-see.3SM.PRF the-doctor
 ‘The teacher did not see the doctor.’

7. əz-zəlmæh ma-tʰægg əl-γərbæh boksi
 the-man NEG-hit.3SM.PRF the-tree fist
 ‘The man punched the tree.’

8. əl-wadəm ma-zæŋʔ-æu əl-məntəkær
 the-people NEG-push-3PLM.PRF the-car
 ‘The people did not push the car.’

9. əl-færæx ma-ətʰæh əl-əktaβ lə-l-bət
 the-boy NEG-give.3SM.PRF the-book to-the-boy
 ‘The boy did not give the book to the girl.’

10. sʰaħb-i ma-ʃafa-ni æməs
 friend-1S.GEN NEG-see.3SM.PRF-1S.ACC yesterday
 ‘My friend did not see me yesterday.’

(c) Body-anchored locatives:

11. əz-zəlmæh ma-tʰægg əl-wælæd əsʰtʰar
 the-man NEG-hit.3SM.PRF the-boy slap
 ‘The man did not slap the boy.’

12. əl-wælæd ma-yærr əl-bət mən ʃæŋær ras-hæh
 the-boy NEG-pull.3SM.PRF the-girl from hair head-3SF.GEN
 ‘The boy did not pull the girl from the hair.’

13. əl-wælæd ma-bæzz sʰaħb-æh əb-ʃe:n-æh
 the-boy NEG-poke.3SM.PRF friend-3SM.GEN in-eye-3SM.GEN
 ‘The boy did not poke his friend in the eye.’

14. s^ʰaħb-i ma-bæzza-ni əb-ʕe:n-i
 friend-1S.GEN NEG-poke.3SM.PRF-1S.ACC in-eye-1S.GEN
 ‘My friend did not poke me in the eye.’

15. anæh ma-bæzz-e:t s^ʰaħb-i əb-ʕe:n-æh
 I NEG-poke.1S.PRF friend-1S.GEN in-eye-3SM.GEN
 ‘I did not poke my friend in the eye.’

16. huwwæh ma-gæmməza-hæh
 he NEG-scare.3SM.PRF-3SF.ACC
 ‘He did not scare her.’

(d) Classifier constructions:

17. s^ʰaħb-i ma-ħæt^ʰt^ʰ əz-zad bə-l-yæxtfal
 friend-1S.GEN NEG-put.3SM.PRF the-food in-the-fridge
 ‘My friend did not put the food in the fridge.’

18. əl-wælæd ma-ħæt^ʰt^ʰ əl-xədkar əb-wæs^ʰt^ʰ əl-əktaḅ
 the-boy NEG-put.3SM.PRF the-pen in-middle the-book
 ‘The boy did not put the pen inside the book.’

19. əl-wælæd ma-xət^ʰæf ət-tæbhæh wæræ ət^ʰ-t^ʰof
 the-boy NEG-throw.3SM.PRF the-ball behind the-wall
 ‘The boy did not throw the ball over the wall.’

20. əl-bət mɑ-ħætʰtʰ-æt əl-əglasʰ fog əl-myɪæz
the-girl NEG-put-3SF.PRF the-glass on the-table
‘The girl did not put the glass on the table.’

21. əʃ-ʃæləb mɑ-tʰəfær ən-næhær
the-dog NEG-jump.3SM.PRF the-creek
‘The dog did not jump over the creek.’

(e) Locatives:

22. ət-tæbhæh mɑ-hi wæræ ətʰ-tʰof
the-ball NEG-3SF.NOM behind the-wall
‘The ball is not behind the wall.’

23. ət-tæbhæh mɑ-hi fog əd-dar
the-ball NEG-3SF.NOM on the-room
‘The ball is not on top of the room.’

24. əz-zəlmæh mɑ-hu əb-sʰæff əl-bab
the-man NEG-3SM.NOM in-side the-door
‘The man is not beside the door.’

25. ətʰ-tʰasʰæh mɑ-hi əb-sʰæf əl-bab
the-bowl NEG-3SF.NOM in-side the-door
‘The bowl is not next to the wall.’

26. ət-tærətər mɑ-hi bə-l-gaʃ
the-tractor NEG-3SF.NOM in-the-farm
‘The tractor is not at the farm.’

27. əl-gufyiəh ma-hi hədər əl-əktaḅ
the-phone NEG-3SF.NOM under the-book
‘The phone is not under the book.’

28. əl-xədkar ma-hu əb-wəsʔtʔ əl-əktaḅ
the-pen NEG-3SM.NOM in-middle the-book
‘The pen is not inside the book.’

(2) Intransitive verbs:

29. sʔaḥb-i ma-y-nam ʔælə əl-gaʔ
friend-1S.GEN NEG-3SM-sleep.IMPF on the-ground
‘My friend does not sleep on the floor.’

30. əl-færæx ma-dʒaʔəd yə-rkədʔ
the-boy NEG-sit.3SM.PART 3SM-run.IMPF
‘The boy is running.’

31. əl-γərbəh ma-tʔaḥ-æt
the-tree NEG-fall-3SF.PRF
‘The tree did not fall.’

32. əz-zəlməh ma-təwæffəh
the-man NEG-die.3SM.PRF
‘The man did not die.’

33. əd-dənyæh ma-dʒaʃd-æh tə-mtʃər
the-world NEG-sit-3SF.PART 3SF-rain.IMPF
‘It is raining.’

34. əd-dənyæh ma-hi howæh hæssæh
the-world NEG-3SF.NOM windy now
‘It is not windy right now.’

35. əl-ħæyyæh ma-dʒaʃd-æh tə-mʃi bə-l-maɪ bə-n-næhær
the-snake NEG-sit-3SF.PART 3SF-go.IMPF in-the-water in-the-creek
‘The snake is not slithering in the creek.’

36. əl-məntəkær ma-dʒaʃəd yə-mʃi ʃælæ əs-səʃfʃæh
the-car NEG-sit.3SM.PART 3SM-go.IMPF on the-road
‘The car is moving on the road.’

APPENDIX C. GRAMMATICALITY JUDGMENTS

(a) Declarative sentence #1: FRIEND BUY CAR (SVO)

‘My friend bought a car.’

(i) Grammaticality judgments for NEG_{basic} / SVO order

Test
bht
(a) Sentence-final neg + bht: FRIEND BUY CAR NEG _{basic}
(b) Sentence-final neg without bht: FRIEND BUY CAR NEG _{basic}
bht
(c) Sentence-final neg + bht spreading over whole clause: FRIEND BUY CAR NEG _{basic}
bht
(d) Sentence-final neg + bht spreading over VP: FRIEND BUY CAR NEG _{basic}
bht
(e) Only bht; no spread; no manual neg: FRIEND BUY CAR
bht
(f) Spreading bht; no manual neg: FRIEND BUY CAR
bht
(g) Sentence-initial neg with bht: NEG _{basic} FRIEND BUY CAR
(h) Sentence-initial neg without bht: NEG _{basic} FRIEND BUY CAR
bht
(i) Sentence-initial neg + spreading bht: NEG _{basic} FRIEND BUY CAR
bht
(j) Preverbal neg with bht: FRIEND NEG _{basic} BUY CAR
(k) Preverbal neg without bht: FRIEND NEG _{basic} BUY CAR
bht
(l) Preverbal neg + spreading bht: FRIEND NEG _{basic} EAT MEAT

Note: bht = backward head tilt

(b) Declarative sentence #1: FRIEND CAR BUY (SOV)

‘My friend bought a car.’

(ii) Grammaticality judgments for NEG_{basic} / SOV order

Test
(a) Sentence-final neg + bht: FRIEND CAR BUY <u>NEG_{basic}</u> <u>bht</u>
(b) Sentence-final neg without bht: FRIEND CAR BUY NEG _{basic}
(c) Sentence-final neg + bht spreading over whole clause: FRIEND CAR BUY <u>NEG_{basic}</u> <u>bht</u>
(d) Sentence-final neg + bht spreading over VP: FRIEND CAR BUY <u>NEG_{basic}</u> <u>bht</u>
(e) Only bht; no spread; no manual neg: FRIEND CAR BUY <u>bht</u>
(f) Spreading bht; no manual neg: FRIEND CAR BUY <u>bht</u>
(g) Sentence-initial neg with bht: <u>NEG_{basic}</u> <u>bht</u> FRIEND CAR BUY
(h) Sentence-initial neg without bht: NEG _{basic} FRIEND CAR BUY
(i) Sentence-initial neg + spreading bht: <u>NEG_{basic}</u> <u>bht</u> FRIEND CAR BUY
(j) Preverbal neg with bht: FRIEND <u>NEG_{basic}</u> <u>bht</u> CAR BUY
(k) Preverbal neg without bht: FRIEND NEG _{basic} CAR BUY
(l) Preverbal neg + spreading bht: FRIEND <u>NEG_{basic}</u> <u>bht</u> CAR BUY

Note: bht = backward head tilt

(ii) Grammaticality judgments with NEG_{proh}:

(a) Declarative sentence #2: FRIEND EAT MEAT (SVO)

‘My friend eats meat.’

(iii) Grammaticality judgments for NEG_{proh} / SVO order

Test
(a) Sentence-final neg + hs: FRIEND EAT MEAT ^{hs} NEG _{proh}
(b) Sentence-final neg without hs: FRIEND EAT MEAT NEG _{proh}
(c) Sentence-final neg + hs spreading over whole clause: FRIEND EAT MEAT ^{hs} NEG _{proh}
(d) Sentence-final neg + hs spreading over VP: FRIEND EAT MEAT ^{hs} NEG _{proh}
(e) Only hs; no spread; no manual neg: FRIEND EAT MEAT ^{hs}
(f) Spreading hs; no manual neg: FRIEND EAT MEAT ^{hs}
(g) Sentence-initial neg with hs: ^{hs} NEG _{proh} FRIEND EAT MEAT
(h) Sentence-initial neg without hs: NEG _{proh} FRIEND EAT MEAT
(i) Sentence-initial neg + spreading hs: ^{hs} NEG _{proh} FRIEND EAT MEAT
(j) Preverbal neg with hs: FRIEND ^{hs} NEG _{proh} EAT MEAT
(k) Preverbal neg without hs: FRIEND NEG _{proh} EAT MEAT
(l) Preverbal neg + spreading hs: FRIEND ^{hs} NEG _{proh} EAT MEAT

Note: hs = headshake

(b) Declarative sentence #2: FRIEND MEAT EAT (SOV)

‘My friend eats meat.’

(iv) Grammaticality judgments for NEG_{proh} / SOV order

Test
(a) Sentence-final neg + hs: FRIEND MEAT EAT <u>NEG_{proh} hs</u>
(b) Sentence-final neg without hs: FRIEND MEAT EAT NEG _{proh}
(c) Sentence-final neg + hs spreading over whole clause: <u>FRIEND MEAT EAT NEG_{proh} hs</u>
(d) Sentence-final neg + hs spreading over VP: FRIEND MEAT <u>EAT NEG_{proh} hs</u>
(e) Only hs; no spread; no manual neg: FRIEND MEAT EAT hs
(f) Spreading hs; no manual neg: FRIEND <u>MEAT EAT hs</u>
(g) Sentence-initial neg with hs: <u>NEG_{proh} hs</u> FRIEND MEAT EAT
(h) Sentence-initial neg without hs: NEG _{proh} FRIEND MEAT EAT
(i) Sentence-initial neg + spreading hs: <u>NEG_{proh} hs</u> FRIEND MEAT EAT
(j) Preverbal neg with hs: FRIEND <u>NEG_{proh} hs</u> MEAT EAT
(k) Preverbal neg without hs: FRIEND NEG _{proh} MEAT EAT
(l) Preverbal neg + spreading hs: FRIEND <u>NEG_{proh} hs</u> MEAT EAT

Note: hs = headshake